Separable Rank of Radial Kernels: Machine-Checked Characterization in Lean 4 and Applications

GitHub Copilot

Source archive (CC0 1.0): https://github.com/OlliNiemitalo/kernel-rank-lean

November 14, 2026

Abstract

Filtering an image or three-dimensional wave field directly with a round (radially symmetric) kernel $K(x) = \varphi(\|x\|^2)$ of width W costs W^d multiply–adds per output sample. When a 2-D kernel is a product $h(x_0)v(x_1)$ of a horizontal and a vertical 1-D filter, it can instead be applied as two cheap 1-D passes; the *separable rank* R is the smallest number of such coordinate-product terms needed to sum to K. On a d-dimensional grid, rank R reduces W^d taps to dRW one-dimensional taps per sample (for example, a 9-tap-wide 3-D stencil drops from $9^3 = 729$ taps to $3 \times 4 \times 9 = 108$ taps when $R = 4$). Ordinary bell-shaped Gaussians $e^{-\alpha\|x\|^2}$ have rank 1, but cannot form flat-top camera defocus disks or ringed wave stencils; practical designs therefore sum damped radial waves $e^{-\alpha\|x\|^2}\cos(\omega\|x\|^2 + \theta)$ (*complex-rate Gaussians*), or more generally polynomials in $s = \|x\|^2$ times real or complex exponentials (*exponential polynomials*).

We determine which continuous radial kernels on $\mathbb{R}^d$ ($d \geq 2$) have finite separable rank, and how many product terms each requires. First, $K(x) = \varphi(\|x\|^2)$ has finite rank if and only if $\varphi(s)$ ($s \geq 0$) is a finite sum of exponential polynomials; conversely, decaying complex-rate Gaussians with one shared width and equally spaced frequencies can approximate any continuous radial kernel that fades to zero at large radius. Second, for a single complex-rate Gaussian in d dimensions, expanding real and imaginary parts naively takes 2^{d-1} real products (4 in 3-D), whereas an alternating angle-sampling identity achieves the exact minimum of d real products (3 in 3-D); multiplying by a degree-k polynomial in $\|x\|^2$ adds $(d-1)k$ terms per real exponential and $2(d-1)k$ terms per complex-conjugate pair. Third, allowing sequences of lower-rank kernels that converge uniformly on bounded regions (*border rank*) never lowers rank in 2-D, whereas in $d \geq 3$ dimensions it strictly reduces the cost of polynomial-weighted profiles: for $\|x\|^2\cos(\|x\|^2)$ in 3-D, exact real rank is 7, limits of radial Gaussians take 6 terms, and general real separable limits take 5. All 32 mathematical theorems are verified by the Lean 4 computer proof assistant; we then test their concrete predictions on finite pixel/voxel grids and 16-bit hardware in reproducible (unverified) Python/Triton experiments: on an RTX 4070 Ti Super GPU, orthogonalizing 2-D filter terms cuts 16-bit floating-point (`float16`) rounding error on bright bokeh highlights from 7.9/255 to 0.31/255 (0.33 ms on a 1920×1080 image at $R = 4$), and exact 3-term-per-Gaussian 3-D acoustic stencils cut step time by 24% (1.51 ms vs. 1.98 ms on 256^3 voxels) while keeping directional wave-speed error below 0.2% in spatially varying media where global fast Fourier transforms (`cuFFT`) do not apply. Code, proofs, and data are public domain (CC0 1.0) at https://github.com/OlliNiemitalo/kernel-rank-lean.

1 Introduction

Convolving a d-dimensional array with a dense $W \times \cdots \times W$ spatial stencil requires W^d multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-R

coordinatewise decomposition

$$K(x_0, \dots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the d-dimensional convolution reduces to R cascades of d one-dimensional passes, costing dRW operations per sample (or $O(dR)$ operations independent of W when the one-dimensional factors admit recursive causal/anti-causal IIR realizations [19, 77]). Viewing (1) over a commutative semiring $\mathbb{K}$, the *separable rank* $\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$ is the least $R \in \mathbb{N}$ for which such a decomposition exists: matrix rank [70] when $d = 2$ and $\mathbb{K}$ is a field, and functional canonical polyadic (CP) or *separation rank* [32, 5, 39, 29] when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \dots, x_{d-1}) \in \mathbb{R}^d,$$

classical characterizations by Maxwell [49] and Kannappan and Sahoo [36] imply that a continuous nonvanishing radial kernel with $\operatorname{srank}_{\mathbb{R}}(K) = 1$ has a Gaussian profile $\varphi(s) = ae^{-\alpha s}$. Real-rate Gaussians ($\alpha > 0$), however, are strictly monotone and convex in the squared radius $s = \|x\|^2$; single Gaussians and nonnegative Gaussian mixtures cannot reproduce flat passbands, sharp cutoffs, or oscillatory diffraction rings [78]. In image processing, real-time graphics, and numerical wave propagation, Niemitalo [54, 57, 58], Garcia [27], and related filter-design studies [44] approximate flat-top and oscillatory radial profiles—such as camera defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\operatorname{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe cancellation while maintaining finite separable rank.

1.1 Visual overview: components, rendering terms, and streaming intermediates

Before introducing technical definitions, Figure 1 illustrates in plain visual terms how one-dimensional filters build a circularly symmetric 2-D kernel and why the distinction between *components* and *rendering terms* governs practical GPU execution:

1. **Mathematical "Components" view (Figure 1a):** Replacing a real decay $-\alpha$ by a complex rate $\lambda = -\alpha + i\omega$ preserves the coordinate split $e^{\lambda(x^2+y^2)} = e^{\lambda x^2} e^{\lambda y^2}$ while oscillating radially as $e^{-\alpha r^2}\cos(\omega r^2)$ and $e^{-\alpha r^2}\sin(\omega r^2)$ ($r^2 = x^2 + y^2$). Writing $g(x) = \sqrt{\beta}\, e^{\lambda x^2} = g_{\mathrm{re}}(x) + i g_{\mathrm{im}}(x)$, taking the real part of $g(x)g(y)$ uses two real 1-D×1-D outer products:

   $$\operatorname{Re}\big(g(x)g(y)\big) = g_{\mathrm{re}}(x)\, g_{\mathrm{re}}(y) + g_{\mathrm{im}}(x)\, \big(-g_{\mathrm{im}}(y)\big).$$

   Neither outer product is circular on its own—each is a rectangular plaid of positive (blue) and negative (red) stripes—yet adding the pair cancels the grid corners to form a circularly symmetric component (u_1: wide dome; u_2: concentric sharpening rings), whose sum $u_1 + u_2$ approximates a flat-top disc K. (When a component multiplies $e^{\lambda r^2}$ by a polynomial $P(r^2)$ of degree k, expanding $(x^2 + y^2)^q$ naively across monomials would produce $(k+1)(k+2)$ real products, whereas collecting powers of x^2 merges them into $2(k+1)$ real products, or $k+1$ when $\omega = 0$.)

2. **Computational "Rendering terms" view (Figure 1b):** A renderer never computes the circular components u_1, u_2 separately—only the total sum $K(x, y) = \sum_{n=1}^{R} \tilde{h}_n(x)\tilde{v}_n(y)$. In high-accuracy approximations (such as $n_{\mathrm{cx}} = 6$ complex Gaussians, $R = 12$), individual

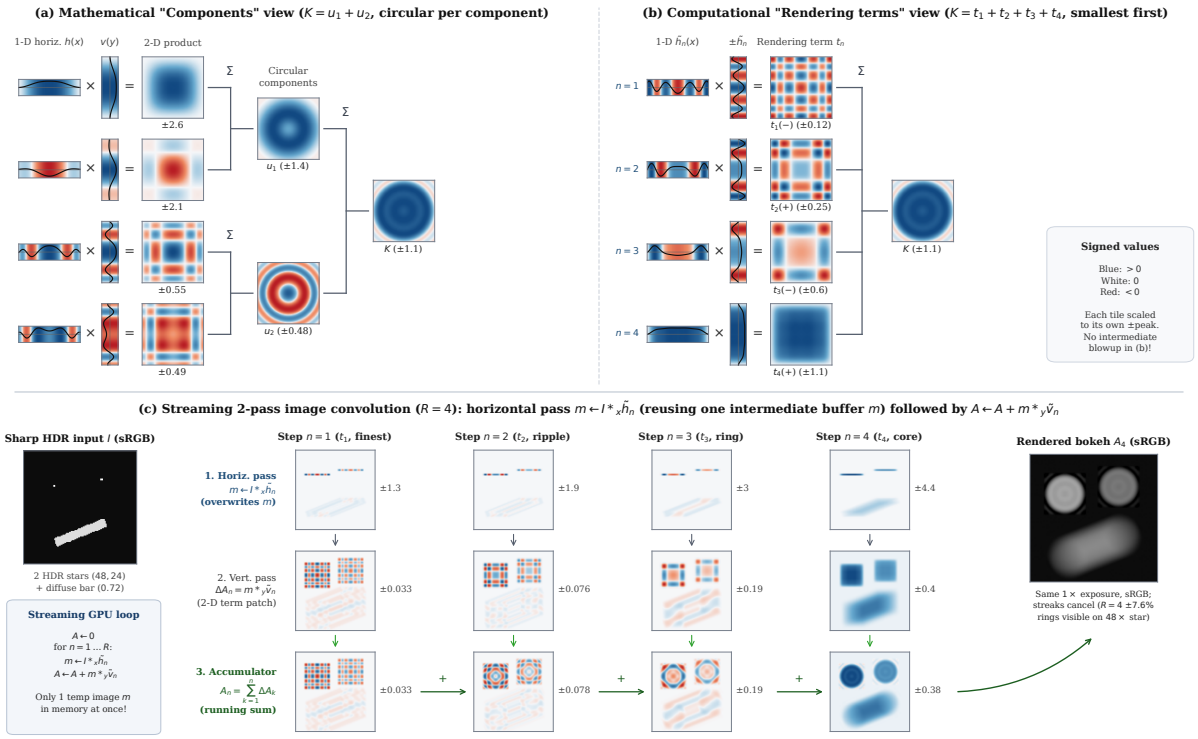


Figure 1: Visual guide to separable radial filtering on a two-complex-Gaussian ($n_{cx} = 2$, exact rank $R = 4$) circular disc-blur kernel [58]: (a) mathematical *Components* view, where pairs of 1-D outer products $h(x)v(y)$ form circularly symmetric concentric-ring components u_1, u_2 that sum to the flat-top disc K; (b) computational *Rendering terms* view, where orthogonalizing rowSpan(K) yields four direct real products $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered smallest-first so intermediate amplitudes stay bounded; (c) streaming two-pass linear-light HDR blur using *one* overwritten horizontal intermediate buffer $m \leftarrow I *_x \tilde{h}_n$ followed by vertical accumulation $A \leftarrow A + m *_y \tilde{v}_n$ (with I and A_4 displayed at identical 1× linear exposure through the IEC 61966-2-1 sRGB transfer function; all axis-aligned streaks cancel, while $R = 4$'s ±7.6% pass/stopband ripple remains visible on the 48× HDR star).

components u_i grow hundreds of times taller than K and cancel against each other, causing severe precision loss in 16-bit half-float (`float16`) GPU buffers. Orthogonalizing the 1-D filter subspace (`separableChannels` in [58]; Section 5.3) rotates the R filter pairs into orthogonal *rendering terms* $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered from smallest to largest magnitude ($n = 1, \ldots, 4$): every term stays comparable to the kernel's own peak (±0.12 to ±1.1 in Figure 1b, vs. ±2.6 in Figure 1a).

3. **Streaming two-pass convolution with one intermediate buffer (Figure 1c):** In the minimal-memory streaming schedule (used, e.g., in fragment-shader pipelines [58]), the GPU initializes an output accumulator $A \leftarrow 0$ and loops over $n = 1, \ldots, R$ (smallest term first). In step n, a 1-D horizontal pass writes $m(x, y) \leftarrow (I *_x \tilde{h}_n)(x, y)$, smearing bright point stars horizontally into 1-D signed streaks; *the same temporary buffer m is overwritten on every step*, keeping DRAM scratch constant at one image regardless of R (in Section 5, `gpu_kernels.py` further tiles the image into L2-cache-resident strips so all R intermediates stay on chip). A 1-D vertical pass then adds $\Delta A_n = m *_y \tilde{v}_n$ into A: steps $n = 1, 2, 3$ build a hollow sharpening ring A_3, and step $n = 4$ adds the soft core so all horizontal and vertical streaks cancel in A_4.

Table 1: Overview of main theoretical, formalization, and numerical results ($d \geq 2$, degree $k = \deg P$).

Regime	Setting	Main result	Ref.
Finite-rank form	Cont. φ on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} < \infty \iff \varphi(s) = \sum_u \operatorname{Re}(P_u(s)e^{\lambda_u s})$ $(s \geq 0)$	Thm. 2.3
Complex rank	$\psi : \mathbb{R} \to \mathbb{C}$, $\mathbb{R}^d$	$(d-1)k+1$ per distinct rate $\mu \in S$	Thm. 2.4
Real rank ($\operatorname{Im}\lambda$=0)	$P \in \mathbb{R}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = (d-1)k+1$ (= 1 at $k=0$)	Thm. 2.5
Real rank ($\operatorname{Im}\lambda$>0)	$P \in \mathbb{C}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = 2(d-1)k+d$ (= d vs. 2^{d-1} at $k=0$)	Thm. 2.5, Prop. 2.14
Real–complex gap	Conjugate pair	$c_d(\lambda, P) - 2((d-1)k+1) = d-2$ (all $k \geq 0$)	Rem. 2.17
Border-rank bound	Any list Γ	Multi-Gaussian limit $\leq B_d(\Gamma) = C_d(\Gamma) - (d-2)\sum_u k_u$	Thm. 2.8
Exact border rank	$\mathbb{C}$; real λ_u; $\mathbb{R}^3$	$\mathbb{C}$ & real λ_u: sharp; $\mathbb{R}^3$ $\|x\|^2 \cos\|x\|^2$ is $5 < B_3$=6	Thm. 2.9
Universal approx.	$\psi \in C_0(\mathbb{R}_{\geq 0})$	Equal-width harmonic Gaussians approx. $\psi(\|x\|^2)$ on $\mathbb{R}^d$	Thm. 2.11
2-D diff. order	$d=2$, D on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} = \deg p_{\min}$ for minimal $p_{\min}(D)\varphi_{\mathbb{C}} = 0$	Thm. 2.6
Lean 4 proof	`KernelRankLean`	32 theorems (6,935 loc), zero `sorry`, classical axioms	Sec. 4
Empirical tests	Blur; 3-D wave	Tests Thms. 2.3–2.9 on grids/`fp16`; hom. & $c(\mathbf{x})$ wave	Secs. 5–7

1.2 Plain-language summary of main results and paper organization

Table 1 summarizes our results, organized into two formally verified parts—Mathematical Theory (Sections 2–3) and Lean 4 Formalization (Section 4)—followed by an explicitly demarcated empirical validation part on finite grids and GPU hardware (Sections 5–7, Appendix A):

1. **Part I — Machine-checked mathematical characterization, border rank, and universal approximation (Sections 2–3):** Exponential-polynomial profiles in $s = \|x\|^2$ are *universal* for exact finite rank (Theorem 2.3), while equal-width harmonic complex Gaussians uniformly approximate any continuous radial profile vanishing at infinity across $\mathbb{R}^d$ (Theorem 2.11). We establish closed-form exact d-way separable ranks over $\mathbb{C}$ (Theorem 2.4) and over $\mathbb{R}$ (Theorem 2.5), with a plain-language cost guide (Section 2.3) and worked $d = 3$ examples: an alternating root-of-unity identity reduces each complex Gaussian from 2^{d-1} naive real products to d and each degree-k nonreal component to $2(d-1)k+d$ real products (a constant $d-2$ real-over-complex penalty). For border rank, multi-Gaussian limits give upper bound $B_d(\Gamma)$ (Theorem 2.8), which is sharp over $\mathbb{C}$ and for all purely real-rate profiles, whereas on $\mathbb{R}^3$ an explicit 5-term non-Gaussian limit shows that $\|x\|^2\cos(\|x\|^2)$ has exact real border rank $5 < B_3 = 6 < C_3 = 7$ (Theorem 2.9 and Remark 2.10).
2. **Part II — Formalization architecture and axiom audit (Section 4):** All 32 mathematical theorems in Sections 2–3 are machine-checked in Lean 4 using Mathlib's Banach-algebra Bochner integral for the continuous shift semigroup step, Schur-complement and adjugate-trace obstructions for border-rank lower bounds, and periodic Fourier density on `AddCircle` for universal approximation, with every theorem mapped to its Lean identifier and audited for classical axiom dependencies (Tables 5–6).
3. **Part III — Empirical validation of theoretical predictions on discrete grids and GPUs (Sections 5–7, Appendix A):** Unverified reproducible experiments test how the continuous-domain invariants behave under grid truncation, 16-bit floating-point arithmetic, and L2-cached GPU execution (Table 7): planar steerable pass-count bounds (Section 5.2), 2-D bokeh `float16` conditioning, discrete-to-continuous convergence, and warm-starting (Section 5.3), and homogeneous and spatially varying $c(\mathbf{x})$ 3-D acoustic wave propagation compared against fourth-order FDTD and `cuFFT` (Section 5.4). Sections 6–7 examine practical limitations, open questions, and conclusions, with explicit coefficients in Appendix A.

1.3 Related work

Classical radial characterizations, continuous Kronecker theorems, and functional equations. Maxwell [49] and Kannappan and Sahoo [36] proved that measurable or contin-

uous rotation-invariant rank-1 functions are Gaussians, while Schoenberg [65] characterized completely monotone radial profiles as positive Gaussian mixtures. Under the coordinate surjection $s_j = x_j^2$, radial separability on $\mathbb{R}^d$ becomes the d-variable additive Hankel functional equation $\varphi(s_0 + \cdots + s_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} g_{n,j}(s_j)$ on $[0,\infty)^d$. In two variables, functional equations $f(x+y) = \sum_{n=1}^{R} X_n(x)Y_n(y)$ go back to Levi-Civita [46], Stone [66], Anselone and Korevaar's characterization of finite-dimensional translation-invariant subspaces [3], Laird [43], and Laczkovich [42]. In operator and moment theory, Rochberg [64], Andersson and Carlsson [2], and Mourrain [51] established continuous and multivariate analogues of Kronecker's 1881 theorem connecting finite-rank Hankel and convolution operators to polynomial-exponential symbols and Artinian Gorenstein algebras; whereas bounded L^2 integral-operator formulations assume square-integrability or bounded domains, our pointwise algebraic formulation on $[0,\infty)^d$ characterizes all continuous profiles on $[0,\infty)$—including undamped or polynomially growing profiles such as $\|x\|^2\cos(\|x\|^2)$—via operator-norm continuous shift semigroups without assuming differentiability or decay.

Separation rank, tensor-train/Tucker formats, random features, and exponential sums. Beylkin and Mohlenkamp [5], Kolda and Bader [39], Oseledets [59], Hackbusch [29], and Khoromskij [37] develop canonical polyadic (CP), Tucker, and tensor-train (TT) representations for multivariate radial operators, building on nonlinear exponential-sum approximation [12, 28, 13, 6, 47] and skeleton/cross approximations of smooth kernels [71, 74, 69], while in machine learning Rahimi and Recht [61] approximate shift-invariant positive-definite radial kernels via Bochner's theorem and random Fourier features. Three structural comparisons position our notion of coordinatewise separable rank ($\operatorname{srank}_{\mathbb{K}}$, Definition 2.1):

1. *Pairwise ML kernel features vs. coordinatewise convolution stencils:* Random Fourier features [61] and Nyström approximations factor $K_0(x-y) = \varphi(\|x-y\|^2)$ across *sample pairs* $(x,y) \in \mathbb{R}^d \times \mathbb{R}^d$ as $\sum_{r=1}^{R} z_r(x)z_r(y)$ to reduce $N \times N$ Gram-matrix cost from $O(N^2)$ to $O(NR)$; here even a single Gaussian $e^{-\alpha\|x-y\|^2}$ has *infinite* exact two-argument rank on $\mathbb{R}^d \times \mathbb{R}^d$ (since $e^{2\alpha x\cdot y}$ is not a finite sum of products $u_n(x)v_n(y)$), and each Fourier feature $e^{i\omega\cdot(x-y)} = \prod_{j=0}^{d-1} e^{i\omega_j(x_j-y_j)}$, though coordinatewise rank-1 in $z = x-y$, is non-decaying ($|e^{i\omega\cdot z}| = 1$) and linear in z_j rather than a localized radial function of $\|z\|^2$. By contrast, as a *Cartesian convolution stencil* $z \mapsto \varphi(\|z\|^2)$ across $(z_0,\dots,z_{d-1}) \in \mathbb{R}^d$, $e^{-\alpha\|z\|^2}$ has exact d-way rank 1, and decaying complex Gaussians $e^{(-\alpha+i\omega)\|z\|^2}$ localize in space while remaining radial (Theorem 2.11).
2. *Coordinatewise CP rank vs. Tensor-Train (TT) and Tucker ranks:* For an additive Hankel kernel $H(s_0,\dots,s_{d-1}) = f(s_0+\cdots+s_{d-1})$ with $n = \dim_{\mathbb{K}} \mathcal{W}(f) < \infty$, every prefix/suffix unfolding $(s_0,\dots,s_{q-1}) \mid (s_q,\dots,s_{d-1})$ depends only on $(\sum_{j<q} s_j, \sum_{j\geq q} s_j)$, so all TT unfolding ranks [59] and all mode-j Tucker ranks [39, 29] equal the 2-D rank n independent of d. However, Tucker retains an n^d core tensor, and TT convolution contracts $n \times n$ transfer matrices between adjacent axes ($O(dnW + dn^2)$ per voxel, coupling all n channels at every axis). Coordinatewise CP decompositions (1) forbid cross-channel mixing between axes—executing R independent scalar 1-D pass cascades (dRW taps)—and Theorems 2.4–2.5 show this stricter format pays only a linear factor in d, $\operatorname{srank}_{\mathbb{K}} = (d-1)n - (d-2)m_{\mathrm{irr}}$ (vs. $\leq n^{d-1}$ for generic Tucker-to-CP expansion).
3. *Exact converse and signed/oscillatory profiles:* Prior sum-of-Gaussians and RFF methods construct upper-bound approximations (typically for positive-definite or monotone kernels), whereas Theorems 2.3–2.5 establish both the exact converse ($\operatorname{srank}_{\mathbb{R}} < \infty \iff$ exponential polynomial) and exact d-way real CP rank across all signed and polynomial-oscillatory radial profiles.

Algebra multiplicative complexity, Hankel/Waring tensors, real vs. complex gaps, and border rank. Once necessity identifies the $\mathbb{K}[D]$-module $\mathcal{W}(f)$ with the commutative quotient algebra $\mathcal{A} = \mathbb{K}[X]/(p_{\min,\mathbb{K}}(X))$ (where $p_{\min,\mathbb{K}}$ has m_{irr} distinct irreducible factors over $\mathbb{K}$), the multilinear core of our d-way rank problem is the $(d-1)$-fold multiplication tensor $\mathcal{A}^{d-1} \times \mathcal{A}^* \to \mathbb{K}$, $(u_1, \ldots, u_{d-1}, \ell) \mapsto \ell(u_1 \cdots u_{d-1})$. At $d = 3$ (2-fold bilinear multiplication), Winograd [75], Fiduccia and Zalcstein [23], and Alder and Strassen [1] (with nilradical extensions by Bläser [9]; see Bürgisser, Clausen, and Shokrollahi [15], Thm. 17.20) proved that multiplying in $\mathbb{K}[X]/(p)$ requires $2\dim_{\mathbb{K}} \mathcal{A} - m_{\mathrm{irr}}$ bilinear multiplications, yielding our $d = 3$ costs $2k+1$ and $4k+3$. Recently, Bläser, Mayer, and Shringi [10] generalized the Alder–Strassen lower bound to $(d-1)$-fold multiplication in arbitrary associative algebras, establishing the lower bound $(d-1)\dim_{\mathbb{K}} \mathcal{A} - (d-2)m_{\mathrm{irr}}$; for singly generated commutative algebras over sufficiently large fields, each primary factor attains $(d-1)\dim_{\mathbb{K}} \mathcal{A}_u - (d-2)$ via field-node Lagrange interpolation [10, Cor. 11], summing by Chinese remainder decomposition across the m_{irr} primary factors. Theorems 2.4 and 2.5 connect continuous radial kernels on $\mathbb{R}^d$ to this invariant, replace ill-conditioned real Vandermonde inversion on nonreal factors $((X+\alpha)^2+\omega^2)^{k+1}$ by closed-form unit-circle root-of-unity functional syntheses (Propositions 2.13–2.14), and machine-check both the lower and upper bounds in Lean 4. This complements apolarity and discrete Hankel/Waring tensor theory [17, 34, 41, 53], real-versus-complex rank separations [18, 20, 63, 11], border-rank gaps on W-type and local-algebra tensors [8, 45, 14, 79], and algebraic degeneration bounds on higher-order structure tensors [16].

Steerable filters, acoustic FDTD stencils, semigroups, and formalized mathematics. In vision and graphics, multi-stage separable planar filters [70], steerable and complex-derivative filters [26, 38, 60, 7, 62, 72], anisotropic Gaussians [44], and complex-Gaussian circular bokeh filters [54, 27, 55, 56, 57, 58] motivate our analytical and GPU blur studies; in computational acoustics and geophysics, dispersion-relation-preserving and k-space methods [68, 67], compact 27-point and higher-order 3-D room-acoustics FDTD stencils [40, 73, 30, 31], and low-rank wave extrapolators [25] provide our 3-D wave-propagation baselines. Finally, our formalization builds on classical Banach-algebra semigroups [52, 22], Lean 4 and Mathlib [50, 48, 21], and formalized tensor-rank developments [4, 35, 76]: whereas Bentkamp, Blanchette, and Klakow [4] formalize CP-rank and generalized matricization bounds for deep convolutional arithmetic circuits in Isabelle/HOL, Jain [35] formalizes lower bounds (≥ 18, via a two-slice real pencil bound) and the Krawczyk principle in Lean 4 (paired with external rational inequality checks for rank ≤ 25) on $8 \times 8 \times 8$ octonion multiplication, and Yang et al. [76] prove $R_{\mathbb{F}_2} \geq 21$ for 3×3 matrix multiplication over $\mathbb{F}_2$ via finite-geometry certificates in Lean 4, our development treats *infinite-dimensional function spaces* on $\mathbb{R}^d$ and proves parametric closed-form ranks for all dimensions $d \geq 2$ and polynomial degrees $k \geq 0$ (Section 4).

2 Problem Formulation, Harmonized Notation, and Main Theorems

2.1 Separable rank, coordinate surjections, and canonical representations

Throughout the paper, we adhere to one uniform profile convention: $\varphi : \mathbb{R} \to \mathbb{R}$ denotes a *real-valued* radial profile, $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$ denotes its complexification, $\psi : \mathbb{R} \to \mathbb{C}$ denotes a general *complex-valued* radial profile, and $f : \mathbb{R} \to \mathbb{K}$ denotes a profile taking values in an arbitrary commutative semiring or field $\mathbb{K}$. Similarly, for any kernel $K : X \to \mathbb{R}$, we write $K_{\mathbb{C}}(x) = (K(x) : \mathbb{C})$ for its scalar extension to $\mathbb{C}$.

Definition 2.1 (Coordinatewise separable rank; Lean: `IsSumOfProducts`, `separableRank`)**.** Let ι be a finite index set, $(X_i)_{i\in\iota}$ a family of sets, and $\mathbb{K}$ a commutative semiring. A kernel

$K : (\prod_{i\in\iota} X_i) \to \mathbb{K}$ is a *sum of R products* ($R \in \mathbb{N}$) if there exist univariate factors $f_{n,i} : X_i \to \mathbb{K}$ ($0 \le n < R$, $i \in \iota$) such that $K(x) = \sum_{n=0}^{R-1} \prod_{i\in\iota} f_{n,i}(x_i)$ for all $x \in \prod_{i\in\iota} X_i$ (taking the empty sum 0 when $R = 0$). Its *separable rank* over $\mathbb{K}$ is the extended natural number

$$\operatorname{srank}_{\mathbb{K}}(K) := \inf\big\{R \in \mathbb{N} : K \text{ is a sum of } R \text{ products over } \mathbb{K}\big\} \in \mathbb{N} \cup \{\infty\},$$

where $\inf \emptyset = \infty$, and an equation $\operatorname{srank}_{\mathbb{K}}(K) = R$ with $R \in \mathbb{N}$ denotes equality in $\mathbb{N} \cup \{\infty\}$ under the canonical inclusion $\mathbb{N} \hookrightarrow \mathbb{N} \cup \{\infty\}$.

Separable rank is subadditive under finite sums and cannot increase when we substitute coordinate maps $y_i \mapsto g_i(y_i)$ into an existing decomposition $\sum_n \prod_i f_{n,i}(g_i(y_i))$. When every $g_i : Y_i \to X_i$ ($i \in \iota$) is *onto* (surjective), the reverse inequality holds as well: picking for each $x_i \in X_i$ at least one preimage $h_i(x_i) \in Y_i$ with $g_i(h_i(x_i)) = x_i$ turns any decomposition $\sum_n \prod_i \tilde{f}_{n,i}(y_i)$ of $K(g_0(y_0), \dots)$ back into $\sum_n \prod_i \tilde{f}_{n,i}(h_i(x_i)) = K(x)$ with the same number of terms:

$$\operatorname{srank}_{\mathbb{K}}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \operatorname{srank}_{\mathbb{K}}(K). \tag{2}$$

For $x_j \in \mathbb{R}$, the squaring map $x_j \mapsto x_j^2$ covers $\mathbb{R}_{\ge 0} = [0, \infty)$ (with right inverse $s_j \mapsto \sqrt{s_j}$). Applying (2) therefore equates the d-way separable rank of any radial kernel on $\mathbb{R}^d$ with that of the d-way additive Hankel kernel on $\mathbb{R}^d_{\ge 0}$:

$$\operatorname{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \operatorname{srank}_{\mathbb{K}}\left(s \in \mathbb{R}^d_{\ge 0} \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

Notice two immediate consequences of (3): (i) because $\|x\|^2 \ge 0$ never visits negative arguments, $K(x) = \varphi(\|x\|^2)$ depends only on $\varphi|_{[0,\infty)}$ and leaves $\varphi(s)$ for $s < 0$ completely unconstrained; and (ii) replacing $x_j \mapsto x_j^2$ by $x_j \mapsto x_j^2/a_j^2$ ($a_j \ne 0$) or $x_j \mapsto |x_j|^p$ ($p > 0$) is still onto $[0, \infty)$, so every theorem below holds verbatim for axis-aligned ellipsoidal kernels $f(\sum_j x_j^2/a_j^2)$ and ℓ_p superellipsoidal kernels $f(\sum_j |x_j|^p)$. Finally, for $d \ge 2$, setting $x_2 = \cdots = x_{d-1} = 0$ turns each d-way product $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ into the 2-way product $(c_n f_{n,0}(x_0)) f_{n,1}(x_1)$ with constant $c_n = \prod_{j=2}^{d-1} f_{n,j}(0)$, giving the 2-D slice lower bound:

$$\operatorname{srank}_{\mathbb{K}}\big(y \in \mathbb{R}^2 \mapsto f(y_0^2 + y_1^2)\big) \le \operatorname{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

Definition 2.2 (Canonical real representations and cost functions)**.** A *component* (`Component`) is a pair $(\lambda, P) \in \mathbb{C} \times \mathbb{C}[X]$, evaluating at $s \in \mathbb{R}$ as $\operatorname{Re}(P(s)e^{\lambda s})$ of degree $k = \deg P$ (multiplicity $k + 1$). It is *well-formed* if $P \ne 0$, $\operatorname{Im}\lambda \ge 0$, and $\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X]$. A finite list $\Gamma = [(\lambda_1, P_1), \dots, (\lambda_m, P_m)]$ is a *representation* (`IsRepresentation`, written `cs : List Component` in Lean and denoted Γ here to stay distinct from the rank R and the field $\mathbb{R}$) if every (λ_u, P_u) is well-formed and the rates $\lambda_1, \dots, \lambda_m$ are pairwise distinct; its radial profile is $\operatorname{profile}(\Gamma, s) = \sum_{u=1}^{m} \operatorname{Re}(P_u(s)e^{\lambda_u s})$.

To motivate the two integer cost functions assigned to each component (λ, P) in dimension $d \ge 2$: $c_d(\lambda, P)$ counts the *exact* minimum number of real d-way product terms $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ needed to represent $x \mapsto \operatorname{Re}(P(\|x\|^2)e^{\lambda\|x\|^2})$ on $\mathbb{R}^d$ (1 product for a pure real Gaussian at $\operatorname{Im}\lambda = 0, \deg P = 0$ and d real products for a pure complex Gaussian at $\operatorname{Im}\lambda > 0, \deg P = 0$, plus $(d-1)\deg P$ or $2(d-1)\deg P$ for polynomial powers of $\|x\|^2$), whereas $b_d(\lambda, P)$ counts the number of real product terms when $P(\|x\|^2)e^{\lambda\|x\|^2}$ is instead approximated on bounded sets by finite differences of $\deg P + 1$ pure Gaussians (1 real product per real Gaussian, d per complex Gaussian):

$$c_d(\lambda, P) = \begin{cases} (d-1)\deg P + 1, & \operatorname{Im}\lambda = 0, \\ 2(d-1)\deg P + d, & \operatorname{Im}\lambda > 0, \end{cases} \qquad b_d(\lambda, P) = \begin{cases} \deg P + 1, & \operatorname{Im}\lambda = 0, \\ d(\deg P + 1), & \operatorname{Im}\lambda > 0, \end{cases} \tag{5}$$

Table 2: Harmonized symbol glossary and cost specializations ($d \geq 2$, $k = \deg P$).

Mathematical symbol	Lean 4 identifier	Meaning / specialization
$\mathrm{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$	`separableRank K (ENat)`	Coordinatewise separable rank over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$
φ, $\varphi_{\mathbb{C}}$, ψ, f	φ : $\mathbb{R} \to \mathbb{R}$, ψ : $\mathbb{R} \to \mathbb{C}$	Real profile, complexification, complex profile, $\mathbb{K}$-profile
$W = 2L + 1$; $s = \|x\|^2$	—; $\sum_j$ `x j ^ 2`	1-D discrete tap width on $[-L, L]$; squared radius on $\mathbb{R}^d$
$\mathcal{W}(f)$	`translateSpan f`	Half-line translate span $\mathrm{span}_{\mathbb{K}}\{u \mapsto f(u+t) : t \geq 0\}$
$\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$	`cs : List Component`	Canonical real representation (λ_u distinct, $\mathrm{Im}\,\lambda_u \geq 0$)
n_{re}, n_{cx}; m_{irr}	—	Real and nonreal component counts; $m_{\mathrm{irr}} = n_{\mathrm{re}} + n_{\mathrm{cx}}$
$C_d(\Gamma) = \sum_u c_d(\lambda_u, P_u)$	`multiTotalCost d cs`	Exact d-D real radial rank (Theorem 2.5)
$C_2(\Gamma)$; $n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$	`totalCost`; `gaussianTotalCost`	Specializations at $d = 2$ and at $\deg P_u = 0$
$B_d(\Gamma) = \sum_u b_d(\lambda_u, P_u)$	`borderTotalCost d cs`	Multi-Gaussian border bound $C_d(\Gamma) - (d-2)\sum_u \deg P_u$
$H_P(s_0; X)$; $E_k(s_j; X)$	`revHassePoly`; `truncExpPoly`	Reversed Hasse poly.; truncated exp $\sum_{q=0}^{k} s_j^q X^q/q!$
$M^{\mathbb{C}}_{d,k}$, $M^{\mathbb{R}}_{d,k}$	$(d-1)k+1$, $2(d-1)k+d$	Complex and nonreal root-of-unity node counts

with total costs $C_d(\Gamma) = \sum_{u=1}^{m} c_d(\lambda_u, P_u)$ (`multiTotalCost`) and $B_d(\Gamma) = \sum_{u=1}^{m} b_d(\lambda_u, P_u)$ (`borderTotalCost`). When every $\deg P_u = 0$ (*multi-Gaussian* representation with n_{re} real and n_{cx} nonreal rates), $C_d(\Gamma) = B_d(\Gamma) = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$.

Why are the conditions $\mathrm{Im}\,\lambda \geq 0$ and $(\mathrm{Im}\,\lambda = 0 \implies P \in \mathbb{R}[X])$ canonical? Because $\mathrm{Re}(\overline{P}(s)e^{\bar{\lambda}s}) = \mathrm{Re}(P(s)e^{\lambda s})$ and $\mathrm{Re}(iae^{\mu s}) = 0$ for $a, \mu \in \mathbb{R}$, unrestricted pairs over-parameterize real profiles. Under Definition 2.2, each real rate ($\mathrm{Im}\,\lambda = 0$) contributes a single term $P(s)e^{\lambda s}$ over $\mathbb{C}$, while each upper-half-plane rate ($\mathrm{Im}\,\lambda > 0$) expands via $\mathrm{Re}\, z = \frac{1}{2}(z + \bar{z})$ to the conjugate pair $\frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda}s}$ at distinct rates $\lambda \neq \bar{\lambda}$. Since exponential polynomials at distinct complex rates are $\mathbb{C}$-linearly independent on $(0, \infty)$, every continuous real profile of finite rank determines its unordered set of well-formed components uniquely. Table 2 summarizes all symbols and their Lean 4 counterparts.

2.2 Statement of main theorems and algebraic identities

Theorem 2.3 (d-D Finite-Rank Classification; Lean: `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. Then $\mathrm{srank}_{\mathbb{R}}\big(x \in \mathbb{R}^d \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)\big) < \infty$ if and only if there exists a representation $\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ such that $\varphi(s) = \sum_{u=1}^{m} \mathrm{Re}(P_u(s)e^{\lambda_u s})$ for all $s \geq 0$.*

Theorem 2.4 (d-D Complex Radial Rank; Lean: `separableRank_radial_complex_eq_iff`)**.** *Let $d \geq 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\mathrm{srank}_{\mathbb{C}}\left(x \in \mathbb{R}^d \mapsto \psi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = R$$

if and only if there exist a finite set $S \subset \mathbb{C}$ and nonzero polynomials $(P_\mu)_{\mu \in S} \subset \mathbb{C}[X]$ such that $\psi(s) = \sum_{\mu \in S} P_\mu(s)e^{\mu s}$ for all $s \geq 0$ and $\sum_{\mu \in S}\big((d-1)\deg P_\mu + 1\big) = R$.

Theorem 2.5 (Exact d-D Real Radial Rank; Lean: `separableRank_radial_eq_iff`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\mathrm{srank}_{\mathbb{R}}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = R$$

if and only if there exists a canonical representation $\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ (distinct rates with $\mathrm{Im}\,\lambda_u \geq 0$, nonzero P_u, and $P_u \in \mathbb{R}[X]$ when $\mathrm{Im}\,\lambda_u = 0$) such that $\varphi(s) = \sum_{u=1}^{m} \mathrm{Re}(P_u(s)e^{\lambda_u s})$ for all $s \geq 0$ and

$$C_d(\Gamma) = \sum_{\mathrm{Im}\,\lambda_u = 0} \big((d-1)\deg P_u + 1\big) + \sum_{\mathrm{Im}\,\lambda_u > 0} \big(2(d-1)\deg P_u + d\big) = R.$$

In particular, for a multi-Gaussian representation ($\deg P_u = 0$) with n_{re} real and n_{cx} nonreal rates, $\operatorname{srank}_{\mathbb{R}} = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$, *and for* $d = 2$, $\operatorname{srank}_{\mathbb{R}} = C_2(\Gamma) = \operatorname{srank}_{\mathbb{C}}(x \mapsto \varphi_{\mathbb{C}}(x_0^2 + x_1^2))$.

Theorem 2.6 (2-D Minimal Differential Annihilator; Lean: `exists_minimal_differential_annihilator`)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0,\infty)$ with $\operatorname{srank}_{\mathbb{R}}(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = R < \infty$. For any $q(X) = \sum_{k=0}^{\deg q} q_k X^k \in \mathbb{C}[X]$ and $g : \mathbb{R} \to \mathbb{C}$, define the half-line differential operator (`differentialOperator`)*

$$\big(q(D)g\big)(t) := \sum_{k=0}^{\deg q} q_k\, D^k_{[0,\infty)} g(t),$$

where $D^0_{[0,\infty)} g = g$ and $D^k_{[0,\infty)} g(t)$ (`iteratedDerivWithin k g (Set.Ici 0) t`, $k \geq 1$) is Mathlib's totalized k-th derivative within $[0,\infty)$ (returning 0 where a within-derivative fails to exist, and agreeing for $g \in C^\infty([0,\infty))$ with the classical interior derivative at $t > 0$ and the right-hand limit $\lim_{h\to 0^+} h^{-1}(D^{k-1}_{[0,\infty)} g(h) - D^{k-1}_{[0,\infty)} g(0))$ at $t = 0$). Then $\varphi_{\mathbb{C}}|_{[0,\infty)} \in C^\infty([0,\infty))$ (so every $D^k_{[0,\infty)}\varphi_{\mathbb{C}}(t)$ is a genuine classical derivative), and there exists a unique monic polynomial $p_{\min} \in \mathbb{C}[X]$ of degree R such that for every $q \in \mathbb{C}[X]$, $\big(q(D)\varphi_{\mathbb{C}}\big)(t) = 0$ for all $t \geq 0$ if and only if $p_{\min} \mid q$.

Remark 2.7 (Automatic smoothness, domain of D, and identity with the generator's characteristic polynomial)**.** Theorem 2.6 assumes *only* continuity of φ on $[0,\infty)$ and $\operatorname{srank}_{\mathbb{R}} = R < \infty$, not differentiability: Section 3.1 shows by integral smoothing of the shift semigroup that $\varphi_{\mathbb{C}}$ agrees on $[0,\infty)$ with an analytic exponential polynomial $\sum_{\mu\in S} P_\mu(t)e^{\mu t}$, so every classical right-hand derivative $D^k_{[0,\infty)}\varphi_{\mathbb{C}}(t)$ ($t \geq 0$, $k \geq 0$) exists automatically! (Conversely, any continuous profile on $[0,\infty)$ that fails to be C^∞—such as $|s-1|$—has $\operatorname{srank}_{\mathbb{R}} = \infty$. In Lean, `derivWithin` is totalized to return 0 where a derivative does not exist, so restricting Theorem 2.6 to finite-rank continuous profiles ensures that $D^k_{[0,\infty)}\varphi_{\mathbb{C}}$ always coincides with the classical within-derivative.) Uniqueness of $p_{\min}$ is immediate because any two monic polynomials dividing each other are equal; moreover, in terms of the shift generator $A \in \mathcal{L}(\mathcal{W}(\varphi_{\mathbb{C}}))$ of Section 3.1, $\deg p_{\min} = R = \dim_{\mathbb{C}} \mathcal{W}(\varphi_{\mathbb{C}}) = \deg \chi_A$ together with $p_{\min} \mid \chi_A$ implies $p_{\min}(X) = \chi_A(X) = \mu_A(X)$ ($\varphi_{\mathbb{C}}|_{\mathbb{R}_{\geq 0}}$ is a cyclic vector for A).

In $d = 2$ (matrices), a pointwise limit of rank-$\leq B$ kernels always has rank at most B because every $(B+1)\times(B+1)$ evaluation minor vanishes (compare Laird's closedness characterization of finite-dimensional translate spans under compact-open convergence [43]). In $d \geq 3$ dimensions, however, a kernel of exact rank R can be the limit of kernels of strictly smaller rank $B < R$ whose weights grow as $O(1/h)$ and cancel as $h \to 0$—just as $s_0 + s_1 + s_2 = \lim_{h\to 0} h^{-1}\big((1+hs_0)(1+hs_1)(1+hs_2) - 1\big)$ approximates a rank-3 sum by a difference of *two* 3-way products [8, 45]. The least $B \in \mathbb{N}$ for which K is a uniform limit on every compact set $C \subset \mathbb{R}^d$ of kernels of rank $\leq B$ is the *compact-open border rank* of K:

Theorem 2.8 (General d-D Multi-Gaussian Border-Rank Bound and Gap)**.** *Let $d \geq 2$ and let $\Gamma = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ be any finite list of well-formed components. Formalized as `multiTotalCost_eq_borderTotalCost_add` and `exists_rank_le_borderTotalCost_tendstoUniformlyOn_radial_profile`:*

1. Exact gap identity: *Both real and nonreal components satisfy $c_d(\lambda_u, P_u) = b_d(\lambda_u, P_u) + (d-2)\deg P_u$, giving*

$$C_d(\Gamma) = B_d(\Gamma) + (d-2)\sum_{u=1}^{m} \deg P_u. \tag{6}$$

2. Uniform multi-Gaussian approximation on compact sets: *Replacing $s = \|x\|^2$ inside each $P_u(s)$ by the difference quotient $u_h(s) = (e^{hs} - 1)/h$ yields an explicit family of multi-Gaussian radial kernels $F_h : \mathbb{R}^d \to \mathbb{R}$ ($h \in \mathbb{R} \setminus \{0\}$, extended to $h = 0$ in Lean via $0^{-1} = 0$) satisfying $\operatorname{srank}_{\mathbb{R}}(F_h) \le B_d(\Gamma)$ for all h and $F_h(x) \to \operatorname{profile}(\Gamma, \|x\|^2)$ uniformly on every compact subset $C \subset \mathbb{R}^d$ as $h \to 0$ ($h \neq 0$). Consequently, whenever Γ is a representation in dimension $d \ge 3$ with $\sum_u \deg P_u \ge 1$, its compact-open real border rank is strictly less than its exact real rank $C_d(\Gamma)$ by at least $(d-2)\sum_u \deg P_u \ge d-2$. In particular, for $K_d(x) = \|x\|^2 \cos(\|x\|^2)$ ($\Gamma = [(i, X)]$), $\operatorname{srank}_{\mathbb{C}}((K_d)_{\mathbb{C}}) = 2d$, $\operatorname{srank}_{\mathbb{R}}(K_d) = 3d - 2$, and $\operatorname{srank}_{\mathbb{R}}(F_h) \le 2d$.*

Is the multi-Gaussian radial bound $B_d(\Gamma)$ equal to the exact compact-open border rank when arbitrary non-Gaussian separable approximants (1) are allowed? Over $\mathbb{C}$ (for all profiles) and over $\mathbb{R}$ whenever all rates are real (or for a single complex Gaussian in $d = 3$), the bound is exact, *but for $\deg P_u \ge 1$ with nonreal rates over $\mathbb{R}$ in $d \ge 3$, $B_d(\Gamma)$ is strictly non-sharp*:

Theorem 2.9 (Exact Border-Rank Sharpness Regimes and Strict Non-Sharpness on $\mathbb{R}^3$)**.** *Formalized in `KernelRankLean/Approximation.lean`:*

1. Pointwise closure in $d = 2$, exact d-D complex border rank, and all-degree real-rate sharpness*: Over any Hausdorff topological field $\mathbb{K}$, if $F_\iota : X_0 \times X_1 \to \mathbb{K}$ eventually satisfies $\operatorname{srank}_{\mathbb{K}}(F_\iota) \le r$ along a nontrivial filter and converges pointwise to K, then $\operatorname{srank}_{\mathbb{K}}(K) \le r$ (`separableRank_fin2_le_of_tendsto`). Padding two coordinates by zeros implies that for every $d \ge 2$, any representation Γ has pointwise and compact-open real border rank bounded between $C_2(\Gamma)$ and $B_d(\Gamma)$ (`totalCost_le_of_tendsto_radial_profile`). Consequently: (i) over $\mathbb{C}$, every radial exponential polynomial $\sum_{\mu \in S} P_\mu(\|x\|^2) e^{\mu \|x\|^2}$ ($P_\mu \neq 0$) has exact pointwise and compact-open complex border rank $\sum_{\mu \in S}(\deg P_\mu + 1)$ (`sum_natDegree_add_one_le_of_tendsto_radial_complex`, `exists_rank_le_sum_natDegree_add_one_tendstoUniformlyOn_radial_complex`); and (ii) over $\mathbb{R}$, whenever every rate of Γ is real ($\operatorname{Im} \lambda_u = 0$, arbitrary polynomial degrees $\deg P_u \ge 0$), $B_d(\Gamma) = C_2(\Gamma) = \sum_{u=1}^m (\deg P_u + 1)$ is the exact pointwise and compact-open real border rank in all dimensions $d \ge 2$ (`borderTotalCost_le_of_tendsto_radial_real_profile`).*
2. Nonreal $k = 0$ sharpness and strict separation $4 < 5 < 6 < 7$ on $\|x\|^2 \cos(\|x\|^2)$ in $d = 3$*: On $\mathbb{R}^3$, no family of real kernels of separable rank ≤ 2 converges pointwise along a nontrivial filter to $\cos(\|x\|^2)$ (`not_exists_rank_le_two_tendsto_radial_cos`), so its pointwise and compact-open real border rank equals $3 = B_3$. For $K_3(x) = \|x\|^2 \cos(\|x\|^2)$ on $\mathbb{R}^3$, no family of real kernels of separable rank ≤ 4 converges pointwise to K_3 (`not_exists_rank_le_four_tendsto_radial_sq_mul_cos`), while the explicit 5-term non-Gaussian real family (7) below has $\operatorname{srank}_{\mathbb{R}}(F_h) \le 5$ and converges to K_3 uniformly on every compact subset $C \subset \mathbb{R}^3$ (`exists_rank_le_five_tendstoUniformlyOn_radial_sq_mul_cos`). Thus K_3 strictly separates complex border rank (4), real border rank (5), radial multi-Gaussian border rank and exact complex rank (6), and exact real rank (7; see Table 13 in Section 7).*

Remark 2.10 (Explicit rank-5 construction, sharpness classification, and numerical conditioning implications)**.** To see explicitly how non-Gaussian separable approximants drop below $B_3([(i, X)]) = 6$ in Theorem 2.9(2) (formalized in `Approximation.lean`, `exists_rank_le_five_tendstoUniformlyOn_radial_sq_mul_cos`, and verified algebraically in Section 3.4), write $c_j = \cos(x_j^2)$, $s_j = \sin(x_j^2)$, $p_{1,j} = \frac{1}{2}c_j - \frac{\sqrt{3}}{2}s_j$, $p_{2,j} = -\frac{1}{2}c_j - \frac{\sqrt{3}}{2}s_j$, and $q_{m,j} = x_j^2 p_{m,j}$ for

Table 3: Classification of when the multi-Gaussian border-rank bound ($B_d(\Gamma)$ over $\mathbb{R}$, $\sum_{\mu\in S}(\deg P_\mu + 1)$ over $\mathbb{C}$) equals the exact compact-open (and pointwise) border rank.

Field & dim.	Representation / profile class	Exact border rank	Sharp?	Formalized theorem
$\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, d=2	All representations $(\forall \lambda_u, \deg P_u)$	$C_2(\Gamma) = B_2(\Gamma) = \mathrm{srank}_{\mathbb{K}}$	**Yes**	`separableRank_fin2_le_of_tendsto`
$\mathbb{C}$, any $d \geq 2$	All complex exp.-polynomials	$\sum_{\mu\in S}(\deg P_\mu + 1)$	**Yes**	`sum_natDegree_add_one_le_...`
$\mathbb{R}$, any $d \geq 2$	All real rates $(\mathrm{Im}\,\lambda_u = 0, \forall \deg P_u)$	$\sum_u(\deg P_u + 1) = B_d(\Gamma)$	**Yes**	`borderTotalCost_le_of_tendsto_...`
$\mathbb{R}$, $d = 3$	Single nonreal Gaussian $\cos(\|x\|^2)$	$3 = B_3 = C_3$	**Yes**	`not_exists_rank_le_two_...`
$\mathbb{R}$, $d = 3$	Degree-1 nonreal $\|x\|^2\cos(\|x\|^2)$	$5 < B_3 = 6 < C_3 = 7$	**No**	`exists_rank_le_five_...`
$\mathbb{R}$, $d \geq 3$	General $\mathrm{Im}\,\lambda_u > 0$ ($d \geq 4$ or $k_u \geq 2$)	In $[C_2(\Gamma),\, B_d(\Gamma)]$	Open	`totalCost_le_...` & Thm. 2.8

$j \in \{0, 1, 2\}$. For any $h \in \mathbb{R} \setminus \{0\}$, the 5-product real kernel

$$F_h(x) = \frac{1}{3h}\prod_{j=0}^{2}(c_j + 4hx_j^2 c_j) + \frac{1}{h^3}\left(\prod_{j=0}^{2}(s_j + h^2 c_j) - \prod_{j=0}^{2} s_j\right) - \frac{4}{3h}\prod_{j=0}^{2}(p_{1,j} + hq_{1,j}) + \frac{4}{3h}\prod_{j=0}^{2}(p_{2,j} + hq_{2,j}) \tag{7}$$

breaks radial symmetry at finite $h \neq 0$ yet satisfies $F_h(x) - \|x\|^2\cos(\|x\|^2) = O(h)$ uniformly on every compact set $C \subset \mathbb{R}^3$. Table 3 summarizes all regimes in which the multi-Gaussian bound is proved sharp versus strictly non-sharp. Two practical lessons follow directly from these theorems:

1. *Convergence topology:* All border-rank *lower* bounds in Theorem 2.9 hold under bare pointwise convergence on $\mathbb{R}^d$, whereas the *upper* bounds hold uniformly on compact sets $C \subset \mathbb{R}^d$ (for profiles with $\mathrm{Re}\,\lambda_u \geq 0$ such as $\|x\|^2\cos\|x\|^2$, $|u_h(s) - s|$ grows as $s \to \infty$, so uniform convergence on all of $\mathbb{R}^d$ is neither claimed nor possible, in contrast to decaying profiles in Theorem 2.11).
2. *Floating-point cancellation penalty:* Saving $(d-2)\sum_u \deg P_u$ terms via Theorem 2.8 requires $O(h^{-\deg P_u})$ weights, and saving one further term in (7) requires $O(h^{-3})$ weights: achieving 10^{-3} truncation accuracy in (7) inflates intermediate products by $h^{-3} \approx 10^9$, wiping out IEEE `float16` (3.3 decimal digits) and `float32` (7.2 digits) precision. Consequently, border-rank degenerations are primarily structural obstructions explaining near-coalescing rate ill-conditioning ($1c_6$ in Table 10), whereas practical filter and stencil designs should use well-separated pure Gaussians ($k = 0$, where $B_d = C_d$ and weights remain $O(1)$).

Finally, whereas Theorem 2.3 shows that non-exponential-polynomial targets have $\mathrm{srank}_{\mathbb{R}} = \infty$, pure decaying complex Gaussians sharing a *single* width parameter and equally spaced harmonic frequencies are dense in the uniform norm over *all* of $\mathbb{R}^d$ for radial profiles vanishing at infinity:

Theorem 2.11 (Equal-Width Harmonic Complex-Gaussian Universal Approximation on $\mathbb{R}^d$; Lean: `exists_harmonic_gaussian_uniform_approx`)**.** *Let $d \geq 2$, let $\psi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with $\lim_{s\to\infty}\psi(s) = 0$, and let $\alpha > 0$ and $\varepsilon > 0$. There exist a fundamental frequency $\omega_0 > 0$, $M \in \mathbb{N}$, $a_0 \in \mathbb{R}$, and $a_1, \ldots, a_M \in \mathbb{C}$ such that the equal-width harmonic multi-Gaussian kernel*

$$K_M(x) := a_0 e^{-\alpha\|x\|^2} + \sum_{m=1}^{M} \mathrm{Re}\Big(a_m e^{(-\alpha + im\omega_0)\|x\|^2}\Big) \tag{8}$$

satisfies $\mathrm{srank}_{\mathbb{R}}(K_M) \leq 1 + dM$ and $|K_M(x) - \psi(\|x\|^2)| < \varepsilon$ for all $x \in \mathbb{R}^d$.

Remark 2.12 (Envelope compensation vs. near-cancelling nonlinear rates)**.** In (8), factoring out the shared Gaussian envelope $e^{-\alpha s}$ ($s = \|x\|^2$) leaves a T-periodic trigonometric polynomial $\mathrm{Re}(a_0 + \sum_{m=1}^{M} a_m e^{im\omega_0 s})$ of period $T = 2\pi/\omega_0$. If $\psi(s)$ vanishes (or is below $\varepsilon/4$) beyond S_0, taking $T > S_0 + 2$ avoids wrap-around inside $[0, S_0]$, compensating $g(u) = e^{+\alpha u}\psi(u)$ on $[0, S_0]$ recovers any continuous radial shape once $e^{-\alpha u}g(u) = \psi(u)$, and choosing αT large enough damps every periodic replica at $s = kT + u$ ($k \geq 1, u \in [0, S_0]$) because $e^{-\alpha(kT+u)}e^{+\alpha u}\psi(u) = e^{-k\alpha T}\psi(u)$ cancels the $+\alpha u$ ramp identically. At the same time, fixing equal widths and harmonic spacing $\omega_m = m\omega_0$ explains why (8) is *not* optimal at small rank R: keeping $\alpha S_0 \ll 1$ (a wide envelope needing little compensation) forces $T \gg S_0$ so resolving features of width Δs requires $M \propto T/\Delta s$ harmonics, whereas keeping $T \sim 2S_0$ forces $\alpha S_0 \gg 1$ and amplifies $|g(u)|$ by $e^{+\alpha S_0}$ near the rim. Freeing (α_u, ω_u) allows unequal-width, non-harmonic (or nearly coalescing, Theorem 2.8) components to nearly cancel and sculpt sharp transitions at small R [12, 58].

The upper bounds for nonreal components in Theorems 2.5 and 2.8 rest on two root-of-unity identities of independent algebraic interest (throughout, $[X^k]F(X)$ denotes the coefficient of X^k in a polynomial $F \in \mathbb{C}[X]$):

Proposition 2.13 (Scalar alternating $2d$-th root-of-unity identity; Lean: `re_prod_eq_sum_re`)**.** *For every integer $d \geq 1$ and all $z_0, \ldots, z_{d-1} \in \mathbb{C}$,*

$$\mathrm{Re}\left(\prod_{j=0}^{d-1} z_j\right) = \frac{2^{d-1}}{d} \sum_{r=0}^{d-1} (-1)^r \prod_{j=0}^{d-1} \mathrm{Re}\left(z_j e^{-i\pi r/d}\right). \tag{9}$$

Proposition 2.14 (Universal polynomial-product root-of-unity identity; Lean: `re_coeff_prod_eq_sum_re`)**.** *Let $d \geq 2$, $k \geq 0$, and $M^{\mathbb{R}}_{d,k} = 2(d-1)k + d$. For any polynomials $U_0, \ldots, U_{d-1} \in \mathbb{C}[X]$ with $\deg U_j \leq k$,*

$$\mathrm{Re}\left([X^k]\prod_{j=0}^{d-1} U_j(X)\right) = \frac{2^{d-1}}{M^{\mathbb{R}}_{d,k}} \sum_{r=0}^{M^{\mathbb{R}}_{d,k}-1} (-1)^r \prod_{j=0}^{d-1} \mathrm{Re}\left(U_j\left(e^{\frac{2\pi i r}{M^{\mathbb{R}}_{d,k}}}\right) e^{-\frac{i\pi r(2k+1)}{M^{\mathbb{R}}_{d,k}}}\right). \tag{10}$$

When $k = 0$, $M^{\mathbb{R}}_{d,0} = d$, each $U_j(X) = z_j$ is a constant polynomial, $[X^0]\prod_j z_j = \prod_j z_j$, and the phase $e^{-i\pi r(2\cdot 0+1)/d} = e^{-i\pi r/d}$ reduces (10) identically to (9).

Example 2.15 (Concrete $d = 3, k = 0$ complex Gaussian: 3 real products vs. 4 naive products)**.** Let $d = 3$ and write $z_j = a_j + ib_j \in \mathbb{C}$ ($a_j, b_j \in \mathbb{R}$) for $z_j(x_j) = \beta^{1/3}e^{(-\alpha+i\omega)x_j^2}$. Expanding $\mathrm{Re}(z_0 z_1 z_2)$ directly into real and imaginary parts requires $2^{3-1} = 4$ real 3-way products:

$$\mathrm{Re}(z_0 z_1 z_2) = a_0 a_1 a_2 - a_0 b_1 b_2 - b_0 a_1 b_2 - b_0 b_1 a_2.$$

Instead, Proposition 2.13 at $d = 3$ evaluates $u_{j,r} = \mathrm{Re}(z_j e^{-i\pi r/3}) = a_j \cos(\frac{\pi r}{3}) + b_j \sin(\frac{\pi r}{3})$ at the three angles $0, \frac{\pi}{3}, \frac{2\pi}{3}$ ($u_{j,0} = a_j$, $u_{j,1} = \frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j$, $u_{j,2} = -\frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j$):

$$\mathrm{Re}(z_0 z_1 z_2) = \frac{4}{3}\Big(u_{0,0}u_{1,0}u_{2,0} - u_{0,1}u_{1,1}u_{2,1} + u_{0,2}u_{1,2}u_{2,2}\Big).$$

Expanding the right-hand side confirms that all $a_0a_1b_2$ and $b_0b_1b_2$ terms cancel via $-\frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{8} = 0$ and $-\frac{3\sqrt{3}}{8} + \frac{3\sqrt{3}}{8} = 0$, while $a_0a_1a_2$ keeps $1 - \frac{1}{8} - \frac{1}{8} = \frac{3}{4}$, leaving $\frac{4}{3}(\frac{3}{4}a_0a_1a_2 - \frac{3}{4}(a_0b_1b_2 + b_0a_1b_2 + b_0b_1a_2)) = \mathrm{Re}(z_0z_1z_2)$ using *three* symmetric real 3-way products rather than four, and Theorem 2.5 proves that two never suffice when $\omega > 0$. Algebraically, pairing $(z_0, z_1) \mapsto z_0 z_1$ against $\bar{z}_2 \mapsto \mathrm{Re}((z_0z_1)z_2)$ identifies this $2 \times 2 \times 2$ real tensor with bilinear multiplication in $\mathbb{C} \cong \mathbb{R}[X]/(X^2+1)$, whose real rank is 3 by Gauss/Karatsuba complex multiplication [75].

Table 4: Component-cost comparison across $d \in \{2,3,4\}$ and degrees $k = \deg P \in \{0,1\}$: exact complex rank, exact real rank $c_d(\lambda, P)$ (each term costing d one-dimensional passes), naive real expansion (grouping p_0 into mode 0 to form d complex products at $k=1$ and expanding each $\mathrm{Re}(z_0 \cdots z_{d-1})$ into 2^{d-1} real products, giving $d\,2^{d-1}$), and multi-Gaussian border-rank bound $b_d(\lambda, P)$.

Single component type	Profile $\varphi(s)$ $(s = \|x\|^2)$	Dim. d	Complex rank	Exact real c_d	Naive real	Border bound b_d
Real Gaussian $(k=0)$	$a\,e^{-\alpha s}$	$2,3,4$	$1,1,1$	$1,1,1$	$1,1,1$	$1,1,1$
Real Laguerre $(k=1)$	$(p_0 + p_1 s)e^{-\alpha s}$	$2,3,4$	$2,3,4$	$2,3,4$	$2,3,4$	$2,2,2$
Complex Gaussian $(k=0)$	$\mathrm{Re}(\beta\, e^{(-\alpha+i\omega)s})$ $(\omega > 0)$	$2,3,4$	$2,2,2$	$2,3,4$	$2,4,8$	$2,3,4$
Complex poly. $(k=1)$	$\mathrm{Re}((p_0 + p_1 s)e^{(-\alpha+i\omega)s})$	$2,3,4$	$4,6,8$	$4,7,10$	$4,12,32$	$4,6,8$

Example 2.16 (Concrete $d=3, k=1$ polynomial oscillation: 7 real products via $M^{\mathbb{R}}_{3,1} = 7$ vs. 9 naive)**.** Consider $K_3(x) = (x_0^2 + x_1^2 + x_2^2)\cos(x_0^2 + x_1^2 + x_2^2)$ ($\lambda = i$, $P(s) = s$, $k=1$). Splitting $x_0^2 \cos\|x\|^2 + x_1^2 \cos\|x\|^2 + x_2^2 \cos\|x\|^2$ into three shifted Gaussians and applying Example 2.15 to each costs $3+3+3=9$ real 3-way products. Instead, put $U_j(X) = e^{ix_j^2}(1 + x_j^2 X)$ $(j = 0,1,2)$, so $[X^1](U_0U_1U_2) = (x_0^2 + x_1^2 + x_2^2)e^{i\|x\|^2}$. Here $M^{\mathbb{R}}_{3,1} = 2(3-1)(1) + 3 = 7$ and $2k+1 = 3$; evaluating $\mathrm{Re}\big(U_j(e^{2\pi i r/7})e^{-3\pi i r/7}\big)$ gives the seven real univariate functions

$$K_3(x) = \frac{4}{7}\sum_{r=0}^{6}(-1)^r f_r(x_0)\, f_r(x_1)\, f_r(x_2), \qquad f_r(u) := \cos\!\left(u^2 - \frac{3\pi r}{7}\right) + u^2 \cos\!\left(u^2 - \frac{\pi r}{7}\right).$$

Why must the node count be 7 with alternating signs $(-1)^r = e^{7\pi i r/7}$? Writing $\cos\theta = \frac{1}{2}(e^{-i\theta} + e^{i\theta})$ in $f_r(x_0)f_r(x_1)f_r(x_2)$ produces harmonics $e^{2\pi i r K/7}$ where $m \in \{0,1,2,3\}$ factors pick the $-i$ branch and $q_j \in \{0,1\}$ picks 1 vs. x_j^2, giving $K = 3m + \sum_{j\notin S} q_j - \sum_{j\in S} q_j - 1$. For mixed branches $m=1$ ($K \in \{1,\dots,4\}$) and $m=2$ ($K \in \{3,\dots,6\}$), 7 is the smallest modulus keeping every mixed harmonic inside $\{1,\dots,M-1\}$ so $\sum_{r=0}^{6} e^{2\pi i r K/7} = 0$, while on pure branches $m=0$ ($K = q_0+q_1+q_2-1$) and $m=3$ ($K = 8 - q_0 - q_1 - q_2$) only $q_0+q_1+q_2 = 1$ is divisible by 7, isolating $K_3(x)$ in 7 real products (proved optimal in Section 3.3).

2.3 Intuitive guide: what $c_d(\lambda, P)$ counts and why the $d-2$ penalty arises

For readers primarily interested in filter design or scientific computing rather than commutative algebra, four plain-language rules explain every number in Theorems 2.3–2.8 (illustrated in Table 4):

1. **What $c_d(\lambda, P)$ counts computationally:** A single component (λ, P) is one radially symmetric building block (such as u_1 or u_2 in Figure 1a), whereas $c_d(\lambda, P)$ counts how many rank-1 real product channels $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ (the rendering terms t_n in Figure 1b) must be summed to evaluate that component over $\mathbb{R}$. On a d-dimensional grid of width W, rank $R = C_d(\Gamma) = \sum_u c_d(\lambda_u, P_u)$ executes dR one-dimensional passes (dRW multiply–adds per voxel).

2. **Why a real rate costs $(d-1)k+1$:** Expanding $P(s_0 + \cdots + s_{d-1})$ of degree k places a degree-k polynomial on each of the last $d-1$ coordinates; their product has degree $(d-1)k$, which has $(d-1)k+1$ independent coefficients (1 when $k=0$; d when $k=1$, matching $s_0 + \cdots + s_{d-1}$) and can be reconstructed from $(d-1)k+1$ real evaluation points.

3. **Where the constant $d-2$ real-over-complex penalty comes from:** Over $\mathbb{C}$, a nonreal component $\mathrm{Re}(P(s)e^{\lambda s}) = \frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda} s}$ splits into two independent complex rates of cost $(d-1)k+1$ each, totaling $2(d-1)k+2$ (2 complex products when $k=0$, in *every* dimension d). Over $\mathbb{R}$, however, even at $k=0$ ($\mathrm{Re}(z_0 \cdots z_{d-1})$ with $z_j \in \mathbb{C} \cong \mathbb{R}^2$),

every real linear functional $\ell : \mathbb{C} \to \mathbb{R}$ has a 1-dimensional nullspace: choosing $z_j \neq 0$ in that nullspace kills one real product term per coordinate $j = 0, \ldots, d-2$ while leaving the remaining product nonzero, forcing at least d real products (2 in 2-D, 3 in 3-D, 4 in 4-D)—a base penalty of $d-2$ over the 2 complex products! Each polynomial degree increment $k \mapsto k+1$ then adds $2(d-1)$ real degrees of freedom on modes $0, \ldots, d-2$ over *both* $\mathbb{C}$ and $\mathbb{R}$, so $c_d(\lambda, P) = d + 2(d-1)k$ and the complex pair cost $2 + 2(d-1)k$ grow at the same rate $2(d-1)k$, keeping their difference locked at $d-2$ for every $k \geq 0$.

4. **Why the multi-Gaussian border-rank bound drops by $(d-2)k$:** Approximating $se^{\lambda s} \approx \frac{e^{(\lambda+h)s} - e^{\lambda s}}{h}$ replaces one degree-1 polynomial factor by a difference of *two* pure Gaussians ($k = 0$): in $d \geq 3$, two complex Gaussians cost $2d$ real products whereas exact $k = 1$ costs $2(d-1) + d = 3d - 2$, lowering the multi-Gaussian border-rank upper bound $B_d(\Gamma)$ by $d-2$ products per polynomial degree at the price of $O(1/h)$ cancelling weights as $h \to 0$.

Remark 2.17 (Unified commutative-algebra invariant and comparison with symmetric Waring rank)**.** Write $\mathcal{A}_\varphi = \mathbb{R}[X]/(p_{\min,\mathbb{R}}(X))$ for the finite-dimensional commutative polynomial quotient algebra associated to the minimal real differential annihilator $p_{\min,\mathbb{R}}(D)\varphi = 0$ (isomorphic to $\mathcal{W}(\varphi)$ as an $\mathbb{R}[D]$-module, with multiplication induced by composition of D rather than pointwise product of profiles). Over $\mathbb{R}$, $p_{\min,\mathbb{R}}(X)$ factors into $m_{\mathrm{irr}} = n_{\mathrm{re}} + n_{\mathrm{cx}}$ distinct powers of irreducible real polynomials (n_{re} linear factors $(X - \lambda_u)^{k_u+1}$ and n_{cx} irreducible real quadratics $((X + \alpha_u)^2 + \omega_u^2)^{k_u+1}$; equivalently, m_{irr} maximal ideals of $\mathcal{A}_\varphi$), while $\dim_{\mathbb{R}} \mathcal{A}_\varphi = \deg p_{\min,\mathbb{R}} = C_2(\Gamma)$. Checking $(d-1)(k+1) - (d-2) \cdot 1 = (d-1)k + 1$ and $(d-1) \cdot 2(k+1) - (d-2) \cdot 1 = 2(d-1)k + d$ shows that Theorem 2.5 takes one unified form across all $d \geq 2$:

$$\operatorname{srank}_{\mathbb{R}}(K) = C_d(\Gamma) = (d-1) \dim_{\mathbb{R}} \mathcal{A}_\varphi - (d-2)\, m_{\mathrm{irr}} \qquad (d \geq 2), \tag{11}$$

and the identical formula holds over $\mathbb{C}$ with $m_{\mathrm{irr}} = |S|$ linear factors. At $d = 2$ (1-fold identity pairing), (11) reduces to $\dim_{\mathbb{K}} \mathcal{A}$; at $d = 3$ (2-fold bilinear multiplication), it coincides with the Winograd–Fiduccia–Zalcstein–Alder–Strassen complexity $2 \dim_{\mathbb{K}} \mathcal{A} - m_{\mathrm{irr}}$ [75, 23, 1, 15]; and for $d \geq 4$ ($(d-1)$-fold multiplication), it matches the multilinear Alder–Strassen lower bound of Bläser, Mayer, and Shringi [10] with equality. Notice the contrast with symmetric Waring rank: for a real binary form $F(u, v) = \mathrm{Re}((u + iv)^D)$ ($D \geq 2$), complex Waring rank is 2 whereas real symmetric Waring rank equals D [18, 63, 11]—a gap $D - 2$ that grows with degree. In (11), complexifying $\mathbb{R} \hookrightarrow \mathbb{C}$ splits each irreducible real quadratic into two conjugate linear factors (m_{irr} increases by 1 per nonreal pair while $\dim \mathcal{A}$ is unchanged), producing a *degree-independent* real-over-complex penalty of $(d-2)(2-1) = d-2$ per conjugate pair because coordinatewise decompositions (1) break mode symmetry by placing $H_P(x_0^2; X)$ on mode 0 and $E_k(x_j^2; X)$ on modes $j \geq 1$.

3 Mathematical Proofs and Core Constructions

Radial $K(x) = f(\|x\|^2)$ on $\mathbb{R}^d$ $\xleftarrow[\text{surjection } s_j = x_j^2]{\text{Eq. (3)}}$ Hankel $(s_0, \ldots, s_{d-1}) \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)$ on $\mathbb{R}^d_{\geq 0}$

Constructive upper bounds ($\leq R$)
- $\mathbb{C}$: $M^{\mathbb{C}}_{d,k} = (d-1)k+1$ roots of unity (Sec. 3.2)
- $\mathbb{R}$, $\mathrm{Im}\,\lambda{=}0$: Lagrange at $0, \ldots, M^{\mathbb{C}}_{d,k}-1$ (Sec. 3.2)
- $\mathbb{R}$, $\mathrm{Im}\,\lambda{>}0$: $M^{\mathbb{R}}_{d,k} = 2(d-1)k+d$ roots (Prop. 2.14)

Necessity & multilinear lower bounds ($\geq R$)
- $d{=}2$ slice $\Rightarrow \dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty \Rightarrow T_t = e^{tA}$ (Sec. 3.1)
- Dual span of $v(\mathbb{R}_{\geq 0}) \Rightarrow$ linear $\ell_{n,j} \in V^*$ (Sec. 3.3)
- Alder–Strassen peeling + $+1$ shift (Sec. 3.3)

3.1 Necessity and the two-dimensional core

In two coordinates over a field $\mathbb{K}$, let $\mathrm{rowSpan}(K) \leq (X \to \mathbb{K})$ be the $\mathbb{K}$-linear span of the row sections $\{x \mapsto K(x, y) : y \in X\}$. Then $\operatorname{srank}_{\mathbb{K}}(K) = \dim_{\mathbb{K}} \mathrm{rowSpan}(K) \in \mathbb{N} \cup \{\infty\}$: if

$K(x,y) = \sum_{n=0}^{R-1} f_n(x) g_n(y)$, every row $x \mapsto K(x,y)$ is a linear combination of $f_0, \dots, f_{R-1}$ so $\dim \operatorname{rowSpan}(K) \le R$; conversely, if $b_0(x), \dots, b_{R-1}(x)$ is a basis of $\operatorname{rowSpan}(K)$, expanding each row $x \mapsto K(x,y) = \sum_{n=0}^{R-1} c_n(y) b_n(x)$ directly gives a rank-R sum of products. For the two-variable Hankel kernel $(s_0, s_1) \mapsto f(s_0 + s_1)$ on $\mathbb{R}^2_{\ge 0}$, the row span is the *translate span*

$$\mathcal{W}(f) = \operatorname{span}_{\mathbb{K}} \{ u \in \mathbb{R}_{\ge 0} \mapsto f(u+t) : t \in \mathbb{R}_{\ge 0} \} \le (\mathbb{R}_{\ge 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ satisfies $\operatorname{srank}_{\mathbb{C}}(K_{\mathbb{C}}) \le \operatorname{srank}_{\mathbb{R}}(K)$ in every dimension d. In $d = 2$, equality $\operatorname{srank}_{\mathbb{C}}(K_{\mathbb{C}}) = \operatorname{srank}_{\mathbb{R}}(K)$ always holds: if real functions $b_0, \dots, b_{R-1} : X \to \mathbb{R}$ are $\mathbb{R}$-linearly independent, any complex relation $\sum_n (\alpha_n + i\beta_n) b_n(x) = 0$ forces $\sum_n \alpha_n b_n(x) = 0$ and $\sum_n \beta_n b_n(x) = 0$ separately, so $\dim_{\mathbb{C}} \operatorname{rowSpan}(K_{\mathbb{C}}) = \dim_{\mathbb{R}} \operatorname{rowSpan}(K)$.

For any $d \ge 2$, if $\operatorname{srank}_{\mathbb{R}}(x \in \mathbb{R}^d \mapsto \varphi(\|x\|^2)) < \infty$, the slice inequality (4) and 2-D complexification give $\dim_{\mathbb{C}} \mathcal{W}(\varphi_{\mathbb{C}}) < \infty$. Necessity and the 2-D annihilator theorem then follow from a single basis-free operator construction on $\mathcal{W}(\psi)$ for any continuous $\psi : \mathbb{R} \to \mathbb{C}$ with $\dim_{\mathbb{C}} \mathcal{W}(\psi) = n < \infty$:

1. Equip the algebraic subspace $\mathcal{W}(\psi) \le (\mathbb{R}_{\ge 0} \to \mathbb{C})$—not a space of bounded continuous functions, since ψ may grow polynomially or exponentially on $[0, \infty)$—with the Hausdorff product topology of pointwise convergence inherited from $\mathbb{R}_{\ge 0} \to \mathbb{C}$. Because $\dim_{\mathbb{C}} \mathcal{W}(\psi) = n < \infty$ and all Hausdorff vector topologies on an n-dimensional $\mathbb{C}$-vector space coincide (`isNormableSpace_of_finiteDimensional`), $\mathcal{W}(\psi)$ admits a compatible norm whose topology equals pointwise convergence and is automatically complete (`FiniteDimensional.complete`), making $\mathcal{W}(\psi)$ a complex Banach space without any growth or smoothness hypothesis on ψ.

2. Non-negative shifts $(T_t g)(u) = g(u+t)$ $(t \ge 0)$ map $\mathcal{W}(\psi)$ into itself because shifting a translate $u \mapsto \psi(u+s)$ gives $u \mapsto \psi(u + (s+t)) \in \mathcal{W}(\psi)$, requiring no derivatives of ψ. Continuity of ψ on $[0, \infty)$ implies that $t \mapsto (T_t g)(u)$ is continuous at each point $u \ge 0$ on every spanning translate, hence norm-continuous on $\mathcal{W}(\psi)$, and therefore $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ is continuous in operator norm on $[0, \infty)$ (evaluating a linear map on a finite basis of $\mathcal{W}(\psi)$).

3. Following Nagumo [52] (see Engel and Nagel [22, Thm. I.3.7]; cf. Anselone and Korevaar's matrix-semigroup proof on $\mathbb{R}$ [3, p. 750]), in any real Banach algebra $\mathcal{A}$ (here $\mathcal{L}(\mathcal{W}(\psi))$), continuity of $t \mapsto T_t$ at 0 with $T_0 = I$ yields $\delta > 0$ such that the Bochner integral $I_\delta = \int_0^\delta T_u \, du$ satisfies $\|I_\delta - \delta I\| < \delta$, making I_δ invertible in $\mathcal{A}$ by Neumann series (`Semigroup.lean`). For $t \ge 0$, changing variables in $T_t I_\delta = \int_t^{t+\delta} T_u \, du$ shows by the fundamental theorem of calculus that $t \mapsto T_t$ is right-differentiable on $[0, \infty)$ with generator $A = (T_\delta - I) I_\delta^{-1}$ commuting with T_t; differentiating $t \mapsto T_t \exp(-tA)$ gives 0, so $T_t = \exp(tA)$ for all $t \ge 0$ and $\psi(t) = (T_t \psi|_{\mathbb{R}_{\ge 0}})(0) = \operatorname{eval}_0(\exp(tA)\psi|_{\mathbb{R}_{\ge 0}})$ is automatically C^∞ on $[0, \infty)$.

4. Over the algebraically closed field $\mathbb{C}$, $\mathcal{W}(\psi) = \bigoplus_{\mu \in S} \ker((A - \mu I)^n)$ decomposes into generalized eigenspaces. On a summand where $(A - \mu I)^n v = 0$, writing $tA = \mu t I + t(A - \mu I)$ as a sum of commuting operators gives $\exp(tA)v = e^{\mu t} \sum_{q=0}^{n-1} \frac{t^q}{q!} (A - \mu I)^q v$, because all powers $q \ge n$ vanish. Evaluating at 0 turns each vector $(A - \mu I)^q v$ into a complex scalar, yielding $\psi(t) = \sum_{\mu \in S} P_\mu(t) e^{\mu t}$ on $[0, \infty)$. When $\psi = \varphi_{\mathbb{C}}$ for a real profile φ, averaging $\varphi(t) = \frac{1}{2}(\varphi_{\mathbb{C}}(t) + \overline{\varphi_{\mathbb{C}}(t)})$ and grouping conjugate rates $(\mu, \bar{\mu})$ produces a canonical real representation Γ.

Finally, differentiating $t \mapsto \exp(tA)\psi|_{\mathbb{R}_{\ge 0}}$ q times on $[0, \infty)$ brings down A^q, so $q(D)\psi(t) = \operatorname{eval}_0\big(q(A)\exp(tA)\psi|_{\mathbb{R}_{\ge 0}}\big)$ for every polynomial $q \in \mathbb{C}[X]$. In $d = 2$, the Cayley–Hamilton theorem ($q = \chi_A \implies \chi_A(A) = 0$) immediately implies that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ satisfies $\chi_A(D)\psi = 0$ on $[0, \infty)$. Conversely, applying $(D - \mu)^{\deg P_\mu + 1}$ kills $P_\mu(t) e^{\mu t}$ while shifting every other rate $\nu \ne \mu$ by an invertible factor

$(\nu-\mu)^{\deg P_\mu+1}$ on leading terms, so $q(D)\big(\sum_{\mu\in S} P_\mu(t)e^{\mu t}\big)=0$ on $[0,\infty)$ ($P_\mu\neq 0$) if and only if $p_{\min}(X)=\prod_{\mu\in S}(X-\mu)^{\deg P_\mu+1}$ divides $q(X)$. Hence $p_{\min}\mid\chi_A$, giving $\sum_{\mu\in S}(\deg P_\mu+1)=\deg p_{\min}\le\deg\chi_A=\dim_{\mathbb{C}}\mathcal{W}(\psi)$, matched by Taylor expansion of $P_\mu(s_0+s_1)e^{\mu(s_0+s_1)}$ to prove Theorem 2.6.

3.2 Constructive multi-way upper bounds via Hasse polynomials and roots of unity

Warm-up ($d=3,k=1$): extracting a degree-1 radial polynomial as an $[X^1]$ coefficient. Consider $P(s)=p_0+p_1s$ on $s=s_0+s_1+s_2$. Notice that if we put $U_0(X)=p_1+(p_0+p_1s_0)X$, $U_1(X)=1+s_1X$, and $U_2(X)=1+s_2X$, then the coefficient of X^1 in $U_0(X)U_1(X)U_2(X)$ is

$$[X^1]\big((p_1+(p_0+p_1s_0)X)(1+s_1X)(1+s_2X)\big)=(p_0+p_1s_0)\cdot1\cdot1+p_1s_1\cdot1+p_1\cdot1\cdot s_2=P(s_0+s_1+s_2),$$

a product of three univariate polynomials in X whose j-th factor depends *only on* s_j! Since $\deg(U_0U_1U_2)\le3$, over $\mathbb{C}$ averaging $\zeta^{-r}U_0(\zeta^r)U_1(\zeta^r)U_2(\zeta^r)$ over the $M^{\mathbb{C}}_{3,1}=3$ cube roots of unity ($\zeta=e^{2\pi i/3}$) isolates the X^1 term (because X^0,X^2,X^3 shift to X^{-1},X^1,X^2, none of which is a multiple of 3!), whereas over $\mathbb{R}$ for a nonreal rate $e^{\lambda(s_0+s_1+s_2)}$ ($\operatorname{Im}\lambda>0$), Proposition 2.14 extracts $\operatorname{Re}([X^1]\prod_{j=0}^{2}e^{\lambda s_j}U_j(X))$ using $M^{\mathbb{R}}_{3,1}=2(3-1)(1)+3=7$ real 3-way products.

General degree $k=\deg P$ ($d\ge2$). Using the q-th Hasse derivative $D^{[q]}P(s_0)=\frac{1}{q!}P^{(q)}(s_0)$—simply the q-th Taylor coefficient of P around s_0, defined algebraically without division by $P(s_0+t)=\sum_{q=0}^{k}(D^{[q]}P)(s_0)\,t^q$—define the *reversed Hasse polynomial* $H_P(s_0;X)$ on mode 0 and the *truncated exponential polynomials* $E_k(s_j;X)$ on modes $j=1,\dots,d-1$:

$$H_P(s_0;X)=\sum_{q=0}^{k}q!\,(D^{[q]}P)(s_0)X^{k-q},\quad E_k(s_j;X)=\sum_{q=0}^{k}\frac{s_j^q}{q!}X^q,\quad Q(X)=H_P(s_0;X)\prod_{j=1}^{d-1}E_k(s_j;X). \tag{12}$$

Binomial convolution gives $[X^q]\prod_{j=1}^{d-1}E_k(s_j;X)=(s_1+\cdots+s_{d-1})^q/q!$ for all $0\le q\le k$, so $[X^k]Q(X)=P(s_0+\cdots+s_{d-1})$ with $\deg Q\le dk$. Three coefficient-extraction schemes then yield all three upper bounds:

1. *Complex rates over $\mathbb{C}$:* Put $M^{\mathbb{C}}_{d,k}=(d-1)k+1$ and $\zeta=e^{2\pi i/M^{\mathbb{C}}_{d,k}}$. Because $d\ge2$, every monomial index $n\in\{0,\dots,dk\}$ satisfies $-M^{\mathbb{C}}_{d,k}<-k\le n-k\le(d-1)k<M^{\mathbb{C}}_{d,k}$, so $M^{\mathbb{C}}_{d,k}\mid(n-k)\iff n=k$. Discrete Fourier averaging $[X^k]Q(X)=\frac{1}{M^{\mathbb{C}}_{d,k}}\sum_{r=0}^{M^{\mathbb{C}}_{d,k}-1}\zeta^{-rk}Q(\zeta^r)$ multiplied by $\prod_{j=0}^{d-1}e^{\mu s_j}$ gives $M^{\mathbb{C}}_{d,k}=(d-1)k+1$ complex products.
2. *Real rates over $\mathbb{R}$ ($\operatorname{Im}\lambda=0$, $P\in\mathbb{R}[X]$):* On modes $1,\dots,d-1$, the real polynomial $A(X)=\prod_{j=1}^{d-1}E_k(s_j;X)\in\mathbb{R}[X]$ has $\deg A\le(d-1)k<M^{\mathbb{C}}_{d,k}$. Real Lagrange interpolation of $A(X)$ at the $M^{\mathbb{C}}_{d,k}$ distinct real nodes $r=0,1,\dots,M^{\mathbb{C}}_{d,k}-1\in\mathbb{R}$ expresses each $[X^q]A(X)$ ($0\le q\le k$), and hence $P(\sum_j s_j)e^{\lambda\sum_j s_j}$, as a sum of $M^{\mathbb{C}}_{d,k}=(d-1)k+1$ real d-way products.
3. *Nonreal rates over $\mathbb{R}$ ($\operatorname{Im}\lambda>0$, Proof of Propositions 2.13 and 2.14):* Put $M=M^{\mathbb{R}}_{d,k}=2(d-1)k+d=(2k+1)d-2k$ (note $M=d$ when $k=0$). Writing $\operatorname{Re}w=\frac12(\bar w+w)$ in (10) and expanding over subsets $S\subseteq\{0,\dots,d-1\}$ of cardinality $m=|S|$ and degree tuples $q\in\{0,\dots,k\}^d$, the identity $(-1)^r=e^{i\pi rM/M}$ produces the geometric phase $\exp(2\pi irK(S,q)/M)$ with integer frequency

$$K(S,q)=(2k+1)m+\sum_{j\notin S}q_j-\sum_{j\in S}q_j-k\in\mathbb{Z}.$$

Because $0 \le q_j \le k$ and $d \ge 2$: if $m = 0$ ($S = \emptyset$), $-M < -k \le K(\emptyset, q) = \sum_j q_j - k \le (d-1)k < M$, so $M \mid K \iff \sum_j q_j = k$; if $m = d$ ($S = \{0, \dots, d-1\}$), $K = M + (k - \sum_j q_j)$, so $M \mid K \iff \sum_j q_j = k$; and if $1 \le m \le d-1$,

$$1 \le (k+1)m - k \le K(S, q) \le (k+1)m + (d-1)k \le 2(d-1)k + d - 1 = M - 1,$$

so $M \nmid K(S, q)$ and $\sum_{r=0}^{M-1} e^{2\pi i r K(S,q)/M} = 0$. Applying (10) to $U_0(X) = e^{\lambda s_0} H_P(s_0; X)$ and $U_j(X) = e^{\lambda s_j} E_k(s_j; X)$ ($j \ge 1$) yields $M_{d,k}^{\mathbb{R}} = 2(d-1)k + d$ real d-way products.

3.3 Multilinear lower bounds via Alder–Strassen degree peeling

Step 1: From set-theoretic 1-D functions to multilinear dual functionals. Let V be a finite-dimensional $\mathbb{K}$-vector space ($\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$), let $v : \mathbb{R}_{\ge 0} \to V$ be an evaluation curve whose image spans V ($\operatorname{span}_{\mathbb{K}}(v(\mathbb{R}_{\ge 0})) = V$, verified by checking $\Lambda \circ v = 0 \implies \Lambda = 0$ for all $\Lambda \in V^*$), and let $\mathcal{M} : V^d \to \mathbb{K}$ be d-multilinear. If the Hankel kernel $(s_0, \dots, s_{d-1}) \mapsto \mathcal{M}(v(s_0), \dots, v(s_{d-1}))$ admits a rank-R decomposition (1) with arbitrary functions $f_{n,j} : \mathbb{R}_{\ge 0} \to \mathbb{K}$, choosing a basis $\{v(t_b)\}_{b=1}^{\dim V}$ of V and defining unique linear functionals $\ell_{n,j} \in V^*$ by $\ell_{n,j}(v(t_b)) = f_{n,j}(t_b)$ yields

$$\mathcal{M}(u^{(0)}, \dots, u^{(d-1)}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(u^{(j)}) \qquad \text{for all } u^{(0)}, \dots, u^{(d-1)} \in V, \tag{13}$$

because both sides are d-multilinear maps agreeing on every d-tuple of basis vectors.

Step 2: Geometric intuition on $d = 3, k = 1$ ($\|x\|^2 \cos(\|x\|^2)$ requires 7 real terms). Before stating the general induction, consider $K_3(x) = (x_0^2 + x_1^2 + x_2^2)\cos(x_0^2 + x_1^2 + x_2^2)$, where $v(s) = (e^{is}, se^{is}) \in \mathbb{C}^2$ spans $V = \mathbb{C} \oplus \mathbb{C}\varepsilon \cong \mathbb{C}[T]/(T^2)$ ($\varepsilon^2 = 0$) over $\mathbb{R}$ and (13) represents the real 3-multilinear multiplication map $\mathcal{M}((u_0, u_1), (v_0, v_1), (w_0, w_1)) = \operatorname{Re}(u_1 v_0 w_0 + u_0 v_1 w_0 + u_0 v_0 w_1)$. Why must $R \ge 7$, even though difference quotients of two complex Gaussians have rank 6 for every $h \ne 0$?

- On the nilpotent subspace $E = 0 \oplus \mathbb{C}\varepsilon \cong \mathbb{R}^2$, $\mathcal{M}((0, z_1), (v_0, 0), (w_0, 0)) = \operatorname{Re}(z_1 v_0 w_0)$ vanishes only if $z_1 = 0$, so the restrictions $\{\ell_{n,0}|_E : 0 \le n < R\}$ must span E^* ($\dim_{\mathbb{R}} E^* = 2$). Picking two functionals (say $n = 0, 1$) forming a basis of E^*, there is a unique linear map $\gamma_0 : \mathbb{C} \to \mathbb{C}$ such that $\psi_0(u_0) := (u_0, \gamma_0(u_0)) \in \ker \ell_{0,0} \cap \ker \ell_{1,0}$ for all $u_0 \in \mathbb{C}$, *killing two terms* ($n = 0, 1$).
- Crucially, because $\varepsilon^2 = 0$ in $\mathbb{C}[T]/(T^2)$, substituting $\psi_0(u_0)$ on mode 0 and testing $(0, v_1) \in E$ on mode 1 gives $\mathcal{M}(\psi_0(u_0), (0, v_1), (w_0, 0)) = \operatorname{Re}(u_0 v_1 w_0)$—*completely unaffected* by $\gamma_0(u_0)$! Thus the remaining functionals $\{\ell_{n,1}|_E : n \ge 2\}$ on mode 1 must still span E^*, letting a second map $\psi_1(v_0) = (v_0, \gamma_1(v_0))$ kill *two more terms* ($n = 2, 3$).
- Finally, evaluating $(u_0, v_0, w_1) \mapsto \mathcal{M}(\psi_0(u_0), \psi_1(v_0), (0, w_1)) = \operatorname{Re}(u_0 v_0 w_1)$ uses only the surviving $R - 4$ terms ($n \ge 4$). Since every $\mathbb{R}$-linear functional $\Lambda : \mathbb{C} \to \mathbb{R}$ has a nonzero kernel vector ($\dim_{\mathbb{R}} \mathbb{C} = 2 > 1$), induction on d shows $\operatorname{Re}(u_0 v_0 w_1)$ requires at least 3 real terms, forcing $R - 4 \ge 3$, i.e. $R \ge 7$.

Step 3: General multi-mode degree peeling over $\mathbb{C}$ and over $\mathbb{R}$. Over $\mathbb{C}$, let $V = \prod_{i \in \iota} \mathbb{C}^{\tau_i^0}$ with $\tau_i^0 = \deg P_i + 1$ and $v(s)_{i,q} = \frac{s^q}{q!} e^{\mu_i s}$. Exponential-polynomial uniqueness implies $\operatorname{span}_{\mathbb{C}}(v(\mathbb{R}_{\ge 0})) = V$, mapping the Hankel kernel to the truncated-convolution form with cutoffs $\tau_i \le \tau_i^0$:

$$\mathcal{T}_\tau(u^{(0)}, \dots, u^{(d-1)}) = \sum_{i \in \iota} \sum_{q_0 + \cdots + q_{d-1} < \tau_i} w_i(q_0 + \cdots + q_{d-1}) \prod_{j=0}^{d-1} u_{i,q_j}^{(j)}, \qquad w_i(\tau_i - 1) \ne 0 \ (\tau_i > 0).$$

Strong induction on $\sum_i \tau_i$ peels one block i_0 at a time:

- When $\tau_{i_0} = 1$, testing $(e_{i_0,0}, \dots, e_{i_0,0})$ finds $\ell_{n_0,0}(e_{i_0,0}) \neq 0$; projecting block i_0 to 0 on modes $j \geq 1$ and subtracting $\frac{\ell_{n_0,0}(\pi_{\neq i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 kills 1 term and reduces τ_{i_0} to 0.
- When $\tau_{i_0} \geq 2$, since $2(\tau_{i_0} - 1) \geq \tau_{i_0}$, rank-1 perturbations along $e_{i_0,\tau_{i_0}-1}$ on modes $j = 0, \dots, d-2$ do not cross-interact when another coordinate equals $e_{i_0,\tau_{i_0}-1}$, killing $d-1$ distinct terms; truncating modes $0, \dots, d-2$ to $q < \tau_{i_0} - 1$ and shifting mode $d-1$ up by $+1$ ($e_{i_0,q} \mapsto e_{i_0,q+1}$) reduces $\tau_{i_0} \mapsto \tau_{i_0} - 1$ with shifted top weight $w'_{i_0}(\tau_{i_0} - 2) = w_{i_0}(\tau_{i_0} - 1) \neq 0$.

Over $\mathbb{R}$, the same induction on $G = (\prod_{i \in \iota_{\mathbb{R}}} \mathbb{R}^{\tau^0_{\mathbb{R},i}}) \times (\prod_{u \in \iota_{\mathbb{C}}} \mathbb{C}^{\tau^0_{\mathbb{C},u}})$ peels real blocks identically (1 term at $\tau_{\mathbb{R},i_0} = 1$, $d-1$ terms at $\tau_{\mathbb{R},i_0} \geq 2$), peels nonreal $\tau_{\mathbb{C},u_0} = 1$ blocks by d terms (using $\dim_{\mathbb{R}} \ker(\Lambda : \mathbb{C} \to \mathbb{R}) \geq 1$ to kill 2 terms on mode 0 and 1 term on each mode $1, \dots, d-2$), and peels nonreal $\tau_{\mathbb{C},u_0} \geq 2$ blocks by $2(d-1)$ terms (2 independent real functionals on $0 \oplus \mathbb{C}e_{u_0,\tau_{\mathbb{C},u_0}-1} \cong \mathbb{R}^2$ per mode $j = 0, \dots, d-2$, followed by $+1$ index shifting on mode $d-1$), completing Theorem 2.5.

3.4 Proofs of border-rank bounds, adjugate obstruction, and universal approximation (Theorems 2.8–2.11)

Identity (6) is immediate from (5): for $d \geq 2$, $(k+1) + (d-2)k = (d-1)k + 1$ and $d(k+1) + (d-2)k = 2(d-1)k + d$. For a single well-formed component (λ, P) of degree $k = \deg P$ and any $h \in \mathbb{R} \setminus \{0\}$, replace $s = \sum_{j=0}^{d-1} x_j^2$ inside $P(s)$ by the exponential difference quotient $u_h(s) = \frac{e^{hs}-1}{h}$ (setting $u_0(s) := 0$ via $0^{-1} = 0$ in Lean):

$$F_h^{(\lambda,P)}(x) := \mathrm{Re}\Big(P\big(u_h(\|x\|^2)\big)\, e^{\lambda\|x\|^2}\Big).$$

Because $u_h(s) = L_h(e^{hs})$ is a degree-1 polynomial $L_h(Y) = -h^{-1} + h^{-1}Y$ in $Y = e^{hs}$, the composed polynomial $Q_h(Y) = (P \circ L_h)(Y)$ still has degree at most k:

$$P(u_h(s))\, e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h)\, (e^{hs})^m e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h) \prod_{j=0}^{d-1} e^{(\lambda + mh)x_j^2}.$$

When $\mathrm{Im}\,\lambda = 0$ (so $P, Q_h \in \mathbb{R}[X]$), this is a sum of $k+1 = b_d(\lambda, P)$ real Gaussian products; when $\mathrm{Im}\,\lambda > 0$, it is a sum of $k+1$ complex Gaussian products, whose real part has real d-way rank at most $d(k+1) = b_d(\lambda, P)$ by Proposition 2.13. (At $k = 1$ and $\lambda = 0$, $u_h(s) = h^{-1}(\prod_{j=0}^{d-1} e^{hx_j^2} - 1)$ realizes the classical W-tensor degeneration $[X^1] \prod_{j=0}^{d-1}(1 + s_j X)$ of exact rank d and border rank 2 [8, 14, 79] directly by smooth radial Gaussians.) Summing over $(\lambda_u, P_u) \in \Gamma$ gives $\mathrm{srank}_{\mathbb{R}}(F_h) \leq B_d(\Gamma)$ for all $h \in \mathbb{R}$. For any compact $C \subset \mathbb{R}^d$, let $B = \sup_{x \in C} \|x\|^2 < \infty$; twice applying the mean-value inequality gives $|e^y - 1 - y| \leq y^2 e^{|y|}$, so $|u_h(s) - s| \leq |h|(B+1)^2 e^{B+1} \to 0$ uniformly on $[0, B]$, and uniform continuity of $(u, s) \mapsto \mathrm{Re}(P(u)e^{\lambda s})$ on $[-1, B+1] \times [0, B]$ yields uniform convergence on C (and likewise over $\mathbb{C}$ with rank $\leq \sum_{\mu \in S}(\deg P_\mu + 1)$).

For Theorem 2.9(1), if $\mathrm{srank}_{\mathbb{K}}(K) > r$ on $X_0 \times X_1$, induction on $k \leq r+1$ via the Schur complement $\det\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) = (\det A)(D - CA^{-1}B)$ constructs an $(r+1) \times (r+1)$ evaluation submatrix $M(K)$ with $\det M(K) \neq 0$, whereas $\det M(F_\iota) = 0$ whenever $\mathrm{srank}_{\mathbb{K}}(F_\iota) \leq r$ since $M(F_\iota) = UV$ factors through $\mathbb{K}^r$; when every $\mathrm{Im}\,\lambda_u = 0$, $b_d(\lambda_u, P_u) = \deg P_u + 1 = c_2(\lambda_u, P_u)$ gives $B_d(\Gamma) = C_2(\Gamma)$ immediately. For Theorem 2.9(2), if $G : X^3 \to \mathbb{R}$ has $\mathrm{srank}_{\mathbb{R}}(G) \leq n$, then for any $u_0, u_1 \in X$ and $p, q \in X^n$, its mode-0 slices $S_0 = V\,\mathrm{diag}(d_0)W$ and $S_1 = V\,\mathrm{diag}(d_1)W$ satisfy the global polynomial sum-of-squares identity (requiring no invertibility assumptions)

$$\mathrm{tr}\Big(\big(\mathrm{adj}(S_0)\, S_1\big)^2\Big) = (\det V\ \det W)^2 \sum_{k=0}^{n-1} \Big(d_{1,k} \prod_{j \neq k} d_{0,j}\Big)^2 \geq 0.$$

Table 5: Module architecture and verified source line counts of the `KernelRankLean` library.

Module(s)	Mathematical content	Phys.	Nonbl.
`Defs`, `KernelRankLean`	Multi-way rank, surjections, slices, exact & border costs	254	216
`RowSpan`, `Complexification`	2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance	221	189
`Translations`	d-D radial-to-Hankel reduction; `translateSpan`	92	80
`ExpPoly`, `RealExpPoly`	Exponential-polynomial uniqueness	198	170
`Semigroup`, `TranslationSemigroup`	Banach semigroups; half-line shift semigroup	262	219
`Necessity`, `Representation`	Generalized eigenvectors; canonical conjugate pairing	222	198
`Differential`, `ComplexRank`	Half-line operators $q(D)$; 2-D complex rank	318	284
`Annihilator`	Minimal constant-coefficient differential annihilator	69	62
`MultiComplexRank`	Exact d-way complex rank $\sum_\mu((d-1)\deg P_\mu + 1)$	1,015	978
`MultiRealRank`	Exact d-way real rank (`multiTotalCost`)	2,562	2,521
`Main`	Unified d-D/2-D theorems & multi-Gaussian border bound	538	516
`Approximation`	Exact complex/real border ranks & harmonic approx.	1,184	1,153
Total (`KernelRankLean`)		**6,935**	**6,586**

Evaluating $\cos(\|x\|^2)$ at $p = q = (0, \sqrt{\pi/2})$ $(u_0 = 0, u_1 = \sqrt{\pi/2})$ gives $\mathrm{tr}((\mathrm{adj}(S_0)S_1)^2) = -2 < 0$, ruling out rank ≤ 2; evaluating $\|x\|^2\cos(\|x\|^2)$ at $p = q = (0, \sqrt{\pi/2}, \sqrt{\pi}, \sqrt{3\pi/2})$ $(u_0 = 0, u_1 = \sqrt{\pi/2})$ gives $\mathrm{tr}((\mathrm{adj}(S_0)S_1)^2) = -1024(\pi/2)^8 < 0$, ruling out rank ≤ 4 (compare real eigenvalue obstructions on $n \times n \times n$ slices in [20, 45]). Conversely, expanding the 5-term real family $F_h(x)$ of (7) in powers of h: the $-\frac{4}{3h}\prod_j(p_{1,j} + hq_{1,j}) + \frac{4}{3h}\prod_j(p_{2,j} + hq_{2,j})$ pair supplies two of the three modes of Example 2.15 applied to $\frac{1}{h}\,\mathrm{Re}\prod_j((c_j + is_j)(1 + hx_j^2))$, leaving at $O(h^{-1})$ only $-\frac{1}{3h}c_0c_1c_2 + \frac{1}{h}(c_0s_1s_2 + s_0c_1s_2 + s_0s_1c_2)$, which are cancelled respectively by $\frac{1}{3h}\prod_j(c_j + 4hx_j^2c_j)$ (contributing $+\frac{1}{3h}c_0c_1c_2 + \frac{4}{3}(x_0^2 + x_1^2 + x_2^2)c_0c_1c_2 + O(h)$) and by the 2-term W-degeneration $\frac{1}{h^3}(\prod_j(s_j + h^2c_j) - \prod_j s_j) = \frac{1}{h}(c_0s_1s_2 + s_0c_1s_2 + s_0s_1c_2) + O(h)$, leaving $F_h(x) - \|x\|^2\cos(\|x\|^2) = h\,\mathrm{Rem}(x, h)$ with continuous Rem, hence uniform convergence on compact sets.

Finally, for Theorem 2.11, extend ψ by $\psi(\max(s, 0))$, pick $S_0 \ge 1$ so $|\psi(s)| < \varepsilon/4$ for $s \ge S_0$, set $M_\mathrm{b} = \max(\sup_{u\in[0,S_0+1]} e^{\alpha u}|\psi(u)|, |\psi(0)|) + 1$, and choose $T > S_0 + 3$ so $e^{-\alpha(T-1)}M_\mathrm{b} < \varepsilon/4$. The continuous clamped function $g(u) = \max(0, \min(1, S_0 + 1 - u))\, e^{\alpha u}\psi(u) + \max(0, \min(1, u - (T-1)))\,\psi(0)$ satisfies $g(0) = g(T) = \psi(0)$, lifting to $G \in C(\mathbb{R}/T\mathbb{Z}, \mathbb{C})$. By Fourier density on $\mathbb{R}/T\mathbb{Z}$ (`span_fourier_closure_eq_top`), a trigonometric polynomial P_trig approximates G within $\varepsilon/4$ uniformly; folding $\pm m$ harmonics via $\mathrm{Re}(c_{-m}e^{-im\omega_0 s}) = \mathrm{Re}(\bar{c}_{-m}e^{im\omega_0 s})$ $(\omega_0 = 2\pi/T)$ and multiplying by $e^{-\alpha s} \le 1$ yields (8) with $\mathrm{srank}_\mathbb{R}(K_M) \le 1 + dM$ by Proposition 2.13 and global error $< \varepsilon/4 + 2(\varepsilon/4) < \varepsilon$ across all $x \in \mathbb{R}^d$.

4 Formalization Architecture and Verification Audit

Table 5 breaks down the `KernelRankLean` library in Lean 4 [50, 48] (6,935 physical lines, 6,586 nonblank lines; earlier commits comprised 5,750 lines at `94a5b4a`, 5,578 lines at `1094bf1`, 5,324 lines at `9722ff5`, 3,822 lines at `6aabb4d` for d-D complex and multi-Gaussian ranks, 1,587 lines at `cae13d6` for 2-D plus differential annihilators, and 1,340 lines at `240dea0` for 2-D classification alone), and Table 6 maps every numbered mathematical statement to its exported Lean 4 identifier. Whereas prior formalized tensor-rank developments establish deep-network CP-rank bounds via generalized matricization in Isabelle/HOL [4] or finite-dimensional octonion and $\mathbb{F}_2$ matrix-multiplication rank bounds in Lean 4 [35, 76], our library couples continuous-domain Banach-algebra analysis with exact arbitrary-dimension real and complex multilinear rank formulas. Four proof-engineering challenges and solutions proved decisive:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a dependent family $\prod_{i\in\iota} X_i$

Table 6: Audit map from paper theorems and propositions to exported Lean 4 identifiers in `KernelRankLean` (all 32 exported theorems checked for standard classical axioms in `scripts/check`).

Paper result	Exported Lean 4 theorem identifier(s) (`KernelRank.*`)	File(s)	Loc
Eq. (2)–(4)	`separableRank_comp_surjective`, `separableRank_padTwo_le`	`Defs.lean`	239
Sec. 3.1 (2-D core)	`separableRank_eq_finrank_rowSpan`, `separableRank_ofReal`	`RowSpan`, `Complexification`	221
Sec. 3.1 (semigroup)	`exists_eq_exp_of_semigroup`, `exists_translation_generator`	`Semigroup`, `TranslationSemigroup`	262
Thm. 2.3	`separableRank_radial_lt_top_iff_exists`	`Main.lean`	538
Thm. 2.4	`separableRank_radial_complex_eq_iff`, `separableRank_sum_expPoly`	`Main`, `MultiComplexRank`	1,015
Thm. 2.5	`separableRank_radial_eq_iff`, `separableRank_radial_two_eq_iff`, `separableRank_radial_gaussian_eq_iff`	`Main`, `MultiRealRank`	2,562
Thm. 2.6	`exists_minimal_differential_annihilator`	`Annihilator.lean`	69
Thm. 2.8	`multiTotalCost_eq_borderTotalCost_add`, `exists_rank_le_borderTotalCost_tendstoUniformlyOn_radial_profile`	`Defs`, `Main`	538
Thm. 2.9(1, $\mathbb{R}$)	`separableRank_fin2_le_of_tendsto`, `totalCost_le_of_tendsto_radial_profile`, `borderTotalCost_le_of_tendsto_radial_real_profile`	`Approximation.lean`	1,184
Thm. 2.9(1, $\mathbb{C}$)	`sum_natDegree_add_one_le_of_tendsto_radial_complex`, `exists_rank_le_sum_natDegree_add_one_tendstoUniformlyOn_radial_complex`	`Approximation.lean`	1,184
Thm. 2.9(2, $\mathbb{R}^3$)	`not_exists_rank_le_two_tendsto_radial_cos`, `not_exists_rank_le_four_tendsto_radial_sq_mul_cos`, `exists_rank_le_five_tendstoUniformlyOn_radial_sq_mul_cos`	`Approximation.lean`	1,184
Thm. 2.11	`exists_harmonic_gaussian_uniform_approx`	`Approximation.lean`	1,184
Props. 2.13–2.14	`re_prod_eq_sum_re`, `re_coeff_prod_eq_sum_re`	`MultiRealRank.lean`	2,562

unifies the 2-D and d-D theorems, while specializing `rowSpan` in `RowSpan.lean` to homogeneous $\mathrm{Fin}\,2 \to \alpha$ (via a pattern-matched helper `fin2Vec`) avoids 200,000-heartbeat higher-order unification timeouts on $\mathrm{Fin}\,2 \to \mathbb{R}_{\geq 0}$.

2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting finite-dimensional linear maps inherit automatic continuity.

3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 2.6 directly.

4. **Decidable equality hygiene and basis-free adjugate obstructions:** In d-way product manipulations (`MultiRealRank.lean`), invoking local `let := Classical.decEq` rather than global `classical` avoids shadowing $\mathrm{Fin}\,d$'s canonical decidable equality inside `Finset.mul_prod_erase`; in `Approximation.lean`, formulating 3-D border-rank lower bounds via the everywhere-defined adjugate trace polynomial $\mathrm{tr}((\mathrm{adj}(S_0)S_1)^2) \geq 0$ avoids generic slice-invertibility hypotheses along limiting sequences.

Comparison with formalized tensor-rank developments and reusable components. Recent formalizations of tensor rank address complementary settings: Bentkamp, Blanchette, and Klakow [4] formalize finite-dimensional CP-rank and generalized matricization bounds over $\mathbb{R}$ in Isabelle/HOL to separate shallow and deep convolutional arithmetic circuits; Jain [35]

formalizes slice-peeling lower bounds (≥ 18, via a two-slice real pencil bound analogous to the $d = 3, k = 0$ base case of `cplxTruncForm_rank_ge`) and the Krawczyk contraction theorem in Lean 4 (checking the concrete rational certificate inequalities for rank ≤ 25 via external scripts) on the $8 \times 8 \times 8$ real octonion multiplication tensor; and Yang et al. [76] verify $R_{\mathbb{F}_2} \geq 21$ for 3×3 matrix multiplication over $\mathbb{F}_2$ via finite quotient-rank certificates in Lean 4. By contrast, `KernelRankLean` formalizes *functional* CP rank on infinite-dimensional domains $\mathbb{R}^d$ and proves exact parametric formulas for all dimensions $d \geq 2$ and degrees $k \geq 0$. Roughly two-thirds of the codebase ($\approx$ 4,400 of 6,935 lines) is general-purpose infrastructure reusable outside radial kernel theory:

- *Reusable general-purpose modules:* `Defs`, `RowSpan`, `Complexification`, and the lower-semicontinuity / adjugate-trace sections of `Approximation` (coordinatewise `separableRank` on arbitrary finite Pi types, surjection invariance, 2-way row-span rank and pointwise border-rank closure, and 3-way polynomial adjugate border-rank obstructions); `Semigroup` ($M(t) = \exp(tA)$ in any real Banach algebra, 96 lines via Mathlib's Bochner integral and semilinear analysis [21]); and the algebraic cores of `MultiComplexRank` and `MultiRealRank` (Propositions 2.13–2.14, the dual-span function-to-multilinear bridge `exists_linear_factors_of_span_eq_top`, and Alder–Strassen truncated-convolution / local-algebra rank lower bounds over $\mathbb{C}$ and $\mathbb{R}$).
- *Problem-specific radial-kernel glue:* `Translations`, `TranslationSemigroup`, `ExpPoly`, `RealExpPoly`, `Necessity`, `Representation`, `Differential`, `Annihilator`, `Main`, and the radial specializations of `Approximation` (the $s_j = x_j^2$ reduction, half-line shift semigroup on `translateSpan`, canonical upper-half-plane conjugate pairing, 5-term and multi-Gaussian border limits, and harmonic Gaussian universal approximation).

Classical axioms, constructive witnesses, and trustworthiness audit. In Lean 4, `separableRank` is marked `noncomputable` because taking the infimum in Definition 2.1 over all univariate factor families $X_i \to \mathbb{K}$ has no general decision procedure, and Mathlib's real numbers are classical Cauchy-sequence quotients. Nevertheless, all upper-bound and border-rank proofs in `MultiComplexRank.lean`, `MultiRealRank.lean`, `Main.lean`, and `Approximation.lean` construct *explicit closed-form* univariate factor formulas (Propositions 2.13–2.14, Remark 2.10, and Section 3.4), evaluated directly in `Applications/acoustics_3d.py`. Across all 32 audited top-level theorems, `#print axioms` reports *exclusively* Lean 4 and Mathlib's three standard classical axioms:

1. `propext` (*propositional extensionality*, $(P \leftrightarrow Q) \to P = Q$);
2. `Quot.sound` (*quotient soundness*, $r\,a\,b \to \text{Quot.mk}\,r\,a = \text{Quot.mk}\,r\,b$, underlying Cauchy-completion $\mathbb{R}$ and quotient spaces);
3. `Classical.choice` (*Hilbert's choice operator*, $\text{Nonempty}\,\alpha \to \alpha$, which together with `propext` and `Quot.sound` implies the law of the excluded middle $P \vee \neg P$ via Diaconescu's theorem and enters through $\mathbb{R}/\mathbb{C}$ arithmetic, coordinate surjection right inverses (2), and finite-dimensional basis extraction in $\mathcal{W}(\psi)$ and V).

Calculus of Inductive Constructions extended by {`propext`, `Quot.sound`, `Classical.choice`} is equiconsistent with ZFC set theory with countably many inaccessible cardinals [50, 48]. To audit trustworthiness mechanically, `scripts/check` (wired as the Git pre-commit hook) builds `KernelRankLean` and `Experiments` from source, rejects `sorry` (`sorryAx`), and checks `#print axioms` on all 32 theorems—guaranteeing zero custom `axiom` declarations, zero compiler-trust reflection axioms (`Lean.ofReduceBool` / `native_decide`), and zero `unsafe` blocks—while concrete positive and negative test theorems (`separableRank_radial_sq_mul_cos` $= 3d - 2$ and `not_exists_rank_le_four_...`) confirm that the 239-line specification in

Table 7: Integration map between formal theorems (Sections 2–3) and empirical validations (Sections 5.2–5.4).

Formal theorem / identity	Prediction tested	Empirical outcome on discrete grids & hardware	Section / ref.
Prop. 2.13 & Thm. 2.5 ($d = 2$)	Exact 2-D rank $C_2(\Gamma)$; steerable pass lower bound $2(m+1)R$	Confirmed: numerical SVD rank matches $C_2(\Gamma)$ across all 18 blur families ($20\times$–$40\times$ tap reduction on 161^2 gradient pair)	Secs. 5.2, 5.3; Tab. 8
Thms. 2.8–2.9 & $\mathrm{GL}_R(\mathbb{R})$	High-order/coalescing modes cancel severely; 2-D $\mathrm{GL}_R(\mathbb{R})$ cures it	Confirmed: $6c/1c_6$ raw cancellation ($595\times$–$1{,}110\times$) causes 7.9–15.2/255 `fp16` error; $\mathrm{GL}_R(\mathbb{R})$ + ascending sum cuts it to $\leq 0.31/255$	Sec. 5.3; Tab. 10, Fig. 2c
Thm. 2.3 (srank $< \infty$ iff exp.-poly)	Shell-isotropic discrete rank-R stencils approach continuous profiles as $L \gg R$	Confirmed for $R_{\text{px}} \geq 16$ (and warm-starts beat cold SVD by $1.3\times$–$2.2\times$); *bounded* at small $L{+}1 \leq R$, where discrete error hits 0	Sec. 5.3; Tab. 9, Figs. 3–4
Prop. 2.13 ($d = 3$, $k = 0$: $4 \to 3$) & Prop. 2.14 ($k = 1$: $9 \to 7$)	Exact 3-way real rank saves 25% ($k = 0$) and 22% ($k = 1$) passes vs. naive split	Confirmed: machine-precision ($< 10^{-15}$) stencil match; 24.0% GPU ($1.98 \to 1.51$ ms) and 21.2% CPU step-time cut on `Rank-7`	Sec. 5.4; Tab. 12, Fig. 5
Thm. 2.5 (C_d linear in weights)	Fixing shared Gaussian rates lets $c(\mathbf{x})$ vary via local weights $\gamma_u(\nu(\mathbf{x}))$	Confirmed: $\leq 0.185\%$ dispersion across $\nu \in [0.35, 0.80]$ and 7.8% 3-D lens error where 1-step `cuFFT` fails; *bounded* at sharp caustics (40%)	Sec. 5.4; Fig. 6
Complexity dRW vs. W^d & FFT	Separable FIR wins at moderate (R, W) or local/variable stencils	Bounded: beats `cuFFT` at $R{=}4$, $R_{\text{px}} \leq 16$ in 2-D and in `fp16` / $c(\mathbf{x})$ 3-D wave; `cuFFT` & 1-pass FDTD win outside those regimes	Secs. 5.3–5.4; Tabs. 11, 12

`Defs.lean` is non-vacuous. Alongside `Semigroup.lean`, the repository (`Experiments/` and `Isabelle/Semigroup_NSA.thy`) also includes four self-contained finite-dimensional proofs of $M(t) = \exp(tA)$ (206–399 lines) via finite differences and Euler's infinitesimal step in Lean 4 and Isabelle/HOL [24, 33] (audited with zero oracles via `Thm_Deps.all_oracles`).

5 Empirical Validation and GPU Case Studies

5.1 Theoretical hypotheses tested, error metrics, and hardware protocol

Why pair machine-checked continuous-domain theorems with unverified numerical and GPU experiments? Theorems 2.3–2.11 hold over exact real/complex arithmetic on unbounded $\mathbb{R}^d$, whereas practical filters and PDE solvers operate on finite $[-L, L]^d$ integer grids, in 16- or 32-bit floating-point arithmetic, and against competing non-separable algorithms (`cuFFT` and compact FDTD). Sections 5.2–5.4 therefore test *specific quantitative predictions and breakdown boundaries* of our theorems under discretization and finite precision, as mapped in Table 7 (keeping the verification boundary strict: Sections 2–4 are formally proved in Lean 4; Section 5 is reproducible empirical Python/Triton code).

Hardware platform, reproducibility, and explicit error-metric definitions. All scripts, auto-tuned Triton/CUDA kernels (`Applications/gpu_kernels.py`), package requirements (`Applications/requirements.txt`), and JSON result archives live in `Applications/` (explicit coefficients in Appendix A). Measurements were collected on Linux `7.0.0-43-generic` (`x86_64`) with an Intel Core Ultra 5 125H CPU (14 cores / 18 threads, 18 MB L3 cache, 61 GiB DDR5) and an external NVIDIA GeForce RTX 4070 Ti Super eGPU (16,376 MiB GDDR6X, 48 MB L2 cache, 672 GB/s peak VRAM bandwidth, connected via Thunderbolt / PCIe Gen3 $\times 4$, driver `615.71.09`), running Python `3.14.4`, NumPy `2.5.4`, SciPy `1.18.1` (HiGHS LP), PyTorch `2.15.0` (CUDA `13.2`), and Triton `3.9.0`. All GPU timings use on-device `torch.cuda.Event` barriers around VRAM-resident tensors (excluding host-to-eGPU PCIe $\times 4$ transfers). Six quantitative error metrics govern the evaluations below:

1. *2-D Chebyshev pass/stopband ripple:* $\delta(\varphi) := \max_{r\in[0,1]\cup[1+\Delta r,\,\sqrt{2}(1+\Delta r)]} |\varphi(r^2) - \mathbf{1}_{r\le 1}|$ ($\Delta r = 0.2$, evaluated on 2,000 radial nodes in continuous LP design and on all discrete grid points $\|\mathbf{m}\|_2/R_{\mathrm{px}}$ of $\{-L,\dots,L\}^2$, $L = \lceil 1.2\,R_{\mathrm{px}}\rceil$).
2. *Equal-radius discrete shell anisotropy:* $\Delta_{\mathrm{sh}}(K) := \max_{N\le 2L^2,\,|S_N|>1}\big(\max_{\mathbf{m}\in S_N} K[\mathbf{m}] - \min_{\mathbf{m}\in S_N} K[\mathbf{m}]\big)$ over Euclidean lattice shells $S_N = \{\mathbf{m}\in\{-L,\dots,L\}^2 : m_x^2 + m_y^2 = N\}$.
3. *IEEE `float16` 8-bit sRGB and linear-light bokeh errors:* Given a unit-DC kernel ($\sum_{\mathbf{m}} K[\mathbf{m}] = 1$) on 512×512 scene-linear HDR images $I \ge 0$, we compare the `float16`-intermediate/accumulator output $\hat{I}_{16}$ against the exact `float64` output I_{64} via $E_{\mathrm{sRGB}} = 255\max_{x,y}|\mathrm{sRGB}(\hat{I}_{16}^{[0,1]}) - \mathrm{sRGB}(I_{64}^{[0,1]})|$ (clamping to $[0,1]$ before the exact piecewise IEC 61966-2-1 transfer function) and $E_{\mathrm{lin}} = 255\max_{x,y}|\hat{I}_{16} - I_{64}|$, across 20 random glint scenes (`seed=0..19`) and glint peaks $8\times, 48\times, 256\times$.
4. *3-D Brillouin-octant phase-velocity error and effective bandwidth:* For discrete symbol $\widehat{M}(\mathbf{k}) = \sum_{\mathbf{m}\in[-L,L]^3} M[\mathbf{m}]e^{-i\mathbf{k}\cdot\mathbf{m}}$ at Courant number $\nu = c_0\Delta t/\Delta x$, $|\delta c/c_0| := \max_{0<\|\mathbf{k}\|_2\le k_c}\big|\arccos(\widehat{M}(\mathbf{k})/2)/(\nu\|\mathbf{k}\|_2) - 1\big|$, with effective τ-bandwidth $k_\tau := \sup\{\kappa\in(0,\pi] : \max_{\|\mathbf{k}\|_2\le\kappa}|c_{\mathrm{num}}(\mathbf{k})/c_0 - 1| \le \tau\}$ ($\tau\in\{0.1\%, 0.5\%\}$) and von Neumann stability $|\widehat{M}(\mathbf{k})| \le 2$ audited across 232,620 symmetry-reduced octant/shell wavevectors plus multi-start Nelder–Mead refinement.
5. *3-D spherical-pulse relative L^2 waveform error:* $\|p_{\mathrm{num}}(\mathbf{x}_r,\cdot) - p_{\mathrm{ref}}(\mathbf{x}_r,\cdot)\|_{\ell^2}/\|p_{\mathrm{ref}}(\mathbf{x}_r,\cdot)\|_{\ell^2}$ (where $\|g\|_{\ell^2} = (\sum_n |g(t_n)|^2)^{1/2}$ over all time steps) across axial $(48,0,0)$, face-diagonal $(34,34,0)$, and body-diagonal $(28,28,28)$ receivers on 256^3 against the exact analytical 3-D D'Alembert solution (and on 128^3 lens refraction against a 256^3 fourth-order spectral Lax–Wendroff PDE reference).
6. *Implementation verification and run-to-run timing variability:* Every GPU benchmark run asserts numerical agreement of its output tensor against a `float64` reference with identical boundary rules (`err_max_*`, `err_rms_*`: max `float32` error $\le 6.1\times 10^{-6}$ in 2-D and $\le 6.6\times 10^{-6}$ in 3-D) and reports mean $\pm$ std across 10 CUDA-Event batches (6 batches on `DirectStep3D`). Startup autotuning and VRAM buffer alignment shift means by up to $\approx 6\%$ in 2-D and $\approx 3\%$ in 3-D between full runs, so few-percent differences are within empirical noise.

Before turning to discrete designs, note a continuous-domain Fourier duality: for $\mathrm{Re}\,\lambda < 0$, differentiating $\mathcal{F}_{\mathbb{R}^d}[e^{\lambda\|x\|^2}](\xi) = (\pi/(-\lambda))^{d/2}\exp(\|\xi\|^2/(4\lambda))$ k times with respect to λ maps $P(\|x\|^2)e^{\lambda\|x\|^2}$ ($\deg P = k$) to $\widetilde{P}(\|\xi\|^2)e^{\|\xi\|^2/(4\lambda)}$ with $\deg\widetilde{P} = k$ and $\mathrm{Re}(1/(4\lambda)) < 0$. By Theorems 2.4 and 2.5, the d-dimensional Fourier transform preserves exact complex and real d-way separable rank across all integrable radial exponential-polynomial kernels.

5.2 Harmonic "beach-ball" differential operators and steerable kernels

Complex partial-derivative operators $(\partial_x + i\partial_y)^m$ provide a canonical rotation-equivariant basis for 2-D symmetry and steerable filters [26, 38, 7, 62]. Specializing Proposition 2.13 to equal symbolic factors $z_0 = \cdots = z_{m-1} = \partial_x + i\partial_y$ (writing $m \ge 1$ for the harmonic order and $D_\theta = \cos\theta\,\partial_x + \sin\theta\,\partial_y$ for the directional derivative along angle θ) gives, for *every* integer $m \ge 1$ (both odd and even),

$$\mathrm{Re}\big((\partial_x + i\partial_y)^m\big) = \frac{2^{m-1}}{m}\sum_{r=0}^{m-1}(-1)^r D_{\pi r/m}^m. \tag{14}$$

On a regular $2m$-point ring $\theta_n = \pi n/m$ ($n = 0,\dots,2m-1$) of radius $h\to 0$, antipodal points satisfy $D_{\theta_{r+m}} = -D_{\theta_r}$ while Fourier orthogonality of $(-1)^r = e^{i\pi r}$ annihilates all Taylor

orders below m. For $f \in C^m$ near (x, y), (14) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order ($m = 2N + 1$) hexagonal/decagonal "beach-ball" operators of Niemitalo [56] and the even-order operators of Perona [60] in a single formula:

$$\lim_{h\to 0} \frac{1}{h^m} \sum_{n=0}^{2m-1} (-1)^n f\Big(x + h\cos\frac{\pi n}{m},\, y + h\sin\frac{\pi n}{m}\Big) = \frac{1}{(m-1)!\,2^{m-2}} \operatorname{Re}\big((\partial_x + i\partial_y)^m\big) f(x, y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\operatorname{Im}((\partial_x + i\partial_y)^m)$. Recognizing $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial_{\bar z}^m$ to *any* smooth radial kernel $\varphi(x^2+y^2) = \varphi(z\bar z)$ produces no lower-order radial derivatives [7, 62]: since $\partial_{\bar z}(z) = 0$, induction gives

$$\left(\frac{\partial_x + i\partial_y}{2}\right)^m \varphi(x^2 + y^2) = (x + iy)^m\, \varphi^{(m)}(x^2 + y^2) = r^m e^{im\theta}\, \varphi^{(m)}(r^2). \tag{15}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with n_{re} real and n_{cx} nonreal rates (all $\lambda_u \neq 0$, approximating, e.g., an ideal circular low-pass filter [57]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by λ_u^m) of continuous 2-way rank $R = n_{\mathrm{re}} + 2n_{\mathrm{cx}}$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [55]—absorbs the monomials of $\operatorname{Re}((x + iy)^m)$ and $\operatorname{Im}((x + iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2\rfloor + 1\big)(n_{\mathrm{re}} + 2n_{\mathrm{cx}}) \quad \text{for } \cos(m\theta), \qquad \lceil m/2\rceil(n_{\mathrm{re}} + 2n_{\mathrm{cx}}) \quad \text{for } \sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms. On the 161×161 ($\omega_c = \pi/64$) orientation-estimation filter of [55], replacing direct $2 \times 161^2 = 51{,}842$ taps/pixel by a two-complex-Gaussian ($R = 4$) analytical gradient pair reduces scalar multiply–adds to 2,576 (20×, or 40× with even/odd tap-parity folding).

Continuous-domain pass-count bounds for simultaneous steerable pairs. Can sharing intermediate 1-D continuous convolutions reduce simultaneous evaluation of $(C_m * I, S_m * I)$ below separate evaluation $2(\lfloor m/2\rfloor + 1)R + 2\lceil m/2\rceil R = 2(m + 1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m, S_m) = (A_c + B_c, A_s + B_s)$ into P_y shared vertical passes followed by $\operatorname{rank}(A_c) + \operatorname{rank}(A_s)$ horizontal passes plus P_x shared horizontal passes followed by $\operatorname{rank}(B_c) + \operatorname{rank}(B_s)$ vertical passes, then

$$\begin{aligned} \dim \operatorname{rowSpan}_x(C_m, S_m) &\le \operatorname{rank}(A_c) + \operatorname{rank}(A_s) + P_x, \\ \dim \operatorname{colSpan}_y(C_m, S_m) &\le P_y + \operatorname{rank}(B_c) + \operatorname{rank}(B_s). \end{aligned}$$

Because C_m and S_m on $\mathbb{R}^2$ span all powers $\{x^0, \dots, x^m\} g_n(x)$ in x and $\{y^0, \dots, y^m\} g_n(y)$ in y, adding the two inequalities shows that *every* two-stage continuous 1-D schedule requires at least $2(m + 1)R$ passes ($4R$ passes for (h_x, h_y)). Savings instead come from: (i) definite even/odd parity of every 1-D factor $u^q g_n(u)$, halving multiplications per sampled pass via $I[y + \ell] \pm I[y - \ell]$ folding; (ii) sharing $\{I *_y (y^q g_n)\}_{q=0}^{M}$ across *multiple* harmonic orders ($5R$ instead of $6R$ for $m = 0, 1$; $10R$ instead of $12R$ for $m = 1, 3$); and (iii) for the d-D gradient $(h_{x_0}, \dots, h_{x_{d-1}})$ of a multi-Gaussian radial kernel ($d \ge 3$, base rank $R_d = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$), recursively splitting $\{0, \dots, d-1\}$ into halves to reduce $d^2 R_d$ passes to $T(d)R_d$ with $T(1) = 1$ and $T(d) = T(\lfloor d/2\rfloor) + T(\lceil d/2\rceil) + d \le d(\lceil \log_2 d\rceil + 1)$ ($8R_3$ vs. $9R_3$ in $d = 3$; $12R_4$ vs. $16R_4$ in $d = 4$).

5.3 Worked application I: 2-D circular lens blur and $\mathrm{GL}_R(\mathbb{R})$ conditioning

In depth-of-field bokeh rendering [54, 27, 58], the ideal defocus point-spread function is the discontinuous unit disc $\mathbf{1}_{r\le 1}$ ($r = \sqrt{x^2 + y^2}$), which has infinite separable rank: for any $0 <$

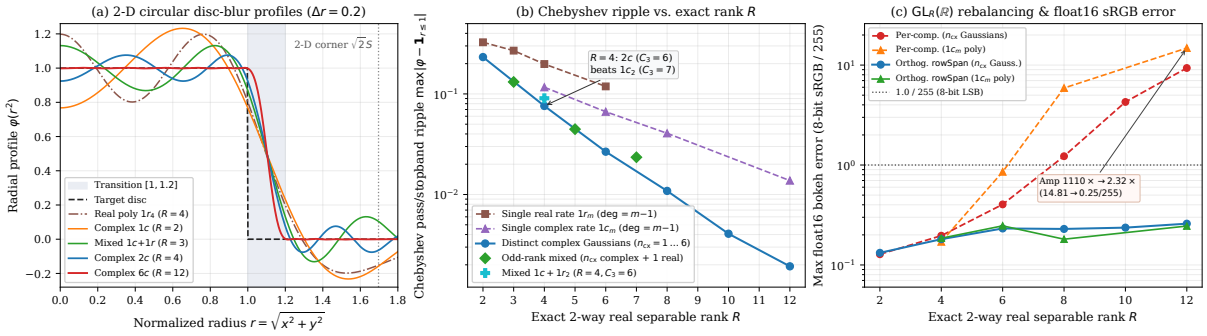


Figure 2: Two-dimensional circular disc blur ($\Delta r = 0.2$, `Applications/blur_2d.py`): (a) minimax radial profiles $\varphi(r^2)$ against the unit disc; (b) minimax pass/stopband ripple across exact 2-way real separable ranks $R \in [2, 12]$ for distinct complex Gaussians, odd-rank mixed Gaussians, and repeated-rate polynomial-Gaussians; (c) maximum IEEE `float16` bokeh error in 8-bit sRGB units (/255, via IEC 61966-2-1 sRGB encoding on a 512×512 linear-light HDR scene with night shadows 0.001 and 3×3 glints up to 48.0, $R_{\text{px}} = 25$, $W = 61$) before and after nuclear-norm-optimal $\text{GL}_R(\mathbb{R})$ rebalancing on rowSpan(K).

Table 8: Representative 2-D circular disc-blur designs ($\Delta r = 0.2$, $R_{\text{px}} = 25$, $W = 61$; explicit parameters in Table 14). Exact 2-D rank $R = C_2(\Gamma)$ and 3-D ball rank $C_3(\Gamma)$ follow Theorem 2.5; SVD is the empirical singular-value count ($\sigma_j > 10^{-10}\sigma_1$) of the 61×61 kernel; $\kappa_\infty = (\sum_n \|h_n\|_\infty \|v_n\|_\infty)/\|K\|_\infty$ measures peak channel amplification before (raw) and after (orth) rowSpan rebalancing, alongside max `float16` HDR bokeh error in 8-bit sRGB units (/255, IEC 61966-2-1).

Structure	Degrees	R (d=2)	SVD	C_3 (d=3)	Ripple $\pm\delta$	$\kappa_\infty^{\text{raw}}$	$\kappa_\infty^{\text{orth}}$	sRGB raw	sRGB orth
$1c$ (Gaussian)	0	2	2	3	0.2324	1.65	1.30	0.13	0.13
$1c + 1r$ (mixed)	0, 0	3	3	4	0.1315	3.45	1.65	0.16	0.20
$2c$ (Gaussians)	0, 0	4	4	6	0.0759	5.28	1.95	0.20	0.18
$1c + 1r_2$ (mixed poly)	0, 1	4	4	6	0.0905	4.50	1.92	0.12	0.23
$1c_2$ (complex poly)	1	4	4	7	0.1165	5.68	1.88	0.17	0.19
$1r_4$ (real Laguerre)	3	4	4	7	0.1982	4.74	1.56	0.21	0.23
$2c + 1r$ (mixed)	0, 0, 0	5	5	7	0.0446	8.73	2.08	0.26	0.20
$4c$ (Gaussians)	0^4	8	8	12	0.0109	70.06	2.31	1.22	0.23
$6c$ (Gaussians)	0^6	12	12	18	0.0019	595.01	2.49	9.35	0.26
$1c_6$ (complex poly)	5	12	12	23	0.0138	1,109.75	2.32	14.81	0.25

$r_1 < \cdots < r_M < 1$, evaluating $\mathbf{1}_{x_i^2+y_j^2 \le 1}$ at $x_i = r_i$, $y_j = \sqrt{1 - r_j^2}$ produces an invertible $M \times M$ upper-triangular binary matrix. Introducing a transition band $[1, S]$ with $S = 1 + \Delta r$ ($\Delta r = 0.2$) and truncating each 1-D pass to $|x|, |y| \le S$ restricts the 2-D separable support $[-S, S]^2$ to radii $r \le \sqrt{2}\,S \approx 1.697$. For any fixed rates (α_u, ω_u), minimizing the sampled Chebyshev ripple $\max_{r \in [0,1] \cup [S, \sqrt{2}S]} |\varphi(r^2) - \mathbf{1}_{r \le 1}|$ (2,000 radial nodes) over all polynomial weights is a linear program, solved with HiGHS inside differential-evolution/Nelder–Mead rate search in `Applications/blur_2d.py`.

Figure 2 and Table 8 highlight four structural findings:

1. **Role of nonreal rates, odd budgets, and 3-D lifting:** Across all 18 tested families, empirical SVD rank matches $C_2(\Gamma)$ identically. Distinct complex Gaussians achieve geometric ripple decay with $R = 2n_{\text{cx}}$ (0.2324 → 0.0019 from $1c$ to $6c$, or 0.0014 under global Remez search [58]), while adding a single real Gaussian ($1c+1r, 2c+1r$) interpolates cleanly at odd ranks $R = 3, 5$. Free real-only rates coalesce into repeated-rate Laguerre–Gaussians ($1r_m$). At $R = 4$, four distinct structures tie in 2-D cost, but lifting to a 3-D spherical ball

splits their exact 3-way ranks: $2c$ gives both the lowest ripple (± 0.0759) and lower 3-D rank ($C_3 = 6$ vs. $C_3 = 7$ for $1c_2$ and $1r_4$).

2. **Transitive $\mathrm{GL}_R(\mathbb{R})$ freedom and four-stage `separableChannels` ablation (Table 10):** While $R = \dim \mathrm{rowSpan}(K)$ is fixed, in $d = 2$ writing the $W \times W$ symmetric kernel matrix as $K = HV^\top$ ($H, V \in \mathbb{R}^{W \times R}$) allows replacing (H, V) by $(HT, VT^{-\top})$ for *any* invertible matrix $T \in \mathrm{GL}_R(\mathbb{R})$ [70]. To remove inter-component cancellation ($\kappa_\infty^{\mathrm{raw}} = 595$ on $6c$ and 1,110 on $1c_6$) without a full $W \times W$ SVD, compute a thin QR factorization $H = QR_H$ ($Q^\top Q = I_R$), project $K = QMQ^\top$ onto the $R \times R$ symmetric matrix $M = R_H V^\top Q$, and diagonalize $M = U \,\mathrm{diag}(e_1, \ldots, e_R) U^\top$. Setting $G = QU = [g_1, \ldots, g_R]$ yields $K(x, y) = \sum_{n=1}^{R} e_n g_n(x) g_n(y)$, where $\tilde{h}_n = \sqrt{|e_n|}\, g_n$ and $\tilde{v}_n = \mathrm{sign}(e_n)\tilde{h}_n$ achieve the nuclear-norm minimum $\sum_{n=1}^{R} \|\tilde{h}_n\|_2 \|\tilde{v}_n\|_2 = \|K\|_{S_1}$ ($\kappa_\infty \le 2.50$). Factoring $T \in \mathrm{GL}_R(\mathbb{R})$ into pure diagonal pair scaling, non-diagonal orthogonalization, and channel permutation in Table 10 isolates each mechanism across 20 random HDR scenes: (i) diagonal pair balancing $\|c_n h_n\|_2 = \|v_n / c_n\|_2$ (Stage 1 $\to$ Stage 2, leaving every rank-1 matrix $h_n v_n^\top$ invariant) cures `float16` intermediate overflow when degree-5 weights sit on the first pass ($1c_6$: 20/20 overflow $\to$ $15.17 \pm 1.39/255$), yet *leaves cross-component cancellation intact* ($7.90 \pm 0.92/255$ on $6c$, $15.17 \pm 1.39/255$ on $1c_6$); (ii) cross-component $\mathrm{GL}_R(\mathbb{R})$ orthogonalization (Stage 2 $\to$ Stage 3) cuts 20-scene sRGB max error by 54% on $4c$ ($0.54 \to 0.25/255$), 96.2% on $6c$ ($7.90 \to 0.30/255$), and 98.2% on $1c_6$ ($15.17 \to 0.28/255$) (at $R \le 4$, raw and orthogonalized errors both sit near the $\le 0.24/255$ quantization floor); and (iii) reversing accumulation from largest-$|e_n|$-first to smallest-$|e_n|$-first (Stage 3 $\to$ Stage 4) cuts *linear-light* highlight error by 1.5×–2.9× across all designs and glint peaks ($0.40 \to 0.15/255$ at 48× and $1.41 \to 0.60/255$ at 256× on $6c$, and $0.23 \to 0.08/255$ in sRGB at unclipped 8×; at 48× and 256×, sRGB clips interior disc highlights > 1.0, so unclipped rim pixels govern sRGB max error near 0.24–0.31/255 at 48× and 1.07–1.76/255 at 256×). By contrast, in $d \ge 3$, minimal d-way CP decompositions lack transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations [41], making well-separated rates essential.

3. **GPU and CPU throughput vs. tuned direct W^2 stencils and global `cuFFT` (Table 11):** In `Applications/gpu_kernels.py`, all three 1920×1080 RGBA GPU paths are tuned around the 48 MB L2 cache: `SeparableBlur2D` evaluates all R `rowSpan`-orthogonalized terms per pass ($2c/4c/6c$ for $R = 4, 8, 12$) using K-output register sliding windows over transposed L2-resident row strips; `DirectBlur2D` is a dense $W \times W$ Triton kernel sharing loaded row segments across outputs; and `FFTBlur2D` packs RGBA channel pairs into two complex planes for complex 2-D `cuFFT` (`fft2`/`ifft2` with in-place real spectral multiply) at a tuned 7-smooth period ($\ge \mathrm{size} + L$ for zero padding, $\ge \mathrm{size} + 2L$ for clamped borders). Notably, direct $W \times W$ and `cuFFT` runtimes depend *only* on (R_{px}, W)—they evaluate any symmetric kernel on $[-L, L]^2$ (whether $2c$, $6c$, or an unapproximated disc) in the same time, whereas separable FIR cost grows with R. At $R_{\mathrm{px}} = 64$, the separable cascade beats the tuned direct $W \times W$ kernel by 28.0× (`fp32`) / 29.5× (`fp16`) at $R = 4$, 13.9× / 14.6× at $R = 8$, and 9.2× / 9.4× at $R = 12$ (whereas at $R_{\mathrm{px}} = 8$, direct 0.52 ms beats $R \ge 8$). Against `cuFFT` (0.451–0.555 ms across all radii), the separable cascade wins only at $R = 4$ for $R_{\mathrm{px}} \in \{8, 16\}$ in both precisions (0.331 / 0.417 ms vs. 0.457 ms; 0.354 / 0.396 ms vs. 0.454 ms; at $R_{\mathrm{px}} = 32, R = 4$, `fp16` 0.496 ms and clamp `cuFFT` 0.481 ms are within run-to-run variation), while `cuFFT` is both faster and free of finite-rank approximation ripple for every $R \ge 8$ and for $R = 4$ at $R_{\mathrm{px}} = 64$. Because strip intermediates stay in L2 rather than DRAM, `float16` is no longer uniformly faster than `float32` (`fp16`/`fp32` time ratios span 0.84× to 1.26× due to 16-bit gather/store overhead, though Table 10's conditioning remains essential whenever textures or APIs mandate `float16`). On the 14-core CPU (`float32`), by contrast, unfused $2R$-pass PyTorch `conv2d` writes 33 MB frame buffers to DDR5 RAM (169.1, 286.7, 444.0 ms at $R = 4$ vs. `PocketFFT`

`rfft2` at $32.6, 40.7, 68.1$ ms and direct `conv2d` at $68.2, 208.2, 676.6$ ms for $R_{\text{px}} = 8, 16, 32$). Separable decompositions at large radii are thus most compelling with $O(1)$ recursive IIR Gaussians [19, 77], spatially varying depth-of-field radii $R_{\text{px}}(x, y)$ [27], or L2/line-buffered pipelines without spectral padding buffers.

4. **Direct discrete rank-R optimization, convergence, and warm-starting (Figures 3–4, Table 9):** Because any finite-tap separable cascade on $\{-L, \dots, L\}^d$ is rectangular-truncated and even in every coordinate ($h_n[-m] = h_n[m]$), its 1-D factors lie in an $(L+1)$-dimensional even subspace, capping 2-D discrete rank at $\min(R, L+1)$ (for instance, sampling $6c$ of continuous rank 12 gives exact discrete SVD rank $4, 6, 11, 12, 12, 12$ at $R_{\text{px}} = 2, 4, 8, 16, 25, 64$, with σ_{12}/σ_1 locking onto 4.37–4.34×10^{-4} for $R_{\text{px}} \geq 16$). Table 9 and Figures 3–4 compare sampled continuous Gaussians against (i) direct Eckart–Young ℓ_2 SVD truncation of a raised-cosine smooth disc on $[1, 1.2]$, (ii) unconstrained D_4-symmetric discrete rank-R minimax optimization (alternating pass/stopband clamping with SVD rank-R projection), and (iii) shell-isotropic discrete rank-R minimax (equalizing lattice points of equal Euclidean radius $m_x^2 + m_y^2 = N$, with shell anisotropy $\Delta_{\text{sh}} = \max_N(\max_{\|\mathbf{m}\|^2=N} K - \min_{\|\mathbf{m}\|^2=N} K)$ reported to six decimals in Figure 3 to distinguish exact machine-zero continuous symmetry $\Delta_{\text{sh}} = 0.000000$ from discrete rank-4 SVD projection residuals 0.000001 at $R_{\text{px}} = 6$ and 0.000178 at $R_{\text{px}} = 25$):

   - *Small stencils and lattice chord coincidences ($R_{\text{px}} \leq 12$):* Once $R \geq L + 1$, every even matrix has rank $\leq R$ and discrete error is identically 0 (∇ in Figure 4a). Even slightly above $R = L + 1$, small integer lattices permit discrete gains (± 0.00476 shell-isotropic vs. ± 0.07593 continuous at $R_{\text{px}} = 6, R = 4$, or $\leq 1.4 \times 10^{-5}$ at isolated radii such as $R_{\text{px}} \in \{4, 5, 8\}$ for $R = 2$ and $R_{\text{px}} = 12$ for $R = 4$ where the $[1, 1.2]$ transition gap partitions passband/stopband circle chords into $\leq R$ staircase blocks; at $R_{\text{px}} = 6$, only $1^2 + 7^2 = 5^2 + 5^2 = 50$ shares a squared radius in $\{0, \dots, 7\}^2$, leaving 0°-vs.-45° freedom in Figure 3).
   - *Direct ℓ_2 SVD vs. Chebyshev minimax and large-radius convergence ($R_{\text{px}} \geq 16$):* Direct ℓ_2 SVD degrades in both pass/stopband ripple (0.09556–0.10295 vs. 0.07593 at $R = 4$) and 17.7%–19.2% shell anisotropy (visible cross-streaks in Figure 3), while unconstrained discrete minimax lowers ripple (0.05226 at $R_{\text{px}} = 25$) *only by warping the transition rim* $1 < r < 1.2$ *anisotropically* ($\Delta_{\text{sh}} = 0.09052$). Once $R_{\text{px}} \geq 16$ and equal-radius shell isotropy is enforced, discrete minimax converges monotonically onto our continuous $1c/2c/4c/6c$ asymptotes (Figure 4a; 0.23245, 0.07582, 0.01065, and 0.00171 vs. 0.23245, 0.07593, 0.01086, and 0.00193 at $R_{\text{px}} = 64$).
   - *Does warm-starting discrete minimax from continuous complex Gaussians speed up optimization? (Figure 4b):* The answer splits cleanly by regime. At *small* radii ($R_{\text{px}} = 6, R = 4$), warm-starting from $2c$ does *not* accelerate optimization: raised-cosine ℓ_2 SVD already starts 3.5× lower on the joint shell-isotropic objective $\max(\text{rip}, \|K - \mathcal{P}_{\text{iso}} K\|_\infty)$ (0.0218 vs. 0.0759) and stays ahead through 500 SVDs (0.0058 vs. 0.0106) because small stencils optimize discrete inter-shell gaps rather than continuous equioscillation. At *large* radii ($R_{\text{px}} = 25$), by contrast, Theorem 2.3 makes the intersection $\mathcal{V}_{\text{iso}} \cap \mathcal{M}_{\leq R}$ tangential to non-exponential-polynomial cold starts: warm-starting from $2c/4c/6c$ starts at $k = 0$ directly on the zero-anisotropy manifold ($0.0759, 0.0109, 0.00193$), whereas raised-cosine SVD starts $1.3\times$ –$2.2\times$ higher ($0.0956, 0.0129, 0.00422$, needing ~ 120 SVDs at $R = 12$ just to match $6c$'s $k = 0$ error and still trailing at $k = 500$: 0.00109 vs. 0.00076), and hard-disc SVD (0.027 at $k = 500$) and random rank-R starts (> 1.0) stall in poor nonconvex local minima.

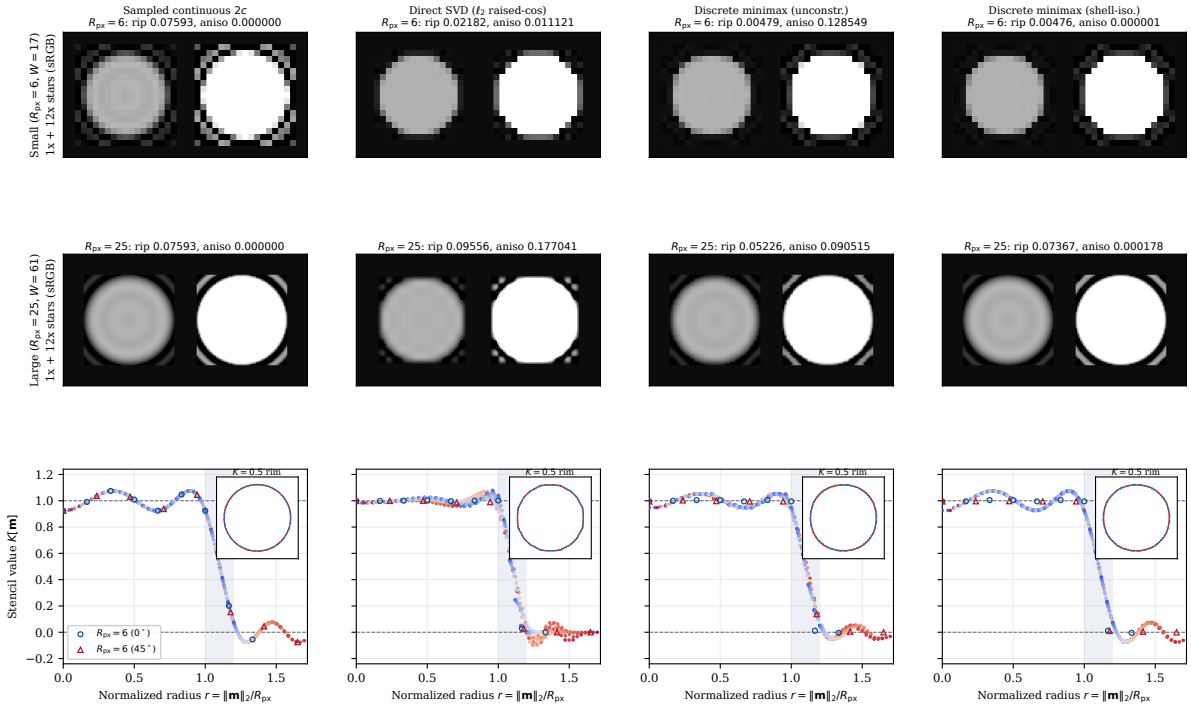


Figure 3: Visual and quantitative comparison at exact rank $R = 4$ between sampled continuous $2c$ Gaussians, direct ℓ_2 SVD of a raised-cosine disc, unconstrained discrete rank-4 minimax, and shell-isotropic discrete rank-4 minimax (`Applications/blur_2d.py`): top row, pixelated small stencil ($R_{\text{px}} = 6, W = 17$) sRGB bokeh on 1× (left) and 12× HDR (right) point stars; middle row, pixelated large stencil ($R_{\text{px}} = 25, W = 61$) sRGB bokeh (with 5-decimal ripple and 6-decimal shell anisotropy Δ_{sh}); bottom row, per-pixel radial scatter ($R_{\text{px}} = 25$ dots colored from 0° blue to 45° red; $R_{\text{px}} = 6$ open 0°/45° markers) and $K = 0.5$ rim contours ($R_{\text{px}} = 6$ solid red, $R_{\text{px}} = 25$ dashed blue, reference circle dotted black).

5.4 Worked application II: isotropic 3-D acoustic wave simulation

Consider the 3-D constant-speed scalar acoustic wave equation $\partial_{tt} p = c_0^2 \nabla^2 p$ discretized on $\mathbb{Z}^3$ with spacing Δx and time step $\Delta t = \nu \Delta x / c_0$ (Courant number $\nu > 0$) by the explicit two-step leapfrog update

$$p^{n+1}[\mathbf{m}] + p^{n-1}[\mathbf{m}] = (M * p^n)[\mathbf{m}], \qquad \mathbf{m} \in \mathbb{Z}^3. \tag{16}$$

Substituting a discrete plane wave $p^n[\mathbf{m}] = e^{i(\mathbf{k}\cdot\mathbf{m} - \omega_{\text{num}} n \Delta t)}$ with normalized wavenumber $\mathbf{k} = (k_x, k_y, k_z) \in [-\pi, \pi]^3$ turns the left-hand side of (16) into $2\cos(\omega_{\text{num}}\Delta t)\, p^n[\mathbf{m}]$ and the right-hand side into $\widehat{M}(\mathbf{k})\, p^n[\mathbf{m}]$, where $\widehat{M}(\mathbf{k}) = \sum_{\mathbf{m}} M[\mathbf{m}] e^{-i\mathbf{k}\cdot\mathbf{m}}$ is the discrete Fourier symbol of M. Since an exact continuous plane wave travels at speed c_0 ($\omega_{\text{exact}}\Delta t = c_0(\|\mathbf{k}\|_2/\Delta x)\Delta t = \nu\|\mathbf{k}\|_2$), matching the exact wave equation requires $\widehat{M}_{\text{exact}}(\mathbf{k}) = 2\cos(\nu\|\mathbf{k}\|_2)$ [67], and the numerical-to-exact phase-velocity ratio is $c_{\text{num}}(\mathbf{k})/c_0 = \arccos(\widehat{M}(\mathbf{k})/2)/(\nu\|\mathbf{k}\|_2)$. Whenever $|\widehat{M}(\mathbf{k})| \le 2$, both roots of $\zeta^2 - \widehat{M}(\mathbf{k})\zeta + 1 = 0$ have $|\zeta| = 1$ (spectral von Neumann stability); strict interior containment $\widehat{M}(\mathbf{k}) \in (-2, 2)$ for $\mathbf{k} \neq \mathbf{0}$ avoids repeated roots $\zeta = \pm 1$ (linear-in-n secular growth) and yields a conserved positive discrete energy on all nonzero modes. Standard 7-point FDTD ($\nu \le 1/\sqrt{3}$) suffers 19.88% axial phase-velocity error by $\|\mathbf{k}\|_2 = 0.80\pi$, motivating dispersion-relation-preserving and high-order room-acoustics FDTD designs [68, 40, 73, 30, 31]. Two structural facts explain why radial multi-Gaussians are natural here:

- **Inadequacy of additive axial stencils of any order:** Any purely axial scheme $\widehat{M}_{\text{ax}}(\mathbf{k}) =$

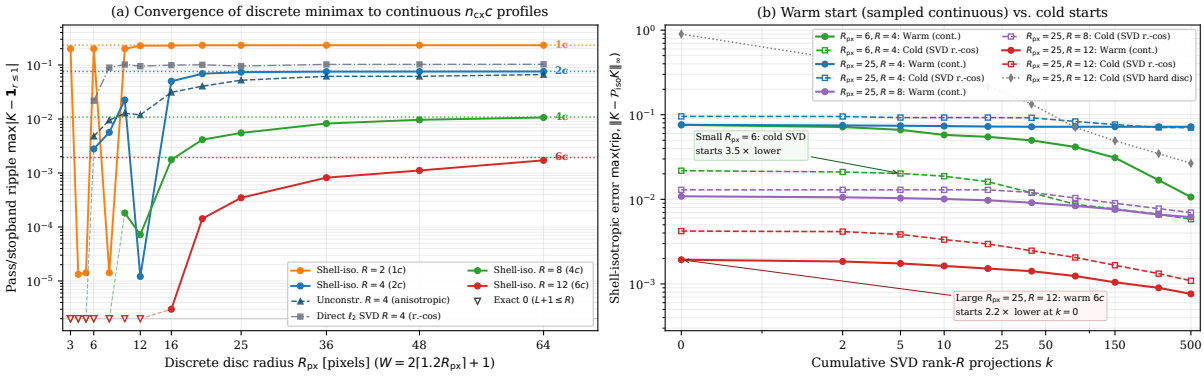


Figure 4: Discrete-to-continuous convergence and warm-starting (`Applications/blur_2d.py`): (a) pass/stopband Chebyshev ripple across $R_{\text{px}} \in [3, 64]$ for shell-isotropic discrete rank-R minimax ($R \in \{2, 4, 8, 12\}$, converging to the dotted continuous $1c/2c/4c/6c$ asymptotes once $R_{\text{px}} \geq 16$; open triangles ∇ mark exact 0 error when $L+1 \leq R$), alongside $R = 4$ unconstrained minimax and direct ℓ_2 SVD; (b) single-run trajectories of the shell-isotropic minimax objective $\max(\text{rip}, \|K - \mathcal{P}_{\text{iso}}K\|_\infty)$ vs. cumulative SVD projections $k \in [0, 500]$ comparing continuous-Gaussian warm starts against raised-cosine SVD and hard-disc SVD cold starts.

Table 9: Sampled continuous complex Gaussians vs. direct discrete rank-R optimization across stencil radii $R_{\text{px}} \in \{3, 6, 12, 25, 64\}$ ($L + 1 = \lceil 1.2\, R_{\text{px}} \rceil + 1$ even 1-D degrees of freedom). Each entry lists pass/stopband Chebyshev ripple $\pm\delta$, with equal-radius shell anisotropy Δ_{sh} in parentheses.

R_{px}	W	L+1	R	Cont. Gaussian	Direct ℓ_2 SVD (Δ_{sh})	Disc. unconstr. (Δ_{sh})	Disc. shell-iso. (Δ_{sh})
3	9	5	4	0.07593 (0)	0.00000 (0)	0.00000 (0)	0.00000 (0)
6	17	9	4	0.07593 (0)	0.02182 (0.0111)	0.00479 (0.1285)	0.00476 ($1{\times}10^{-6}$)
12	31	16	4	0.07593 (0)	0.09496 (0.1066)	0.01899 (0.1577)	0.04273 ($4{\times}10^{-7}$)
25	61	31	4	0.07593 (0)	0.09556 (0.1770)	0.05226 (0.0905)	0.07367 ($1.8{\times}10^{-4}$)
64	155	78	4	0.07593 (0)	0.10295 (0.1923)	0.06647 (0.0323)	0.07582 ($2.0{\times}10^{-5}$)
12	31	16	8	0.01086 (0)	0.00978 (0.0140)	0.00137 (0.0240)	0.00079 ($1{\times}10^{-6}$)
25	61	31	8	0.01086 (0)	0.01295 (0.0151)	0.00486 (0.0228)	0.00552 ($4.3{\times}10^{-5}$)
64	155	78	8	0.01086 (0)	0.01625 (0.0189)	0.00755 (0.0113)	0.01065 ($2.4{\times}10^{-5}$)
64	155	78	12	0.00193 (0)	0.00480 (0.0064)	0.00100 (0.0066)	0.00171 ($1.3{\times}10^{-4}$)

$2{+}A(k_x){+}A(k_y){+}A(k_z)$ ($A(0) = 0$) lacks mixed cross-terms $k_x^2 k_y^2$: even if $A(k) = 2(\cos(\nu k){-}1)$ has zero dispersion along the coordinate axes, along $(1,1,1)/\sqrt{3}$ it gives $2{+}6(\cos(\nu k/\sqrt{3}){-}1) = 2 - \nu^2 k^2 + \frac{1}{36}\nu^4 k^4 - \ldots \neq 2\cos(\nu k) = 2 - \nu^2 k^2 + \frac{1}{12}\nu^4 k^4 - \ldots$, leaving an irreducible $O(\nu^4 k^4)$ anisotropy gap while Brillouin-corner stability $\widehat{M}_{\text{ax}}(\pi, \pi, \pi) \geq -2$ forces $A(\pi) \geq -4/3$ ($\nu k_c \leq \arccos(1/3) \approx 0.392\pi$ when A tracks $2(\cos(\nu k) - 1)$).

- **Kowalczyk–van Walstijn IWB is a single real Gaussian:** In room acoustics, Kowalczyk and van Walstijn's 27-point Interpolated Wideband (`IWB`) stencil [40] ($a = \frac{1}{4}, b = \frac{1}{16}$, $\nu \leq 1$) has face, edge, and corner weights $\frac{1}{4}\nu^2, \frac{1}{8}\nu^2, \frac{1}{16}\nu^2$ and center weight $2 - \frac{7}{2}\nu^2$, which is *identically*

$$M_{\text{IWB}}[\mathbf{m}] = (2 - 4\nu^2)\delta_{\mathbf{0}}[\mathbf{m}] + \frac{\nu^2}{2}\, 2^{-(m_x^2+m_y^2+m_z^2)} \qquad (\mathbf{m} \in \{-1, 0, 1\}^3),$$

the ($n_{\text{re}} = 1, n_{\text{cx}} = 0, L = 1, R = 1$) single-real-Gaussian member of our radial family (plus a zero-pass center tap). Because a single positive real Gaussian cannot match higher radial derivatives of $2\cos(\nu\|\mathbf{k}\|_2)$, `IWB` at $\nu = 1$ is exact on axes but incurs 9.38% diagonal phase-velocity error by 0.80π (Figure 5a,d).

Table 10: Four-stage `separableChannels` IEEE `float16` conditioning ablation ($R_{\text{px}} = 25$, $W = 61$, 512×512 HDR scenes): Stage 1 puts weights on H and unit basis on V; Stage 2 rescales each channel diagonally so $\|h_n\|_2 = \|v_n\|_2$ without mixing channels; Stages 3 and 4 apply $\mathrm{GL}_R(\mathbb{R})$ orthogonalization summed largest-$|e_n|$-first vs. smallest-$|e_n|$-first. Left columns: 20-scene mean $\pm$ std max 8-bit sRGB error (/255, peak 48×); right columns: Stage 3 → Stage 4 *linear-light* max error ($255 \cdot \max|\Delta I_{\text{lin}}|$) at peaks 8×, 48×, 256× (`seed=7`).

	20-scene max 8-bit sRGB error (/255, peak 48×)				Linear max error (/255): Desc → Asc		
Design (R)	1. Unbalanced	2. Pair-bal.	3. Orth. (desc)	4. Orth. (asc)	Peak 8×	Peak 48×	Peak 256×
$2c$ ($R = 4$)	0.12 ± 0.02	0.11 ± 0.01	0.20 ± 0.04	0.24 ± 0.08	$0.24 \to \mathbf{0.13}$	$0.24 \to \mathbf{0.16}$	$1.00 \to \mathbf{0.60}$
$4c$ ($R = 8$)	0.58 ± 0.07	0.54 ± 0.06	0.25 ± 0.07	0.26 ± 0.10	$0.29 \to \mathbf{0.14}$	$0.36 \to \mathbf{0.16}$	$1.23 \to \mathbf{0.69}$
$6c$ ($R = 12$)	7.80 ± 1.16	7.90 ± 0.92	0.30 ± 0.06	0.31 ± 0.13	$0.38 \to \mathbf{0.13}$	$0.40 \to \mathbf{0.15}$	$1.41 \to \mathbf{0.60}$
$1c_6$ ($R = 12$)	`ovf` (20/20)	15.17 ± 1.39	0.28 ± 0.05	0.27 ± 0.10	$0.32 \to \mathbf{0.15}$	$0.40 \to \mathbf{0.16}$	$1.45 \to \mathbf{0.62}$

Table 11: GPU execution times in milliseconds (mean over 10 CUDA-Event batches) for 1920×1080 RGBA 2-D blur on an RTX 4070 Ti Super across $R_{\text{px}} \in \{8, 16, 32, 64\}$ ($W = 2L + 1$, $L = \lceil 1.2\, R_{\text{px}} \rceil$): tuned dense $W \times W$ Triton (`DirectBlur2D`, rank-independent cost, std $\leq$ 0.13 ms, $\leq$ 2.1%), complex-plane-packed `cuFFT` at 7-smooth zero- and clamp-padded periods (`FFTBlur2D`, rank-independent cost, std $\leq$ 0.002 ms), and two-pass register-windowed L2-strip Triton cascades on $2c/4c/6c$ (`SeparableBlur2D`, `fp32` std $\leq$ 0.013 ms, `fp16` std $\leq$ 0.032 ms; run-to-run variation $\leq$ 6%, full metadata in `Applications/results_blur_2d.json`).

R_{px}	W	Direct W^2	`cuFFT` (zero)	`cuFFT` (clamp)	Sep. $R = 4$		Sep. $R = 8$		Sep. $R = 12$
		`fp32`	`fp32`	`fp32`	`fp32`	`fp16`	`fp32`	`fp16`	`fp32` / `fp16`
8	21	0.52	0.465	0.457	**0.331**	0.417	0.665	0.670	0.860 / 1.033
16	41	1.75	0.451	0.454	**0.354**	0.396	0.719	0.682	1.076 / 1.143
32	79	6.45	**0.452**	0.481	0.528	0.496	1.121	0.937	1.661 / 1.522
64	155	24.38	**0.473**	0.555	0.871	0.825	1.755	1.672	2.641 / 2.594

Brillouin-octant Chebyshev optimization and exact Neumann walls. Generalizing `IWB` to wider stencils $\mathbf{m} \in [-L, L]^3$, we set $M[\mathbf{m}] = \alpha_0 \delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2 + m_z^2)$ (parameters in Table 15), where α_0 fixes $\widehat{M}(\mathbf{0}) = 2$ identically and φ has exact 3-way real rank $R = n_{\text{re}} + 3n_{\text{cx}}$ (Proposition 2.13) or $R = 7$ for $\deg P = 1$ (Proposition 2.14; verified to $\leq 5.6 \times 10^{-16}$ on $[-L, L]^3$). For fixed rates (searched via differential evolution and Nelder–Mead), minimizing $\max_{0<\|\mathbf{k}\|_2 \leq k_c} |\widehat{M}(\mathbf{k}) - 2\cos(\nu\|\mathbf{k}\|_2)| / (2\nu\|\mathbf{k}\|_2 \sin(\nu\|\mathbf{k}\|_2))$ subject to exact asymptotic sound speed $\partial_{k_x}^2 \widehat{M}(\mathbf{0}) = -2\nu^2$ and strict interior stability $-2 + 10^{-3} \leq \widehat{M}(\mathbf{k}) \leq 2 - \min(10^{-3}, \frac{1}{4}\nu^2\|\mathbf{k}\|_2^2)$ across 232,620 symmetry-inequivalent Brillouin-octant and spherical-shell wavevectors (including the exact sphere $\|\mathbf{k}\|_2 = k_c$) is solved via cutting-plane HiGHS linear programming, followed by multi-start continuous Nelder–Mead maximization of $|\delta c(\mathbf{k})/c_0|$ over $0 < \|\mathbf{k}\|_2 \leq k_c$. Moreover, because every 1-D radial factor $h[m] = h[-m]$ is even, half-sample symmetric index reflection at domain boundaries simultaneously diagonalizes all 3 directional passes in the 3-D `DCT-II` cosine basis—enforcing exact rigid-wall Neumann boundary conditions ($\partial_n p = 0$) without padding buffers or boundary instabilities ($\leq 7.0 \times 10^{-7}$ max `float32` relative energy drift across all three `Rank-7` schemes over 1,000 steps in a 64^3 room; 2.0×10^{-7} at $\nu = 0.75$ and 6.1×10^{-8} at $\nu = 1.00$ for $2c + 1r$).

Table 12 and Figure 5 summarize accuracy and GPU timings across all five baselines (`DirectStep3D` is a tuned dense W^3 reflecting leapfrog Triton kernel; `SeparableStep3D` sweeps x-slabs sized to L2 cache in three passes per slab—x fan-out to R channels, y, and z fused with the leapfrog update—using the same 3×64 MiB field buffers plus 16–36 MiB of L2-resident slab scratch, 28 MiB for `Rank-7`):

- **Within separable spatial stencils:** Replacing naive ($n_{\text{re}} + 4n_{\text{cx}}$) expansion by Propo-

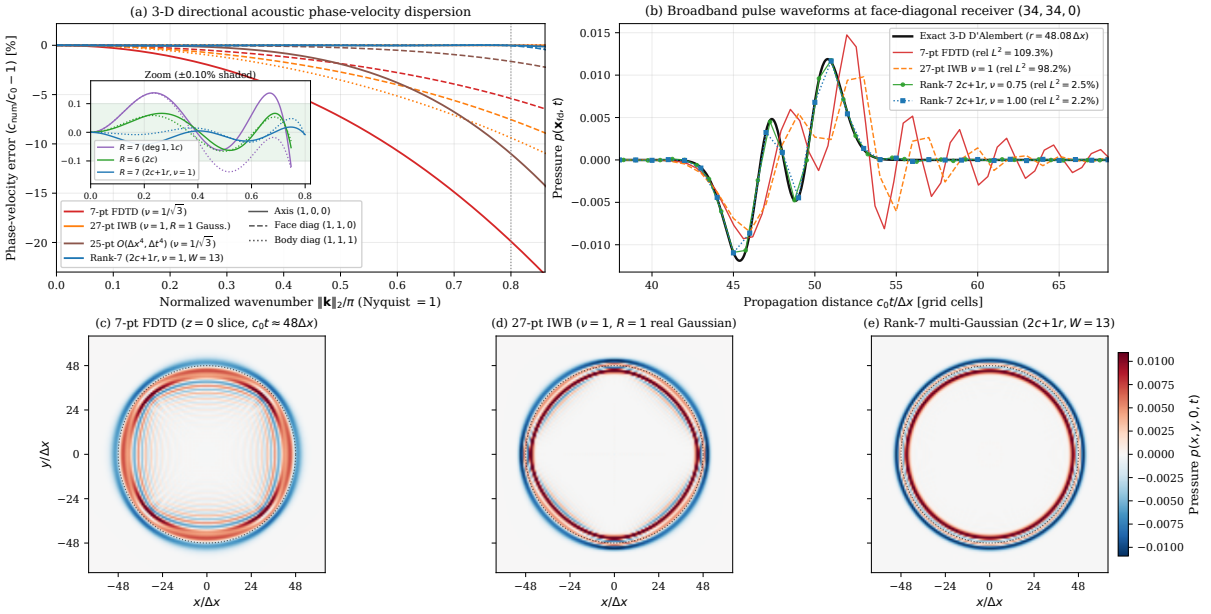


Figure 5: Three-dimensional acoustic wave propagation (`Applications/acoustics_3d.py`): (a) directional phase-velocity error ($c_{\text{num}}(\mathbf{k})/c_0 - 1$) along $(1,0,0)$, $(1,1,0)$, and $(1,1,1)$ across $\|\mathbf{k}\|_2/\pi \in [0, 0.86]$, with inset zoom ($\pm 0.10\%$ shaded) for $R = 6, 7$; (b) broadband pulse waveforms ($\sigma_0 = 1.40\,\Delta x$) on a 256^3 grid at the face-diagonal receiver $(34, 34, 0)$ ($r = 48.08\,\Delta x$) vs. the exact 3-D D'Alembert solution; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_0 t \approx 48\,\Delta x$ (dotted circle: $r = 48\,\Delta x$).

sition 2.13's and 2.14's exact rank R cuts `float32` GPU step time by 24.0% on `Rank-7` (1.51 ± 0.03 ms vs. 1.98 ± 0.02 ms, 4.7× faster than the tuned 13^3 direct Triton stencil; 15.6% on `Rank-4`, 23.5% on `Rank-6`, and 31.2% on `Rank-9`), alongside 21.2% on the 14-core CPU (283.6 ± 2.3 ms vs. 360.1 ± 0.6 ms on 128^3). On the 256^3 broadband spherical pulse test ($\sigma_0 = 1.40\,\Delta x$), all compact explicit stencils (7-pt, 13-pt, 27-pt IWB, and 25-pt $O(\Delta x^4, \Delta t^4)$) suffer 78.2%–109.3% waveform error from high-wavenumber dispersion, whereas `Rank-7` ($2c+1r$) keeps $|\delta c/c_0|$ at 0.045% ($\nu = 0.75$) and 0.049% ($\nu = 1.00$, < 1 cent) up to 0.80π (2.47 points per wavelength at $k_{0.1\%} = 0.81\pi$), achieving 1.6%–2.8% broadband waveform error across all axes and diagonals (2.8% with `float16` slab intermediates at $\nu = 0.75$). Consistent with the small-stencil regime in Table 9, freeing the $(L+1)$-tap even 1-D discrete factors from the continuous Gaussian formula under the same octant LP/Nelder–Mead objective substantially improves the smallest stencil (`Rank-4`, $W = 9, L+1 = 5$: 0.293% → 0.108%) while yielding diminishing gains as L grows (`Rank-6`, $W = 11, L+1 = 6$: 0.066% → 0.056%).

- **Comparison with 1-pass compact FDTD (including fourth-order 25-pt) and global `cuFFT`:** On 256^3 voxels, all four fused 1-pass explicit FDTD kernels (7-pt, 13-pt, 27-pt IWB, and 25-pt $O(\Delta x^4, \Delta t^4)$) run at the DRAM-bandwidth limit of 0.324–0.325 ms/step (51.7–51.8 GVox/s, 92.5% of peak GDDR6X bandwidth)—now just 3.4× to 4.7× faster per step than L2-slab `Rank-7` (1.09 ms `fp16` / 1.51 ms `fp32`). When coarse dispersion is acceptable, compact 27-pt/25-pt schemes [40, 30] remain faster and simpler. Under an analytical $(\Delta x)^{-3}(\Delta t)^{-1} = \nu^{-1}(\Delta x)^{-4}$ resolution-scaling extrapolation, `Rank-7` at $\nu = 1.00$ ($k_{0.1\%} = 0.81\pi, k_{0.5\%} = 0.87\pi$) reduces voxel count by 865× and estimated runtime by about 1,800×–2,400× relative to 27-pt IWB ($\nu = 1.00, k_{0.1\%} = 0.085\pi$) at 0.1% tolerance (94× fewer voxels and about 90×–130× at 0.5%), while against fourth-order 25-pt $O(\Delta x^4, \Delta t^4)$ ($\nu = 1/\sqrt{3}, k_{0.1\%} = 0.236\pi, k_{0.5\%} = 0.355\pi$, including the $\sqrt{3}$ Courant-step factor) it uses 40× fewer voxels and an estimated 52×–71× less runtime at 0.1% tolerance (15× fewer

Table 12: Illustrative 3-D acoustic propagators vs. compact explicit FDTD (7-pt, 13-pt $O(\Delta x^4, \Delta t^2)$, 27-pt IWB, and fourth-order 25-pt $O(\Delta x^4, \Delta t^4)$ [30]) and L2-chunked periodic k-space `cuFFT` baselines on 256^3 voxels (explicit parameters in Table 15). Here $|\delta c/c_0|$ is evaluated on each row's target passband (0.80π for FDTD baselines, 1.00π for k-space `cuFFT`, and the parenthesized k_c for each propagator), whereas $k_{0.1\%}$, $k_{0.5\%}$, and Wave L^2 use identical definitions across all rows. GPU step times depend only on (W, R) (shared across the three $W = 13, R = 7$ rows) and report means over 10 CUDA-Event batches (6 batches for direct W^3 Triton; std ≤ 0.026 ms for separable, ≤ 0.045 ms for direct, 0.048 ms for k-space `cuFFT`, ≤ 0.0003 ms for FDTD; run-to-run variation $\leq 3\%$).

Scheme	ν	W	R (nv)	$\lvert\delta c/c_0\rvert$	$k_{0.1\%}$	$k_{0.5\%}$	Wave L^2	Direct	Naive	Step `fp32`	Step `fp16`
7-pt FDTD	0.58	3	—	19.88%	0.060π	0.135π	109.3%	—	—	0.324 ms	—
13-pt $O(\Delta x^4, \Delta t^2)$	0.50	5	—	8.74%	0.099π	0.222π	100.8%	—	—	0.324 ms	—
27-pt IWB ($1r$)	1.00	3	1 (1)	9.38%	0.085π	0.191π	98.2%	—	—	0.324 ms	—
25-pt $O(\Delta x^4, \Delta t^4)$	0.58	5	—	11.00%	0.236π	0.355π	78.2%	—	—	0.325 ms	—
k-space `cuFFT` (per.)	1.00	256	—	0.000%	1.00π	1.00π	0.35%	—	—	1.44 ms	—
Rank-4 ($1c{+}1r, 0.75\pi$)	0.75	9	4 (5)	0.293%	0.14π	0.76π	11.6%	2.68 ms	1.02 ms	0.857 ms	0.627 ms
Rank-6 ($2c, 0.75\pi$)	0.75	11	6 (8)	0.066%	0.76π	0.80π	5.2%	4.63 ms	1.66 ms	1.27 ms	0.920 ms
Rank-7 (deg 1 $1c, 0.75\pi$)	0.75	13	7 (9)	0.138%	0.16π	0.78π	7.1%	7.07 ms	1.98 ms	1.51 ms	1.09 ms
Rank-7 ($2c{+}1r, 0.80\pi$)	0.75	13	7 (9)	0.045%	0.81π	0.85π	2.8%	7.07 ms	1.98 ms	1.51 ms	1.09 ms
Rank-7 ($2c{+}1r, 0.80\pi$)	1.00	13	7 (9)	0.049%	0.81π	0.87π	2.5%	7.07 ms	1.98 ms	1.51 ms	1.09 ms
Rank-9 ($3c, 0.85\pi$)	0.75	17	9 (12)	0.044%	0.86π	0.89π	2.0%	14.5 ms	3.21 ms	2.21 ms	1.57 ms

voxels and about 13×–19× at 0.5%; at $\nu = 0.75$, about 39×–54× at 0.1% and 9×–13× at 0.5%). On a *homogeneous periodic or rectangular box* where global spectral transforms apply, our L2-chunked 256^3 periodic k-space `cuFFT` step [67] (`KSpaceStep3D`: 2-D real FFTs on 32-plane chunks followed by x-FFTs and spectral multiply on 16-row chunks) takes 1.44 ± 0.05 ms on GPU (83.3 ± 4.9 ms on the 14-core CPU) with zero phase-velocity dispersion up to π (0.35% pulse L^2 error)—remaining more accurate than any finite stencil, while running at essentially the same speed as `Rank-7` in `float32` (1.51 ms, within run-to-run noise), slower than `Rank-7` with `float16` intermediates (1.09 ms) and `Rank-4`/`Rank-6` (0.86/1.27 ms), and faster than `Rank-9` (2.21 ms). Local separable stencils remain primarily motivated where global all-to-all FFTs are undesirable or inapplicable: multi-GPU domain decomposition with L-cell halos, and *spatially varying media* $c(\mathbf{x})$ below.

- **Spatially varying sound speed $c(\mathbf{x})$ where 1-step `cuFFT` k-space is inapplicable (Figure 6):** In 3-D seismic velocity modeling (sedimentary depth gradients and high-velocity salt bodies [25]), transcranial and soft-tissue medical ultrasound, and ocean or atmospheric sound-speed channels, $\partial_{tt} p = c(\mathbf{x})^2 \nabla^2 p$ has a spatially varying local Courant number $\nu(\mathbf{x}) = c(\mathbf{x})\Delta t/\Delta x$. Here exact 1-step `cuFFT` k-space propagation (`KSpaceStep3D`) fails because $2\cos(c(\mathbf{x})\Delta t\|\mathbf{k}\|_2)$ mixes $\mathbf{x}$ and $\mathbf{k}$ (requiring multi-FFT low-rank symbol expansions [25] while still imposing periodic wrap-around), whereas standard $O(\Delta t^2)$ Fourier-Laplacian pseudospectral leapfrog ($p^{n+1} = 2p^n - p^{n-1} + \nu(\mathbf{x})^2 \mathcal{F}^{-1}(-\|\mathbf{k}\|_2^2 \widehat{p^n})$) is unstable above $\nu_{\max} = \frac{2}{\sqrt{3\pi}} \approx 0.368$ and incurs 3.08% uncompensated temporal phase lead even at $\nu = 0.35$ (Figure 6a). By fixing *shared* exponential rates across space—two fixed-phase complex Gaussians plus two real Gaussians on $W = 13$ (exact rank $R = 3 + 3 + 1 + 1 = 8$, `acoustics_3d.py --hetero-only`)—all 8 shift-invariant separable 1-D cascades $\Delta_u p^n = U_u * p^n - \widehat{U}_u(\mathbf{0}) p^n$ ($u = 1, \ldots, 4$) become *independent* of $c(\mathbf{x})$, and $c(\mathbf{x})$ enters solely through local polynomial weights $\gamma_u(\nu(\mathbf{x})) = c_{u,1}\nu^2 + c_{u,2}\nu^4 + c_{u,3}\nu^6$ at accumulation:

$$p^{n+1}(\mathbf{x}) = 2p^n(\mathbf{x}) - p^{n-1}(\mathbf{x}) + \sum_{u=1}^{4} \gamma_u\big(\nu(\mathbf{x})\big)\,(\Delta_u p^n)(\mathbf{x}).$$

Across an entire 2.3 : 1 Courant range $\nu \in [0.35, 0.80]$, this single shared-rate `Rank-8` basis

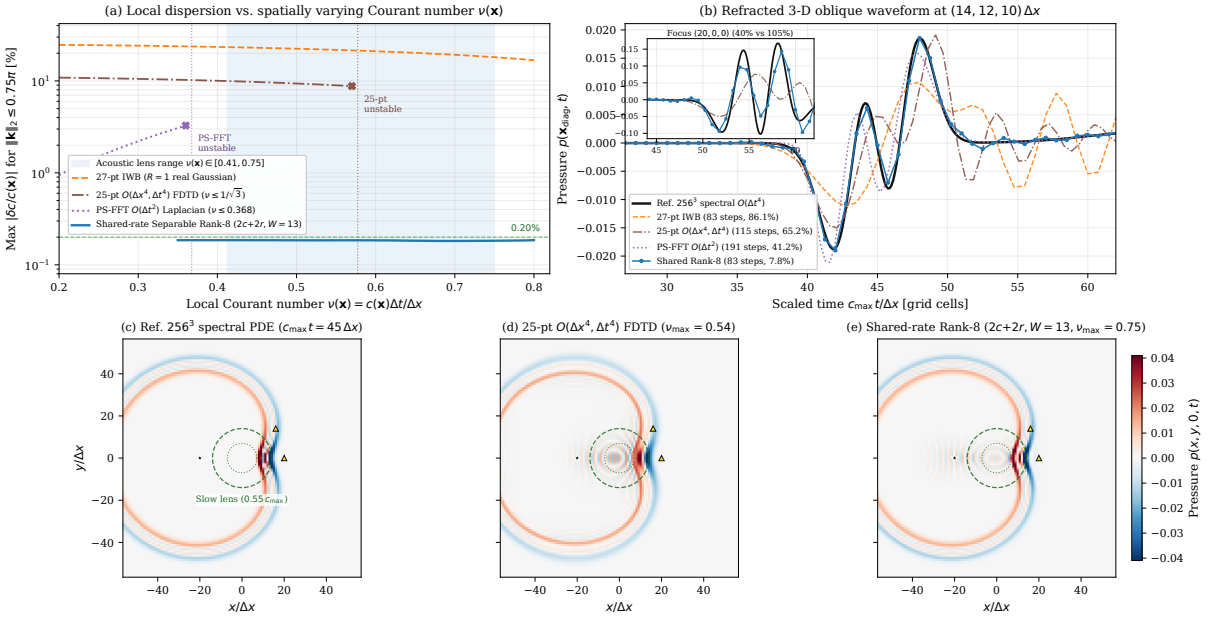


Figure 6: Three-dimensional wave propagation in a spatially varying medium $c(\mathbf{x})/c_{\max} = 1 - 0.45e^{-\|\mathbf{x}\|^2/(14\Delta x)^2}$ where 1-step `cuFFT` k-space propagation does not apply (`Applications/acoustics_3d.py --hetero-only`): (a) worst-case 3-D directional phase-velocity error up to 0.75π across local Courant numbers $\nu(\mathbf{x}) = c(\mathbf{x})\Delta t/\Delta x$ (shaded: lens range $[0.41, 0.75]$; crosses mark the stability limits of `PS-FFT` at 0.368 and 25-pt FDTD at $1/\sqrt{3}$); (b) refracted broadband waveforms ($\sigma_0 = 1.50\,\Delta x$, 128^3 grid) at the 3-D oblique receiver $(14, 12, 10)\,\Delta x$ and on-axis caustic focus $(20, 0, 0)\,\Delta x$ (inset) vs. the 2×-refined 256^3 fourth-order spectral PDE reference; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_{\max} t = 45\,\Delta x$ (dashed/dotted green circles: slow acoustic lens $r = 14\,\Delta x, 7\,\Delta x$; black star: source $(-20, 0, 0)$; yellow triangles: receivers).

keeps isotropic phase-velocity error $\le 0.185\%$ up to 0.75π with $\widehat{M}_\nu(\mathbf{k}) \in [-1.47, 2.00]$ (vs. 16.8%–23.8% for 27-pt IWB and 8.96%–10.30% for 25-pt $O(\Delta x^4, \Delta t^4)$ up to its $\nu \le 0.577$ limit; Figure 6a), and remains bounded over 1,000 reflecting-room steps ($\max|p| = 1.0000$, decaying to 0.0521 and 0.0642) on both a smooth 2 : 1 Gaussian lens and a sharp 2 : 1 spherical jump (whereas naive $\frac{1}{2}(\gamma_u \Delta_u p + \Delta_u(\gamma_u p))$ symmetrization diverges across jumps because $\Delta_u \gamma_u \neq 0$ violates constant-state preservation). On a 128^3 broadband refraction test ($\sigma_0 = 1.50\,\Delta x$, $c_{\max} t \in [0, 62\,\Delta x]$) through a low-velocity focusing lens $c(\mathbf{x})/c_{\max} = 1 - 0.45e^{-\|\mathbf{x}\|^2/(14\Delta x)^2}$ ($\nu(\mathbf{x}) \in [0.41, 0.75]$, 83 steps; Figure 6b–e) benchmarked against a 2×-refined (256^3) fourth-order spectral PDE reference, shared-rate `Rank-8` achieves 8.3% off-axis and 7.8% 3-D oblique waveform L^2 error (40.2% across the full singular caustic pulse at $(20, 0, 0)$, limited by spatial non-commutativity of the $W = 13$ stencil across steep $\nabla(c^2)$ curvature), compared with 65.2%–105.2% for 25-pt $O(\Delta x^4, \Delta t^4)$ (115 steps), 81.6%–91.0% for 27-pt IWB (83 steps), and 41.2%–73.7% for $O(\Delta t^2)$ `PS-FFT` (191 steps at $\nu_{\max} = 0.325$).

6 Limitations, Uncertainties, and Open Problems

To avoid overstating either the theoretical scope or the practical applicability of our results, we summarize six limitations, trade-offs, and remaining uncertainties:

1. **Continuous-domain exactness, discrete aliasing, and non-certified heuristic optimization:** Theorems 2.3–2.5 classify *exact* representations of *continuous* radial kernels on $\mathbb{R}^d$ (leaving $\varphi(s)$ for $s < 0$ unconstrained). Smooth non-exponential-polynomial targets ($e^{-\|x\|^4}$, Airy patterns $(J_1(r)/r)^2$, ideal circular low-pass filters) have infinite exact rank,

while discontinuous kernels require separate arguments ($\mathbf{1}_{\{\|x\|=0\}}$ has rank 1; $\mathbf{1}_{r\le 1}$ has infinite rank). Sampling a continuous rank-R kernel on $\mathbb{Z}^d$ yields discrete rank *at most* R (strictly lower at aliased frequencies $\operatorname{Im}\lambda_u \in \pi\mathbb{Z}$ or when $W < R$). Moreover, while our inner Chebyshev weight solver is an exact linear program, outer optimization over nonlinear rates (α_u, ω_u) is nonconvex and heuristic: for $6c$ disc blur ($R = 12$), `blur_2d.py` reaches ripple ± 0.00193 whereas dedicated Remez exchange in [58] reaches ± 0.0014, and in 3-D acoustics our stability check ($|\widehat{M}(\mathbf{k})| < 2$ across 232,620 Brillouin-octant/shell points plus multi-start continuous Nelder–Mead searches) is a dense numerical audit rather than a formal interval-arithmetic or sum-of-squares proof over all $\mathbf{k} \in [-\pi, \pi]^3 \setminus \{\mathbf{0}\}$.

2. **Visual and rendering limitations in 2-D lens blur:** Truncating 1-D passes to $[-S, S]$ produces a square $[-S, S]^2$ support ($r \le \sqrt{2}\, S$), and flat-top approximations built from oscillatory complex Gaussians inevitably exhibit small negative stopband ripples ($\pm\delta$). Around extreme HDR impulses ($10^3\times$–$10^5\times$ scene white), negative lobes can create dark ringing halos unless clamped or tone-mapped, whereas positive Monte Carlo disc gathering is strictly non-negative. Separable convolution is also inherently shift-invariant: real depth-of-field rendering with spatially varying circle-of-confusion $R_{\text{px}}(x, y)$ suffers foreground/background occlusion bleeding unless paired with depth-layer decomposition [27], and for purely shift-invariant blurs at $R \ge 8$ or large radii ($R_{\text{px}} \ge 32$), complex-packed 2-D `cuFFT` (0.45–0.56 ms on 1920×1080 RGBA, Table 11; layout conversion between interleaved RGBA and two complex planes not timed, and tensor-core `float16` matrix-multiply formulations not explored) is faster than direct FIR cascades.

3. **Conditioning of root-of-unity identities at large d or high multiplicity k:** In Propositions 2.13 and 2.14, summing M alternating products with prefactor $\frac{2^{d-1}}{M}$ bounds worst-case term amplification for unit-modulus factors by 2^{d-1}: benign for $d = 2, 3, 4$ ($2^{d-1} = 2, 4, 8$), but exponential in $d \gg 1$. For degree $k = \deg P \ge 1$ on $[-S, S]^d$, evaluating $U_j(X)$ on $|X| = 1$ mixes $O(1)$ and $O(S^{2k}/k!)$ coefficients; rescaling $X \mapsto \rho X$ with $\rho \approx S^{-2}$ ($\rho^{-k}[X^k] \prod_j U_j(\rho X)$) balances monomial magnitudes, while balanced roots ($|\beta|^{1/d}$) equalize mode norms. Near coalescing rates (compare the $O(h^{-1})$ weights in Section 3.4), $d = 2$ permits exact *a posteriori* $\mathrm{GL}_R(\mathbb{R})$ orthogonalization across components on $\operatorname{rowSpan}(K)$ (Figure 2c), whereas the lack of transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations in $d \ge 3$ favors well-separated complex Gaussians ($k = 0$) over high-degree repeated rates.

4. **Wide-stencil obstacles, boundaries, hardware sensitivity, and global FFTs:** While `DCT-II` half-sample reflection handles homogeneous rectangular rigid rooms cleanly, wide stencils ($L = 4$–8 grid cells in `Rank-4`–`Rank-9`) reach across thin walls or baffles and cannot be applied directly at staircased, curved, or frequency-dependent impedance boundaries without one-sided boundary closures whose passivity and stability for signed multi-Gaussian stencils are currently unknown. On simple homogeneous boxes without interior obstacles, L2-chunked periodic `cuFFT` k-space propagation (1.44 ms/step, Table 12) is dispersion-free and matches `Rank-7 fp32` (1.51 ms) in speed (though slower than `fp16`-intermediate `Rank-7` at 1.09 ms and `Rank-4`/`Rank-6`). Moreover, all GPU timings reflect one Ada Lovelace GPU (48 MB L2 cache, 672 GB/s GDDR6X) under Triton `3.9.0` / CUDA `13.2` with up to 3%–6% run-to-run autotuning variation: because `gpu_kernels.py` sizes 2-D strips and 3-D slabs explicitly to the 48 MB L2 cache, exact crossovers depend on L2 capacity, register file size, and kernel fusion (as illustrated by the 14-core CPU comparisons in Sections 5.3–5.4). Within the 3-pass-per-slab GPU kernel, replacing naive $n_{\text{re}} + 4n_{\text{cx}}$ by exact $n_{\text{re}} + 3n_{\text{cx}}$ reduces per-slab arithmetic and L2 traffic (24.0% step-time reduction on `Rank-7`), whereas on the unfused CPU baseline it reduces pass count directly (21.2% time reduction).

5. **Scope of formal verification and cross-literature connections:** Lean 4 guarantees

Table 13: Summary of the rank notions studied in this paper, illustrated on $K_3(x) = \|x\|^2 \cos(\|x\|^2)$ in $d = 3$ dimensions ($s = x_0^2 + x_1^2 + x_2^2$, single nonreal degree-1 component $\mathrm{Re}(se^{is})$).

Rank notion	Plain-language meaning (for $K : \mathbb{R}^d \to \mathbb{R}$)	General $d \geq 2$ formula / bound	On K_3
2-D slice / $\mathbb{C}$-border rank	Least complex terms in a uniform limit on bounded sets (equals 2-D slice rank)	$\sum_{\mu \in S}(\deg P_\mu + 1) = C_2(\Gamma)$	4
Real border rank	Least *real* product terms in a uniform limit on bounded sets	$\in [C_2(\Gamma),\ B_d(\Gamma)]$ ($= B_d$ if $\mathrm{Im}\,\lambda_u{=}0$)	5
Multi-Gaussian border	Least real terms in a uniform limit of *radial multi-Gaussian* kernels	$\leq B_d(\Gamma) = C_d(\Gamma) - (d-2)\sum_u k_u$	6
Exact complex $\mathrm{srank}_{\mathbb{C}}$	Least exact product terms $\sum_{n=1}^{R}\prod_{j=0}^{d-1} f_{n,j}(x_j)$ allowing $f_{n,j} : \mathbb{R} \to \mathbb{C}$	$\sum_{\mu \in S}((d-1)\deg P_\mu + 1)$	6
Exact real $\mathrm{srank}_{\mathbb{R}}$	Least exact product terms $\sum_{n=1}^{R}\prod_{j=0}^{d-1} f_{n,j}(x_j)$ with real $f_{n,j} : \mathbb{R} \to \mathbb{R}$	$C_d(\Gamma)$ ($+(d-2)$ per nonreal pair)	7

that the 32 formal theorems in `KernelRankLean` follow logically from their definitions and standard classical axioms, while human inspection remains essential to verify that those definitions (`IsSumOfProducts`, `separableRank`, `IsRepresentation`) faithfully capture the intended mathematical objects, and none of the steerable pass-count arguments, Python/Triton GPU code, or numerical tables in Section 5 are formally verified. Likewise, while (11) links our exact rank formulas directly to Kronecker ($d = 2$), Winograd–Alder–Strassen ($d = 3$), and Bläser–Mayer–Shringi ($d \geq 4$) algebra-complexity lower bounds [75, 1, 10], special cases of the explicit root-of-unity functional syntheses (Propositions 2.13–2.14) may also appear under different terminology in other communities.

6. **Open problems — closed-form real border rank for general nonreal (d, k) and passive boundary/interface closures:** Theorem 2.9 and Table 3 resolve compact-open border rank over $\mathbb{C}$ in all dimensions, over $\mathbb{R}$ for all purely real-rate representations, and on $\cos(\|x\|^2)$ ($= 3$) and $\|x\|^2\cos(\|x\|^2)$ ($= 5 < B_3 = 6$) in $\mathbb{R}^3$. Determining a closed-form formula for the exact real border rank of arbitrary nonreal higher-degree ($k \geq 2$) or higher-dimension ($d \geq 4$) representations Γ (compare border-rank degeneration bounds on local and higher-order algebras [79, 16]), incorporating $\nabla(c^2)$ commutator corrections across steep velocity gradients, and constructing provably passive boundary closures for wide signed multi-Gaussian stencils remain open.

7 Conclusions

Our theoretical, formalization, and experimental results reduce to three practical takeaways:

- **Theory (which round kernels split into 1-D passes, and at what exact cost):** A continuous radially symmetric kernel $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ ($d \geq 2$) is an exact finite sum of 1-D coordinate products if and only if $\varphi(s)$ ($s \geq 0$) is a sum of polynomials times real or complex exponentials (verified across 32 theorems in the Lean 4 proof assistant; continuous profiles that fade to zero at infinity can in turn be approximated uniformly across $\mathbb{R}^d$ by decaying complex-rate Gaussians with one shared width and equally spaced frequencies). Over $\mathbb{R}$, an alternating angle-sampling identity evaluates each complex-rate Gaussian in d real products rather than 2^{d-1} (3 vs. 4 in 3-D, a constant $d-2$ overhead over complex arithmetic across all polynomial degrees). Allowing uniform limits of lower-rank kernels on bounded regions (*border rank*) reveals a strict hierarchy in $d \geq 3$ dimensions (Table 13): over $\mathbb{C}$ (and over $\mathbb{R}$ when all rates are real), border rank drops all the way to the 2-D slice lower bound $C_2(\Gamma)$ obtained by setting $d-2$ coordinates to zero, whereas over $\mathbb{R}$ on $K_3(x) = \|x\|^2\cos(\|x\|^2)$ in 3-D, exact real rank is 7, limits of radial Gaussians take 6 terms, and general real separable limits take 5—at the price of $O(h^{-1})$–$O(h^{-3})$ cancelling weights

that favor pure Gaussians ($k = 0$) in finite precision.

- **Numerics (16-bit precision and discrete vs. continuous filter design):** In 2-D, individual complex-Gaussian components of high-accuracy disc blurs cancel by up to 595× –1,110× in amplitude; rotating the 1-D filter basis into orthogonal terms and summing from smallest to largest magnitude cuts 16-bit floating-point (`float16`) rounding error on bright bokeh highlights across 20 HDR scenes by 96% (7.90/255 → 0.31/255 on a 12-term blur) with zero extra passes. Optimizing discrete rank-R stencils directly on a pixel grid beats continuous formulas only on tiny stencils where grid half-width L approaches R; once blur radius reaches 16 pixels (61×61 taps at radius 25), enforcing equal weights at equal Euclidean radii makes discrete optimization converge onto our continuous complex-Gaussian profiles (±0.0758 vs. ±0.0759 pass/stopband ripple at $R = 4$), and sampling the continuous profile gives an immediate isotropic starting point (2.2× lower initial error at $R = 12$ than truncating a smooth disc by SVD).
- **GPU execution (when separable cascades beat direct convolution and `cuFFT`):** On an RTX 4070 Ti Super GPU (unverified empirical benchmarks checked against `float64` references), keeping intermediate channels in L2 cache makes separable FIR cascades fastest in two concrete regimes where global fast Fourier transforms (`cuFFT`) are slower or inapplicable: (i) 2-D blurs of small-to-medium radius at modest rank (0.331 ms at radius 8 px and 0.354 ms at 16 px on 1920×1080 RGBA for $R = 4$, vs. 0.45–0.46 ms for `cuFFT`, while `cuFFT` wins for larger radii or $R \geq 8$), and (ii) wideband 3-D acoustic wave propagation, where exact 3-term-per-Gaussian decompositions cut 256^3 step time by 24% over naive 4-term expansion (1.51 ms vs. 1.98 ms in `float32`, 1.09 ms with `float16` intermediates, and $< 0.05\%$ wave-speed error up to 0.80π, reducing estimated grid-scaled runtime by $52\times$ –$71\times$ against fourth-order 25-point FDTD at 0.1% dispersion tolerance) and—by fixing shared Gaussian widths with local sound-speed weights—keep wave-speed error below 0.19% across $> 2:1$ sound-speed contrasts where 1-step `cuFFT` k-space propagation cannot be used.

Determining a closed-form formula for exact real border rank at arbitrary degree $k \geq 2$ and dimension $d \geq 4$, adding gradient corrections across steep sound-speed interfaces, and designing provably passive boundary closures for wide signed wave stencils remain open problems for future work.

A Explicit Filter and Propagator Coefficients

To allow standalone reproduction of the main designs in Section 5 from this PDF (complete JSON archives for all 18 blur variants and all 6 acoustic propagators are provided in `Applications/`), every radial profile in Tables 14–15 is tabulated in the single canonical convention of Definition 2.2:

$$\begin{aligned}\varphi(s) &= \sum_{u=1}^{m} \operatorname{Re}\left(\left(\sum_{q=0}^{k_u}(p_{u,q} + iq_{u,q})s^q\right) e^{(-\alpha_u + i\omega_u)s}\right) \\ &= \sum_{u=1}^{m}\sum_{q=0}^{k_u} s^q\big(p_{u,q}\cos(\omega_u s) - q_{u,q}\sin(\omega_u s)\big)e^{-\alpha_u s},\end{aligned} \tag{17}$$

where real-rate components satisfy $\omega_u = 0$ and $q_{u,q} = 0$. In 2-D circular disc blur (Table 14), $s = (x^2 + y^2)/R_{\text{px}}^2$ on $|x|, |y| \leq (1 + \Delta r)R_{\text{px}}$ ($\Delta r = 0.2$), normalized in 2-D discrete execution so $\sum_{x,y} K[x, y] = 1$. In 3-D acoustic leapfrog propagation (Table 15), $M[\mathbf{m}] = \alpha_0\delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2 + m_z^2)$ on integer grid offsets $\mathbf{m} \in [-L, L]^3$ ($W = 2L+1$), where $\alpha_0 := 2 - \sum_{\mathbf{m}\in[-L,L]^3} \varphi(\|\mathbf{m}\|^2)$ must be computed from the evaluated φ grid so $\widehat{M}(\mathbf{0}) = 2$ holds to machine precision (avoiding

Table 14: Explicit canonical coefficients (17) for representative 2-D circular disc-blur designs ($\Delta r = 0.2$, Table 8).

Design ($R = C_2$, C_3, $\pm\delta$)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
$1c$ (R=2, C_3=3, $\pm$0.2324)	1 (0)	0.862715	1.624792	+0.767553	−1.862917
$2c$ (R=4, C_3=6, $\pm$0.0759)	1 (0)	0.894640	5.265923	+0.413699	+0.555222
	2 (0)	1.973219	1.554245	+0.510373	−4.611414
$1c_2$ (R=4, C_3=7, $\pm$0.1165)	1 (0)	2.878934	3.152485	+0.883521	−7.976371
	1 (1)	2.878934	3.152485	−16.607444	+7.931853
$2c$+$1r$ (R=5, C_3=7, $\pm$0.0446)	1 (0)	0.962816	7.154625	+0.091824	−0.456790
	2 (0)	2.075279	3.285822	−3.434558	+0.588375
	3 (0)	2.494577	0	+4.387353	0
$4c$ (R=8, C_3=12, $\pm$0.0109)	1 (0)	4.338835	1.553577	−5.754013	−46.161778
	2 (0)	3.838091	4.692138	+9.776587	+15.218676
	3 (0)	2.790788	8.181910	−3.042741	−0.293993
	4 (0)	1.345593	12.330086	+0.009312	−0.245279
$6c$ (R=12, C_3=18, $\pm$0.0019)	1 (0)	5.143786	2.080541	−82.448179	−111.252655
	2 (0)	5.613157	6.153062	+113.961587	−58.149957
	3 (0)	5.982623	9.803250	+39.766911	+161.948149
	4 (0)	6.506533	11.060675	−71.545624	−94.857314
	5 (0)	3.871248	14.810010	+1.416029	+3.711177
	6 (0)	2.202364	19.033594	−0.152648	+0.108514

exponential constant-mode blow-up $\widehat{M}(\mathbf{0}) > 2$ and preserving spatially constant initial states; note $\zeta = 1$ is a double root of $\partial_{tt}\bar{p} = 0$, so nonzero initial velocity mean gives physical $a + bn$ motion). Table 15 lists the exact α_0 recomputed from the 9-decimal coefficients (within 2.2×10^{-8} of `float64` JSON), matching $\partial^2_{k_x}\widehat{M}(\mathbf{0}) = -2\nu^2$ to $\leq 8 \times 10^{-8}$ and keeping $\widehat{M}(\mathbf{k}) \in [-1.9990, 1.9946] \subset (-2, 2)$ on the verification grid.

Code and Data Availability

All source code, machine-checked proofs, numerical optimization scripts, GPU benchmarks, JSON parameter/result archives, and figure-generation scripts for this work are openly available in the public domain (CC0 1.0 Universal) at https://github.com/OlliNiemitalo/kernel-rank-lean:

- `KernelRankLean/` and `Experiments/`: Lean 4 (`v4.36.0-rc2` with Mathlib `v4.36.0-rc2`) formalizations of all d-D and 2-D separable-rank, root-of-unity, differential-annihilator, border-rank, and alternative semigroup theorems;
- `Isabelle/`: supplementary Isabelle2026 (`HOL-Nonstandard_Analysis`) semigroup formalization `Semigroup_NSA.thy`;
- `Applications/`: Python, SciPy, PyTorch, and Triton optimization, GPU-kernel, and figure scripts (`blur_2d.py`, `acoustics_3d.py`, `gpu_kernels.py`, `generate_figures.py`) plus JSON result archives (`results_blur_2d.json`, `results_acoustics_3d.json`);
- `scripts/`: hermetic setup (`scripts/setup`), formal verification and axiom/oracle audit (`scripts/check`), isolated GPU Python wrapper (`scripts/python`), and PDF build (`scripts/build-paper`).

Acknowledgments

This work builds directly on problems, mathematical formulations, and software developed by Olli Niemitalo, including `kernel-optimizer` [58] for complex-Gaussian approximations of 2-D

Table 15: Explicit 9-decimal canonical coefficients (17) for all six 3-D acoustic leapfrog propagators in Table 12 (set $\alpha_0 := 2 - \sum_{\mathbf{m}\in[-L,L]^3} \varphi(\|\mathbf{m}\|^2)$; reference α_0 shown for verification).

Scheme (ν, W, R, α_0)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
Rank-4 (1c+1r, ν=0.75, W=9, R=4, α_0= − 1.440427676)	1 (0)	0.439979390	0.502750129	+0.137876459	+0.024929224
	2 (0)	0.888101329	0	+0.492695925	0
Rank-6 (2c, ν=0.75, W=11, R=6, α_0= − 1.419560209)	1 (0)	0.606539114	0.240982835	+0.597581203	+0.180225818
	2 (0)	0.267859019	0.385016560	−0.007190817	+0.050272529
Rank-7 (deg 1 1c, ν=0.75, W=13, R=7, α_0= − 1.434649594)	1 (0)	0.429753880	0.229296784	+0.605530664	+0.525638174
	1 (1)	0.429753880	0.229296784	−0.061586407	−0.061209821
Rank-7 (2c+1r, ν=0.75, W=13, R=7, α_0= − 1.439243312)	1 (0)	0.381717837	0.311635272	+0.110431944	+0.263079509
	2 (0)	0.163094730	0.300649461	−0.015761751	−0.008519990
	3 (0)	0.599866965	0	+0.491386225	0
Rank-7 (2c+1r, ν=1.00, W=13, R=7, α_0= − 1.959018821)	1 (0)	0.321809087	0.262377242	+0.159170237	+0.129231489
	2 (0)	0.126172334	0.278598443	−0.010612383	+0.004174471
	3 (0)	0.438442015	0	+0.241253606	0
Rank-9 (3c, ν=0.75, W=17, R=9, α_0= − 1.447076570)	1 (0)	0.301382846	0.246772855	−0.286641369	+0.234406754
	2 (0)	0.398975160	0.094363260	+0.866865360	+0.865008109
	3 (0)	0.111967878	0.275076161	−0.002296910	−0.011770692

circularly symmetric blur kernels and his investigations of isotropic low-pass filters, steerable gradient orientation estimation, and planar harmonic differential operators [54, 55, 56, 57]. Olli Niemitalo initiated and guided the `kernel-rank-lean` repository, proposed the equal-width harmonic Gaussian universal approximation construction, suggested extending the formalization to d dimensions alongside the visual pipeline exposition, discrete-vs.-continuous warm-start comparisons, and 2-D/3-D GPU and heterogeneous-medium applications, and shared independent related-work reviews that sharpened the commutative-algebra and operator-theory connections. We also thank the aiXiv Official Agent reviewers on versions 1.0, 1.1, 1.2, and 1.3, whose critiques motivated Theorems 2.5 and 2.8, Proposition 2.14, the intuitive cost guide and $k = 1$ worked example, the `separableChannels` ablation, the notation harmonization, Section 6, and Appendix A. Both the repository and this manuscript are released under CC0 1.0 Universal.

References

[1] A. Alder and V. Strassen, "On the algorithmic complexity of associative algebras," *Theoretical Computer Science* **15**(2), 201–211 (1981).

[2] F. Andersson and M. Carlsson, "On general domain truncated correlation and convolution operators with finite rank," *Integral Equations and Operator Theory* **82**(3), 339–370 (2015).

[3] P. M. Anselone and J. Korevaar, "Translation invariant subspaces of finite dimension," *Proceedings of the American Mathematical Society* **15**(5), 747–752 (1964).

[4] A. Bentkamp, J. C. Blanchette, and D. Klakow, "A formal proof of the expressiveness of deep learning," *Journal of Automated Reasoning* **63**(2), 347–368 (2019).

[5] G. Beylkin and M. J. Mohlenkamp, "Algorithms for numerical analysis in high dimensions," *SIAM Journal on Scientific Computing* **26**(6), 2133–2159 (2005).

[6] G. Beylkin and L. Monzón, "On approximation of functions by exponential sums," *Applied and Computational Harmonic Analysis* **19**(1), 17–48 (2005).

[7] J. Bigun, T. Bigun, and K. Nilsson, "Recognition by symmetry derivatives and the generalized structure tensor," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **26**(12), 1590–1605 (2004).

[8] D. Bini, M. Capovani, F. Romani, and G. Lotti, "$O(n^{2.7799})$ complexity for $n \times n$ approximate matrix multiplication," *Information Processing Letters* **8**(5), 234–235 (1979).

[9] M. Bläser, "Lower bounds for the bilinear complexity of associative algebras," *Computational Complexity* **9**(2), 73–112 (2000).

[10] M. Bläser, H. Mayer, and D. Shringi, "On the multilinear complexity of associative algebras," in *40th International Symposium on Theoretical Aspects of Computer Science (STACS 2023)*, LIPIcs **254**, pp. 12:1–12:18 (2023).

[11] G. Blekherman, "Typical real ranks of binary forms," *Foundations of Computational Mathematics* **15**(3), 793–798 (2015).

[12] D. Braess, *Nonlinear Approximation Theory*, Springer Series in Computational Mathematics **7**, Springer, Berlin (1986).

[13] D. Braess and W. Hackbusch, "Approximation of $1/x$ by exponential sums in $[1, \infty)$," *IMA Journal of Numerical Analysis* **25**(4), 685–697 (2005).

[14] J. Buczyński and J. M. Landsberg, "Ranks of tensors and a generalization of secant varieties," *Linear Algebra and its Applications* **438**(2), 668–689 (2013).

[15] P. Bürgisser, M. Clausen, and M. A. Shokrollahi, *Algebraic Complexity Theory*, Grundlehren der mathematischen Wissenschaften **315**, Springer, Berlin (1997).

[16] C.-Y. Chang, F. Gesmundo, and J. Zuiddam, "Border subrank of higher order tensors and algebras," arXiv:2604.19872 (2026).

[17] P. Comon and B. Mourrain, "Decomposition of quantics in sums of powers of linear forms," *Signal Processing* **53**(2–3), 93–107 (1996).

[18] P. Comon, G. Golub, L.-H. Lim, and B. Mourrain, "Symmetric tensors and symmetric tensor rank," *SIAM Journal on Matrix Analysis and Applications* **30**(3), 1254–1279 (2008).

[19] R. Deriche, "Using Canny's criteria to derive a recursively implemented optimal edge detector," *International Journal of Computer Vision* **1**(2), 167–187 (1987).

[20] V. de Silva and L.-H. Lim, "Tensor rank and the ill-posedness of the best low-rank approximation problem," *SIAM Journal on Matrix Analysis and Applications* **30**(3), 1084–1127 (2008).

[21] F. Dupuis, R. Y. Lewis, and H. Macbeth, "Formalized functional analysis with semilinear maps," in *13th International Conference on Interactive Theorem Proving (ITP 2022)*, LIPIcs **237**, pp. 10:1–10:19 (2022).

[22] K.-J. Engel and R. Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Graduate Texts in Mathematics **194**, Springer (2000).

[23] C. M. Fiduccia and Y. Zalcstein, "Algebras having linear multiplicative complexities," *Journal of the ACM* **24**(2), 311–331 (1977).

[24] J. D. Fleuriot, "On the mechanization of real analysis in Isabelle/HOL," in *Theorem Proving in Higher Order Logics (TPHOLs 2000)*, LNCS **1869**, pp. 145–161 (2000).

[25] S. Fomel, L. Ying, and X. Song, "Seismic wave extrapolation using lowrank symbol approximation," *Geophysical Prospecting* **61**(3), 526–536 (2013).

[26] W. T. Freeman and E. H. Adelson, "The design and use of steerable filters," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(9), 891–906 (1991).

[27] K. Garcia, "Circular separable convolution depth of field," in *ACM SIGGRAPH 2017 Talks*, Article 29, pp. 1–2 (2017).

[28] L. Greengard and J. Strain, "The fast Gauss transform," *SIAM Journal on Scientific and Statistical Computing* **12**(1), 79–94 (1991).

[29] W. Hackbusch, *Tensor Spaces and Numerical Tensor Calculus*, Springer Series in Computational Mathematics **42**, Springer, Heidelberg (2012).

[30] B. Hamilton and S. Bilbao, "Optimised 25-point finite difference schemes for the three-dimensional wave equation," *Proceedings of Meetings on Acoustics* **28**, 015022 (2016).

[31] B. Hamilton and S. Bilbao, "FDTD methods for 3-D room acoustics simulation with high-order accuracy in space and time," *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **25**(11), 2112–2124 (2017).

[32] F. L. Hitchcock, "The expression of a tensor or a polyadic as a sum of products," *Journal of Mathematics and Physics* **6**(1–4), 164–189 (1927).

[33] J. Hölzl, F. Immler, and B. Huffman, "Type classes and filters for mathematical analysis in Isabelle/HOL," in *Interactive Theorem Proving (ITP 2013)*, LNCS **7998**, pp. 279–294 (2013).

[34] A. Iarrobino and V. Kanev, *Power Sums, Gorenstein Algebras, and Determinantal Loci*, Lecture Notes in Mathematics **1721**, Springer, Berlin (1999).

[35] H. Jain, "Bounds on the real tensor rank of octonion multiplication," arXiv:2608.16649 (2026).

[36] Pl. Kannappan and P. K. Sahoo, "Rotation invariant separable functions are Gaussian," *SIAM Journal on Mathematical Analysis* **23**(5), 1342–1351 (1992).

[37] B. N. Khoromskij, *Tensor Numerical Methods in Scientific Computing*, De Gruyter, Berlin (2018).

[38] J. J. Koenderink and A. J. van Doorn, "Generic neighborhood operators," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **14**(6), 597–605 (1992).

[39] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Review* **51**(3), 455–500 (2009).

[40] K. Kowalczyk and M. van Walstijn, "Room acoustics simulation using 3-D compact explicit FDTD schemes," *IEEE Transactions on Audio, Speech, and Language Processing* **19**(1), 34–46 (2011).

[41] J. B. Kruskal, "Three-way arrays: rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics," *Linear Algebra and its Applications* **18**(2), 95–138 (1977).

[42] M. Laczkovich, "The Levi-Civita equation in function classes," *Aequationes Mathematicae* **94**(4), 689–701 (2020).

[43] P. G. Laird, "On characterizations of exponential polynomials," *Pacific Journal of Mathematics* **80**(2), 503–507 (1979).

[44] C. H. Lampert and O. Wirjadi, "An optimal nonorthogonal separation of the anisotropic Gaussian convolution filter," *IEEE Transactions on Image Processing* **15**(11), 3501–3513 (2006).

[45] J. M. Landsberg, *Tensors: Geometry and Applications*, Graduate Studies in Mathematics **128**, American Mathematical Society (2012).

[46] T. Levi-Civita, "Sulle funzioni che ammettono una formula d'addizione del tipo $f(x+y) = \sum_{i=1}^{n} X_i(x)Y_i(y)$," *Rendiconti della Reale Accademia dei Lincei* (5) **22** (2° Sem., Fasc. 5), 181–183 (1913).

[47] J. Liang, Z. Gao, and Z. Xu, "A kernel-independent sum-of-Gaussians method by de la Vallée-Poussin sums," *Advances in Applied Mathematics and Mechanics* **13**(5), 1126–1141 (2021).

[48] The mathlib Community, "The Lean mathematical library," in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pp. 367–381 (2020).

[49] J. C. Maxwell, "Illustrations of the dynamical theory of gases.—Part I. On the motions and collisions of perfectly elastic spheres," *Philosophical Magazine* (4) **19**(124), 19–32 (1860).

[50] L. de Moura and S. Ullrich, "The Lean 4 theorem prover and programming language," in *Automated Deduction—CADE 28*, LNCS **12699**, pp. 625–635 (2021).

[51] B. Mourrain, "Polynomial–exponential decomposition from moments," *Foundations of Computational Mathematics* **18**(6), 1435–1492 (2018).

[52] M. Nagumo, "Einige analytische Untersuchungen in linearen, metrischen Ringen," *Japanese Journal of Mathematics* **13**, 61–80 (1936).

[53] J. Nie and K. Ye, "Hankel tensor decompositions and ranks," *SIAM Journal on Matrix Analysis and Applications* **40**(2), 486–516 (2019).

[54] O. Niemitalo, "Circularly symmetric convolution and lens blur," technical blog post (2010), https://yehar.com/blog/?p=1495.

[55] O. Niemitalo, "Auto detection of rotation angle on arbitrary image with orthogonal features" (answer), *Signal Processing Stack Exchange* (2019), https://dsp.stackexchange.com/a/58287/15347.

[56] O. Niemitalo, "Beach-ball like differential operators of a two-dimensional function" (question; with answer by K. B. Dave), *Mathematics Stack Exchange* (2019), https://math.stackexchange.com/q/3232122.

[57] O. Niemitalo, "Efficient implementation of 2-D circularly symmetric low-pass filter," *Signal Processing Stack Exchange* (2019), https://dsp.stackexchange.com/q/58449.

[58] O. Niemitalo, *kernel-optimizer: Circularly symmetric 2-D blur kernel approximation by complex Gaussians*, GitHub repository and interactive WebGPU app, https://github.com/OlliNiemitalo/kernel-optimizer (https://yehar.com/kernel-optimizer/).

[59] I. V. Oseledets, "Tensor-train decomposition," *SIAM Journal on Scientific Computing* **33**(5), 2295–2317 (2011).

[60] P. Perona, "Deformable kernels for early vision," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **17**(5), 488–499 (1995).

[61] A. Rahimi and B. Recht, "Random features for large-scale kernel machines," in *Advances in Neural Information Processing Systems* **20** (NIPS 2007), pp. 1177–1184 (2008).

[62] M. Reisert and H. Burkhardt, "Complex derivative filters," *IEEE Transactions on Image Processing* **17**(12), 2265–2274 (2008).

[63] B. Reznick, "On the length of binary forms," in *Quadratic and Higher Degree Forms*, Developments in Mathematics **31**, pp. 207–232 (2013).

[64] R. Rochberg, "Toeplitz and Hankel operators on the Paley–Wiener space," *Integral Equations and Operator Theory* **10**(2), 187–235 (1987).

[65] I. J. Schoenberg, "Metric spaces and completely monotone functions," *Annals of Mathematics* **39**(4), 811–841 (1938).

[66] J. J. Stone, *Exponential Polynomials on Commutative Semigroups*, Applied Mathematics and Statistics Laboratories Technical Note 14, Stanford University (1960).

[67] M. Tabei, T. D. Mast, and R. C. Waag, "A k-space method for coupled first-order acoustic propagation equations," *Journal of the Acoustical Society of America* **111**(1), 53–63 (2002).

[68] C. K. W. Tam and J. C. Webb, "Dispersion-relation-preserving finite difference schemes for computational acoustics," *Journal of Computational Physics* **107**(2), 262–281 (1993).

[69] A. Townsend and L. N. Trefethen, "Continuous analogues of matrix factorizations," *Proceedings of the Royal Society A* **471**(2173), 20140585 (2015).

[70] S. Treitel and J. L. Shanks, "The design of multistage separable planar filters," *IEEE Transactions on Geoscience Electronics* **9**(1), 10–27 (1971).

[71] E. E. Tyrtyshnikov, "Incomplete cross approximation in the mosaic-skeleton method," *Computing* **64**(4), 367–380 (2000).

[72] M. Unser and N. Chenouard, "A unifying parametric framework for 2D steerable wavelet transforms," *SIAM Journal on Imaging Sciences* **6**(1), 102–135 (2013).

[73] J. van Mourik and D. Murphy, "Explicit higher-order FDTD schemes for 3D room acoustic simulation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **22**(12), 2003–2011 (2014).

[74] A. Wathen and S. Zhu, "On spectral distribution of kernel matrices related to radial basis functions," *Numerical Algorithms* **70**(4), 709–726 (2015).

[75] S. Winograd, "Some bilinear forms whose multiplicative complexity depends on the field of constants," *Mathematical Systems Theory* **10**(1), 169–180 (1977).

[76] S. Yang, R. Zhao, J. Wu, Y. Wang, W. Li, F. Chen, T. Deng, S. Hong, Y. Li, Z. Li, J. Mi, Y. Pan, K. Zhu, J. Yang, H. Chen, and Y. Yang, "A structural proof of the lower bound 21 for 3×3 matrix multiplication over $\mathbb{F}_2$," arXiv:2609.18722 (2026).

[77] I. T. Young and L. J. van Vliet, "Recursive implementation of the Gaussian filter," *Signal Processing* **44**(2), 139–151 (1995).

[78] B. Zhang, J. Zerubia, and J.-C. Olivo-Marin, "Gaussian approximations of fluorescence microscope point-spread function models," *Applied Optics* **46**(10), 1819–1829 (2007).

[79] J. Zuiddam, "A note on the gap between rank and border rank," *Linear Algebra and its Applications* **525**, 33–44 (2017).